Calculus

Career Avenues



CONTENTS		
Chapter no.	**CHAPTER NAME**	**Page no.**
2.	**Calculus**	3
	2.1 Introduction	
	2.2 Basics of functions	
	2.2.1 Types of functions	
	2.3 Limits	
	2.4 Continuity	
	2.4.1 L' Hospital rule	
	2.5 Extrema of continuous functions (maxima & minima)	
	2.5.1 Conditions for maxima and minima	
	2.5.2 Geometrical illustrations	
	2.5.3 Extrema of discontinuous function	
	2.6 Derivatives:	
	2.6.1 Differentiation and the derivative	
	2.7 Differentiability and Continuity	
	2.7.1 Some standard results on differentiability	
	2.8 Rolle's & Lagrange's Theorem	
	2.8.1 Rolle's Theorem	
	2.8.2 Lagrange's Mean value theorem	
	2.9 The Mean Value Theorem	
	2.10 Partial derivatives:	
	2.10.1 Homogenous function:	
	2.10.2 Euler's Theorem:	
	2.11 Taylor series	
	2.12. Maclaurin series	
	2.13 Integral calculus	
	2.13.1 Introduction to Integration	
	2.13.2 Integrating discrete functions	
	2.14 Definite Integral	
	2.15 Indefinite Integral	
	SET-4	
	SET-5	
	SET-6	
	Past GATE Questions	

Career Avenues

Chapter - 2 Calculus

2.1 Introduction

Calculus is a branch of mathematics involving or leading to calculations dealing with continuously varying functions – such as velocity or acceleration, rates of change or maximum and minimum values of curves. Calculus has widespread applications in science and engineering and is used to solve complicated problems for which algebra alone is insufficient.

Calculus is a subject that falls into two parts:

a) Differential calculus or differentiation
b) Integral calculus or integration.
c) Vector Calculus

2.2 BASICS OF FUNCTIONS

INTRODUCTION

The concept of a function expresses dependence between two quantities, one of which is known and the other which is produced. A function associates a single output to each input element drawn from a fixed set, such as the real numbers, although different inputs may have the same output.

There are many ways to give a function: by a formula, by a plot or graph, by an algorithm that computes it, or by a description of its properties. Sometimes, a function is described through its relationship to other functions (see, for example, inverse function). In applied disciplines, functions are frequently specified by their tables of values or by a formula. Not all types of description can be given for every possible function, and one must make a firm distinction between the function itself and multiple ways of presenting or visualizing it.

2.2.1 TYPES OF FUNCTIONS

In this section some different types of functions are introduced which are particularly useful in calculus.

LINEAR FUNCTIONS:-

The generalized form for a linear function (1 is highest power):

$f(x) = ax+b$, where a and b are constants, and a is not equal to 0

The generalized form for a quadratic function (2 is highest power):

$f(x) = ax^2+bx+c$, where a, b and c are constants, and a is not equal to 0

The generalized form for a cubic function (3 is highest power):

$f(x) = ax^3+bx^2+cx+d$, where a, b, c and d are constants, and a is not equal to 0

The roots of a function are defined as the points where the function $f(x)=0$. For linear and quadratic functions, this is fairly straight–forward, but the formula for a cubic is quite complicated and higher powers get even more involved.

A system of linear equations (or linear system) is a collection of linear equations involving the same set of variables. For example,

$3x + 2y – z = 1$
$2x – 2y + 4z = – 2$
$– x + 1/2\ y – z = 0$

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POLYNOMIAL FUNCTIONS:

Here a few examples of polynomial functions:

$f(x) = 4x^3 + 8x^2 + 2x + 3$

$g(x) = 2.5x^5 + 5.2x^2 + 7$

$h(x) = 3x^2$

$i(x) = 22.6$

Polynomial functions are functions that have this form:

$$f(x) = a_n x^n + a_{n-1}x^{n-1} + ... + a_1 x + a_0$$

The value of 'n' must be an non-negative integer. That is, it must be whole number, it is equal to zero or a positive integer.

The coefficients, as they are called, are a_n, a_{n-1}, ..., a_1, a_0. These are real numbers.

The degree of the polynomial function is the highest value for 'n', where 'n' is not equal to 0.

So, the degree of $g(x) = 2.5x^5 + 5.2x^2 + 7$ is 5.

EXPONENTIAL FUNCTION:

The exponential function is the entire function defined by

$\exp(z) = e^z$

It satisfies the identity

$\exp(x + y) = \exp(x) \exp(y)$

If $z \equiv x + iy$, ,

$e^z = e^{x+iy} = e^x e^{iy} = e^x (\cos y + i \sin y)$

LOGARITHMIC FUNCTION

In mathematics, the logarithm of a number to a given base is the power or exponent to which the base must be raised in order to produce the number.

For example:- the logarithm of 1000 to the base 10 is 3, because 3 is how many 10s you must multiply to get 1000: thus $10 \times 10 \times 10 = 1000$; the base 2 logarithm of 32 is 5 because 5 is how many 2s one must multiply to get 32: thus $2 \times 2 \times 2 \times 2 \times 2 = 32$. In the language of exponents: $10^3 = 1000$, so $\log_{10}1000 = 3$, and $2^5 = 32$, so $\log_2 32 = 5$.

The logarithm of x to the base b is written $\log_b(x)$ or, if the base is implicit, as $\log(x)$. So, for a number x, a base b and an exponent y,

$$\text{If } x = b^y, \text{ then } y = \log_b(x)$$

An important feature of logarithms is that they reduce multiplication to addition, by the formula:

$$\log(xy) = \log x + \log y \quad \text{and} \quad \log(x/y) = \log x - \log y$$

That is, the logarithm of the product of two numbers is the sum of the logarithms of those numbers. The use of logarithms to facilitate complicated calculations was a significant motivation in their original development.

EVEN FUNCTIONS: Consider $f(x) = x^2 + 2$. If we replace x by –x, we get $f(-x) = (-x)^2 + 2 = x^2 + 2 = f(x)$. On replacing x by –x, the function does not change. Such a function is called an Even Function.

A function f is thus said to be an even function if, $f(-x) = f(x)$ for all x and (–x), which belong to the domain of f.

ODD FUNCTION: Consider $f(x) = x^3 + 4x$.

If we replace x by (–x) we get, $f(-x) = (-x)^3 + 4(-x) = -x^3 - 4x = -(x^3 + 4x) = -f(x)$.

In this case, the replacement of x by (–x) changes only the sign of the function. Such a function is called an odd function.

A function f is thus said to be an odd function, if $f(-x) = -f(x)$ for all x and (–x) which belong to the domain of f.

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Note:-
(i) $f(x) = x^2$, $f(x) = x^4$. or in general $f(x) = x^{2n}$, where n is any integer, are even functions.
(ii) f(x) = cos x and f(x) = sec x are even function.
(iii) $f(x) = x^3$, $f(x) = x^5$ or in general $f(x) = x^{2n+1}$ where n is any integer, are odd functions.
(iv) f(x) = sin x, f(x) = tan x, f(x) = cosec x, f(x) = cot x are odd functions.

Maximum and minimum values of functions:

Example:- (i) Find the maximum values of $5 - 2x - x^2$
(i.e find the algebraically greatest value of $5 - 2x - x^2$ for various values of x)
The given expression $5 - 2x - x^2 = 5 + 1 - 1 - 2x - x^2$
$= 6 - (1 + x)^2 = 6 + \{-(1 + x)^2\}$
Since $(1 + x)^2$ cannot be negative, $\{-(1 + x)^2\}$ cannot be positive.
Hence whatever real values x may have, the given expression can never be greater then 6.
Evidently, the given expression = 6, when x + 1 = 0 i.e x = – 1
Hence we notice that the expression can never be greater than 6.
Hence the maximum value of the expression = 6.

Example:- (ii) Find the minimum value of $4x^2 + 12x + 18$
(i.e find the algebraically smallest value of $4x^2 + 12x + 18$ for various values of x).
The given expression = $(2x)^2 + 2(2x)(3) + (3)^2 + 9$
$= (2x + 3)^2 + 9$
Since $(2x + 3)^2$ cannot be negative, the given expression can never be less than 9, but can be equal to 9, when 2x + 3 = 0 i.e when $x = \frac{-3}{2}$.
The smallest value required = 9.

2.3 LIMITS

The *limit of a function f*(*x*) at some point x_0 exists and is equal to *L* if and only if every "small" interval about the limit *L*, say the interval $(L - \varepsilon , L + \varepsilon)$ means you can find a "small" interval about x_0, say the interval $(x_0 - \delta , x_0 + \delta)$ which has all values of *f*(*x*) existing in the former "small" interval about the limit *L*, except possibly at x_0 itself.

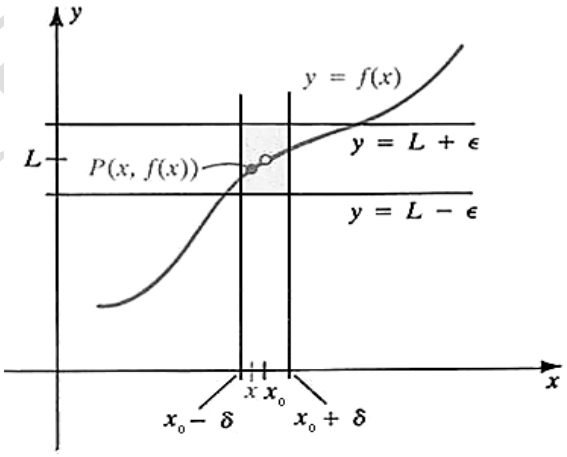


Figure 2. 1

This is a difficult concept to fully appreciate. However, you should be able to grasp the idea through several examples.

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Examples:

1. Consider $f(x) = x^2 - x - 6$. Find the limit as x approaches 1. It is not hard to see from either the graph or from the way you have always evaluated this quadratic function that as x approaches 1, $f(x)$ approaches –6, since $f(1) = -6$.

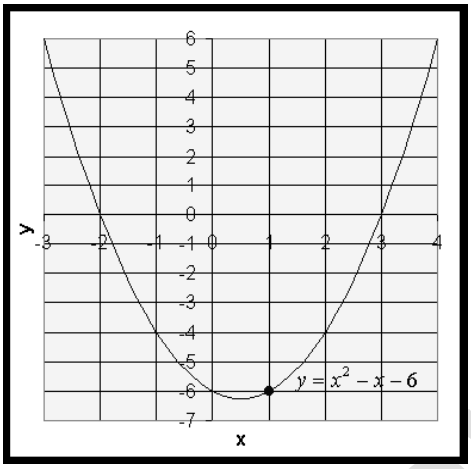


Figure 2. 2

Fact: Any polynomial, $p(x)$, has as its limit at some x_0, the value of $p(x_0)$.

2. Consider the rational function $r(x) = (x^2 - x - 6)/(x - 3)$. Find the limit as x approaches 1. If x is not 3, then this rational function reduces to $r(x) = x + 2$. So as x approaches 1, this function simply goes to 3.

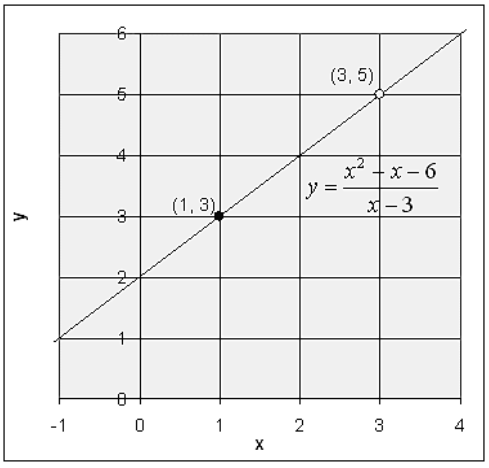


Figure 2. 3

Fact:- Any rational function, $r(x) = p(x)/q(x)$, where $p(x)$ and $q(x)$ are polynomials with $q(x_0)$ not zero, then the limit exists with the limit being $r(x_0)$.

3. Consider the rational function in Example 2. Now find the limit as x approaches 3. Though $r(x)$ is not defined at $x_0 = 3$, we can see that arbitrarily "close" to 3, $r(x) = x + 2$. So as x approaches 3, this function simply goes to 5. Its limit exists though the function is not defined at $x_0 = 3$.

Career Avenues

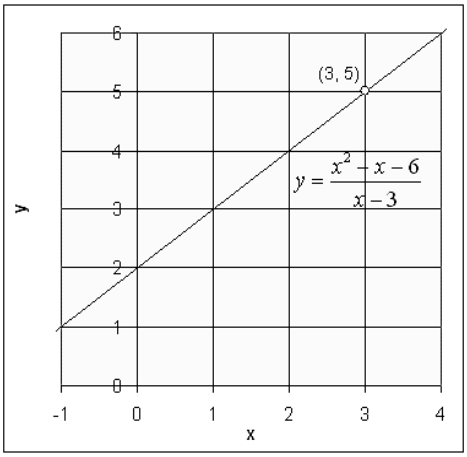


Figure 2. 4

4. Consider the rational function $f(x) = 1/x^2$. Find the limit as x approaches 0, if it exists. From our statement above on rational functions, this function has a limit for any value of x_0 where the denominator is not zero. However, at $x_0 = 0$, this function is undefined. Thus, the graph has a vertical asymptote at $x_0 = 0$. This means that no limit exists for $f(x)$ at $x_0 = 0$.

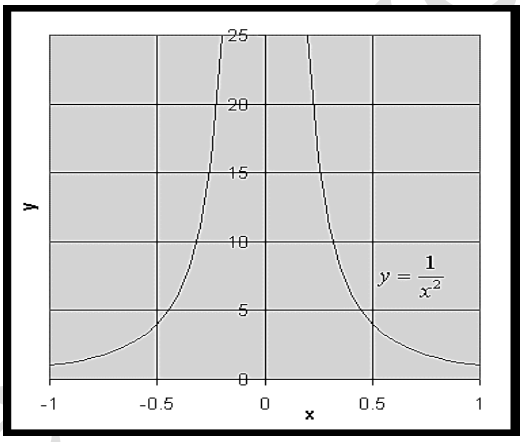


Figure 2. 5

Fact: Whenever you have a vertical asymptote at some x_0, then the limit fails to exist at that point.

2.4 CONTINUITY

Closely connected to the concept of a limit is that of continuity. Intuitively, the idea of a continuous function is what you would expect. If you can draw the function without lifting your pencil, then the function is continuous. Most practical examples use functions that are continuous or at most have a few points of discontinuity.

Definition: A *function* $f(x)$ is continuous at a point x_0 if the limit exists at x_0 and is equal to $f(x_0)$.

Career Avenues

Comparing Limits and Continuity

An example is provided to show the differences between limits and continuity. Below is a graph of a function, $f(x)$, that is defined on the interval [–2, 2], except at $x = 0$, where there is a vertical asymptote.

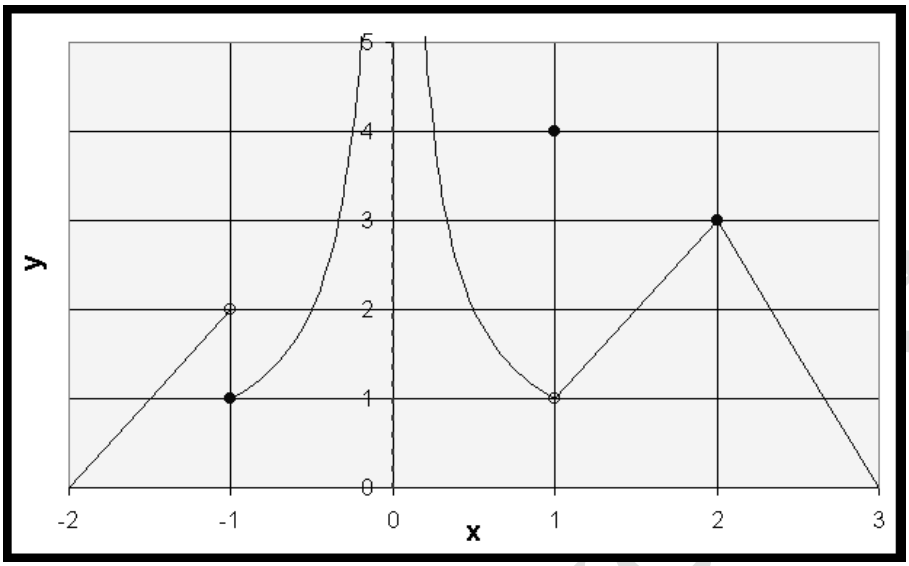


Figure 2. 6

It is clear that the difficulties with this function occur at integer values.

At $x = –1$, the function has the value $f(–1) = 1$, but it is clear that the function is not continuous nor does a limit exist at this point.

At $x = 0$, the function is not defined (not continuous nor has any limits) as there is a vertical asymptote.

At $x = 1$, the function has the value $f(1) = 4$. The function is not continuous at $x = 1$, but the limit does exist with $\lim_{x \to 1} f(x) = 1$

At $x = 2$, the function is continuous with $f(2) = 3$, which also means that the limit exists. At all non–integer values of x the function is continuous (hence its limit exists).

Definition of a Limit:-

Let f be a function defined on an open interval containing c (except possibly at c) and let L be a real number. The statement $\lim_{x \to c} f(x) = L$ means that for each $\varepsilon > 0$, there exists a $\delta > 0$ such that

if $0 < |x - c| < \delta$

then $|f(x) - L| < \varepsilon$.

Explanation of Definition:- "f(x) becomes arbitrarily close to L" means f(x) lies in the interval (L – ε , L + ε) which can be written

| f(x) – L | < ε

Similarly, "x→c" means there exists a positive number δ such that x lies in either the interval (c – δ, c) or (c, c + δ) or 0 < | x – c | . x is within the distance δ of c.

Career Avenues

Notation:-

- The symbol $\rightarrow$ means "approaches" in calculus when we discuss limits.
- $x \rightarrow 2$ is read "x approaches 2".
- As superscripts, the symbol '$^+$'(plus) means 'from the right' (or positive direction), and '$^-$'(minus) means "from the left" when we discuss limits. Thus
  - $x \rightarrow 2^+$ means "x approaches 2 from the right".
  - $x \rightarrow 2^-$ means "x approaches 2 from the left".

Sample Proof:-

Prove: $\lim_{x \to -3}(2x+3) = -3$

We must show that given ε > 0, then there exists a δ > 0 such that | f(x) – L | < ε when 0 < | x – c | < δ. Because δ depends on ε, we need to establish a relationship or connection between | f(x) – L | < ε and | x – c | or in this case: | 2x + 3 – (–3) | and | x – (–3) |

Given ε > 0,

| f(x) – L | < ε → | 2x + 3 – (–3) | < ε
| 2x + 6 | < ε
2 | x + 3 | < ε
| x + 3 | < ε/2 = δ

So let δ = ε/2

Hence, if 0 < | x + 3 | < δ = ε/2, you have
| x + 3 | < ε/2
| 2x + 6 | < ε
| 2x + 3 – (–3) | < ε
| f(x) – L | < ε

Graphical and Numerical Approaches:- Consider the function $f(x) = \dfrac{x^2-1}{x-1}$. Note that the function is *undefined* at the value $x = 1$, owing to the fact that the denominator $x - 1$ is equal to zero there. Otherwise, $f(x)$ behaves like the linear function $x + 1$, since

$\dfrac{x^2-1}{x-1} = \dfrac{(x+1)(x-1)}{x-1} = (x+1), x \neq 1$. So the graph of $f(x)$ would be the same as the graph of the line $y = x + 1$, except at $x = 1$, where there is a 'hole' in the function:

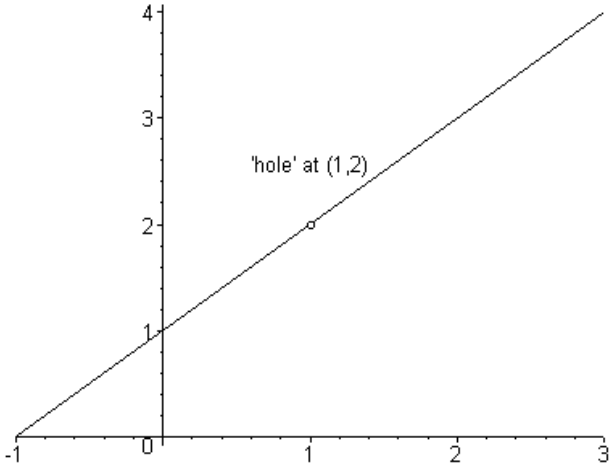


Figure 1

What we would like to do is examine the behavior of the function *near* $x = 1$, and draw some conclusions. There are too ways we might go about this. One is to simply check out values of $f(x)$ for x's near $x = 1$, perhaps in a table:

Career Avenues

x	0.75	0.9	0.99	0.999	1	1.001	1.01	1.1	1.25
$f(x)$	1.75	1.9	1.99	1.999		2.001	2.01	2.1	2.25

Take a look at the top row from left to right until we get to 1 – the values go up slowly *from the left* toward 1: .75, .9, .99, .999, and underneath them the (*y*–) values of $f(x)$ increase toward what appears to be 2: 1.75, 1.9, 1.99, 1.999. Thus, we feel comfortable saying that as

$x \to 1^-, f(x) \to 2.$

Now look in from the right side at the top row – the values of *x* decrease: 1.25, 1.1,1.01, 1.001. The corresponding *y*–values of $f(x)$ also decrease: 2.25, 2.1, 2.01, 2.001, seemingly "zooming in" on the value 2. We say that as

$x \to 1^+, f(x) \to 2.$

And so it seems that as *x* approaches 2 from either side, our function $f(x) = \frac{x^2 - 1}{x - 1}$ has *y*–values that "zoom in", or approach the value $y = 2$. We characterize this behavior as a *limit* of $f(x)$ and introduce the following notation:

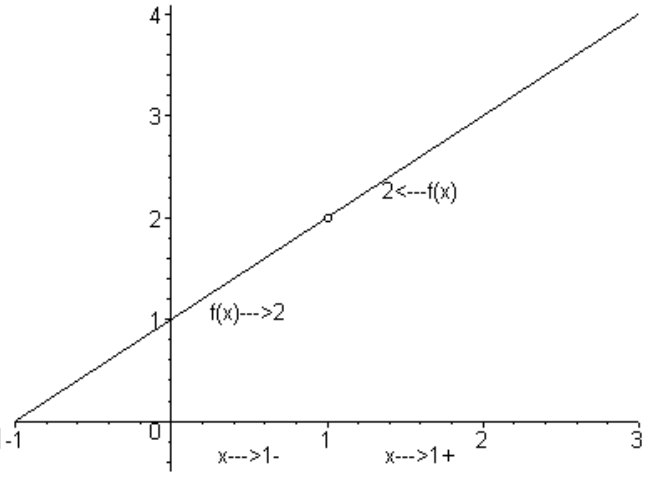
$\lim_{x \to 1} f(x) = 2.$

This is read "the limit of *f* as *x* approaches *x* is 2", and generally:

$\lim_{x \to c} f(x) = L.$

is read "the limit of *f* as *x* approaches *c* is *L*". The formal definition of limit will be given in a later section.

Our deduction of the limit in this case was an example of the use of a *numerical approach*, but this is not the only technique available for finding said limit. For instance, this limit could have been determined by inspecting the graph of the function:



Note that as we approach the *x*–value of 1 from either side, the function "zooms in on" the hole at (1,2), and so the *y*–values ($f(x)$) are approaching $y = 2$.

Limits of some trigonometric functions:

1) $\lim_{x \to 0} \frac{\cos x - 1}{x} = 0$ 2) $\lim_{x \to 0} \frac{\sin x}{x} = 1$

Definition of Limit for a Function of Two Variables:-

Let *f* be defined on the interior of a circle centered at the point (*a*, *b*), except possibly at (*a*, *b*) itself. We say that $\lim_{x,y \to a,b} f(x,y) = L$ if for every $\varepsilon > 0$ there exists a $\delta > 0$ such that $|f(x,y) - L| < \varepsilon$ whenever $0 < \sqrt{(x-a)^2 + (y-b)^2} < \delta$.

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Limits of combinations of functions:-

1) The limit of a sum or difference is the sum or difference of the limits.

$$\lim_{x,y \to a,b} \left[f(x,y) \pm g(x,y) \right] = \lim_{x,y \to a,b} f(x,y) \pm \lim_{x,y \to a,b} g(x,y)$$

2) The limit of a product is the product of the limits

$$\lim_{x,y \to a,b} \left[f(x,y)\, g(x,y) \right] = \left[\lim_{x,y \to a,b} f(x,y) \right] \left[\lim_{x,y \to a,b} g(x,y) \right]$$

3) The limit of a quotient is the quotient of the limits (except where the denominator is zero)

$$\lim_{x,y \to a,b} \left(\frac{f(x,y)}{g(x,y)} \right) = \frac{\lim_{x,y \to a,b} f(x,y)}{\lim_{x,y \to a,b} g(x,y)}$$

4) The limit of a polynomial always exists and is found simply by substitution.

$$\lim_{x,y \to a,b} P_n(x,y) = P_n(a,b)$$

Notes on disproving limits:-

For a limit to exist, the function must approach that limit for every possible path of (x, y) approaching (a, b). Thus, it is usually very hard to prove a limit exists, and easier to show a limit does not exist.

So, if a function $f(x, y)$ approaches L_1 as (x, y) approaches (a, b) along a path P_1 and $f(x, y)$ approaches $L_2 \neq L_1$ as (x, y) approaches (a, b) along a different path P_2, then $\lim_{x,y \to a,b} f(x,y)$ **does not exist (DNE)**.

Some simple paths to try are the lines along $x = a$, $y = b$, or any other line through the point.

How can you show a limit *does* exist? One way is with the squeeze theorem.

The **Squeeze Theorem** allows us to find the limit of a function at a certain point, c, so long as the function is squeezed between two other functions whose limits at c are the same. The trick, of course, is finding those two functions in between which our function exists. The Squeeze Theorem has been used to define the following important trigonometric limits, which should be committed to memory:

$$\lim_{x \to 0} \frac{\sin x}{x} = 1 \text{ and } \lim_{x \to 0} \frac{1 - \cos x}{x} = 0$$

Squeeze Theorem

Let $A \subseteq R$, let f, g and h be functions on A to R, and let $x_o \in R$. If $f(x) \leq g(x) \leq h(x) \quad \forall x \in A,\ x \neq x_o$ and if $\lim_{x \to x_o} f(x) = L = \lim_{x \to x_o} h(x)$, then $\lim_{x \to x_o} g(x) = L$

Continuity of a Function of Two Variables:-

Suppose $f(x, y)$ is defined in the interior of a circle centered at the point (a, b). We say that f is **continuous** at (a, b) if $\lim_{x,y \to a,b} f(x,y) = f(a,b)$. If $f(x, y)$ is not continuous at (a, b), then we call (a, b) a **discontinuity** of f.

Definition of Limit for a Function of Three Variables:-

Let $f(x, y, z)$ be defined on the interior of a sphere centered at the point (a, b, c) except possibly at (a, b, c) itself. We say that $\lim_{x,y,z \to a,b,c} f(x,y,z) = L$ if for every $\varepsilon > 0$ there exists a $\delta > 0$ such that $\left| f(x,y,z) - L \right| < \varepsilon$ whenever $0 < \sqrt{(x-a)^2 + (y-b)^2 + (z-c)^2} < \delta$.

Continuity of a Function of Three Variables:-

Suppose $f(x, y)$ is defined in the interior of a sphere centered at the point (a, b, c). We say that f is **continuous** at (a, b, c) if $\lim_{x,y,z \to a,b,c} f(x,y,z) = f(a,b,c)$. If $f(x, y, z)$ is not continuous at (a, b, c), then we call (a, b, c) a **discontinuity** of f.

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One sided limits:-

As the name implies, with one–sided limits we will only be looking at one side of the point in question. There are right handed and left handed limits.

Right handed limit:- $\lim x \to a^+ f(x)=L$ provided we can make f(x) as close to L as we want for all x sufficiently close to a and x>a without actually letting x be a.

Left handed limit:- we say $\lim x \to a^- f(x)=L$ provided we can make f(x) as close to L as we want for all x sufficiently close to a and x<a without actually letting x be a.

Limits to Infinity: - occur when f(x) approaches either positive or negative infinity as it approaches a.

CONTINUITY AND ONE–SIDED LIMITS:-

A function, *f*, is **continuous at a point**, *c*, if three conditions are met:

(1) *f(c)* is defined, (2) $\lim_{x\to c} f(x)$ exists, and (3) $\lim_{x\to c} f(x) = f(c)$.

If *f* is not continuous at *c*, it is said to be **discontinuous at *c***. This can occur if any of the three conditions above are not met. If the limit exists at *c*, then the **discontinuity is removable**, i.e., defining or redefining the function at *c* will make the function continuous. If the limit does not exist at *c*, then the **discontinuity is non–removable**.

A function, *f*, is **continuous over an open interval**, (*a*, *b*), if for every point within the interval, the three conditions above are met. Polynomial, rational, radical and trig functions are continuous on every point in their domains. A function that is continuous over the entire real number line, i.e., (−∞, ∞), is said to be **everywhere continuous**. When asked to describe the continuity of a function, you must (1) identify any points of discontinuity, (2) categorize those discontinuities as removable or non–removable, and (3) define all intervals of continuity.

Continuity is a property and it is preserved when functions are combined through addition, subtraction, multiplication, division and composition.

One–sided limits, i.e., $\lim_{x\to c^+} f(x)$ and $\lim_{x\to c^-} f(x)$, can be evaluated. In order for $\lim_{x\to c} f(x)$ to exist, the right–handed and left–handed limits must exist and be the same.

One–sided limits allow us to extend **continuity to closed intervals**: [a, b]. In this case, the function must be shown continuous over the open interval, (a, b), and then one–sided limits of each edge must be shown to be equal to the function's values at those edge points,

i.e., $\lim_{x\to a^+} f(x) = f(a)$ and $\lim_{x\to b^-} f(x) = f(b)$.

INFINITE LIMITS:-

Infinite limits occur as *x* approaches a specific *c* value when the value of a function grows without bounds, i.e., $\lim_{x\to c} f(x) = \infty$ or declines without bounds, i.e., $\lim_{x\to c} f(x) = -\infty$. These statements do not mean that the limit exists. On the contrary, they indicate that the limit does not exist and the function's value is growing or declining without bounds.

One sided limits can also be evaluated to determine whether they are infinite, i.e., $\lim_{x\to c^+} f(x) = \pm\infty$.

Example:-

f(x)= $\dfrac{x^2-3x}{4x+3}$ is a closed form function, and x = 2 is in its domain.

Therefore we can obtain $\lim_{x\to 2}$ by substitution: $\lim_{x\to 2} \dfrac{x^2-3x}{4x+3} = f(2) = -\dfrac{2}{11}$

Continuity at a point:-

A function is **continuous at "c"** if each of these three (3) conditions is met:

1. $f(c)$ is defined 2. $\lim_{x\to c} f(x)$ exists 3. $\lim_{x\to c} f(x) = f(c)$

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Continuity on an Open Interval:-
A function is **continuous on an open interval (a, b)**, if it is continuous at every point within the interval.

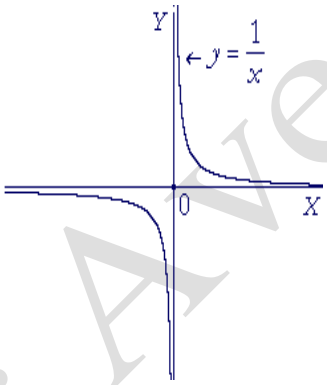
Continuity Everywhere:-
A function is **continuous everywhere**, if (a, b) is $(-\infty, \infty)$.

Point of discontinuity :- A function is said to have a point of discontinuity at x = a, or the graph of the function has a hole at x = a, if the original function is undefined for x = a, whereas the related rational expression of the function in simplest form is defined for x = a.

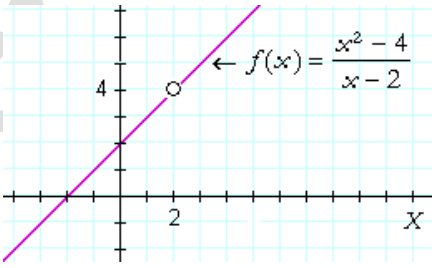
Discontinuous:- A point at which a function is discontinuous is called a discontinuity, or sometimes a jump.

Jump discontinuity:- If the left– and right–hand limits at x=a exist but disagree, then the graph jumps at x=a.

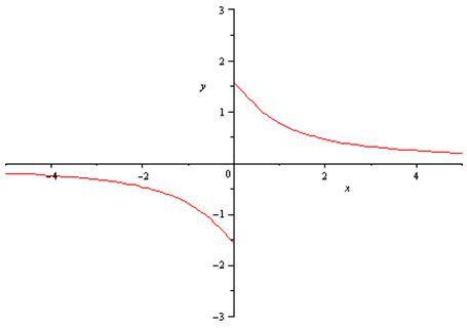
Example of discontinuity :-
Asymptote



Point discontinuity

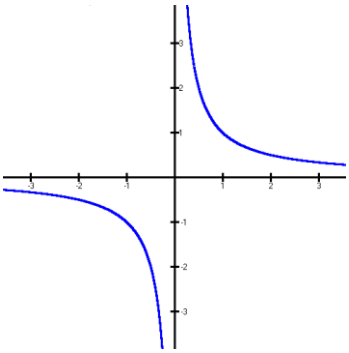


Jump Discontinuity



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EXAMPLE:-

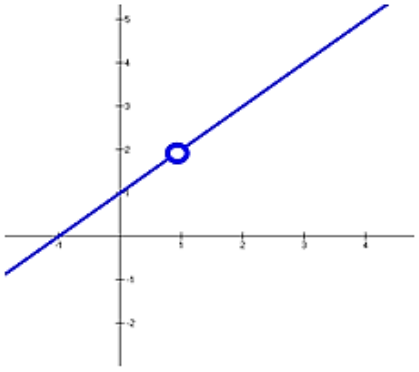


Discuss the continuity of each function:

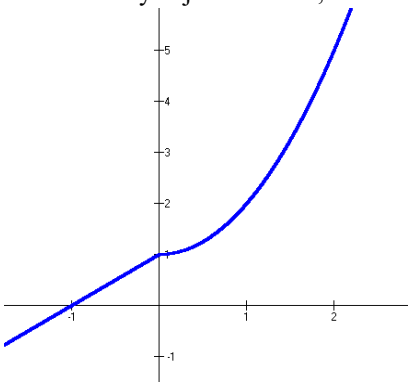
a. $f\ x = \frac{1}{x}$

Since at $x = 0$, the function is not defined, there is a discontinuity at $x = 0$. Therefore f is continuous for $-\infty, 0 \cup 0, \infty$. Considering the familiar graph we can see it is a NONREMOVABLE – INFINTE type.

b. $f\ x = \frac{x^2 - 1}{x - 1}$



Again, we can see that $x = 1$ will produce a 0 in the denominator. All other x values are in the domain, so f is continuous for $-\infty, 0 \cup 0, \infty$. However, due to analytic cancellation, $g(x) = x + 1$ is a function identical to f except for the "hole" at (1, 2). Since the discontinuity is just a "hole", this is the REMOVABLE type.

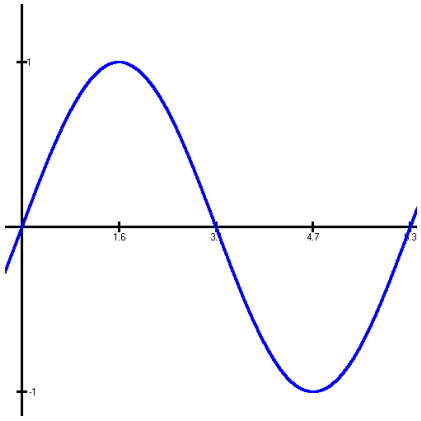


c. $f\ x = \begin{cases} x+1 & x \leq 0 \\ x^2+1 & x > 0 \end{cases}$

This is a special type of function which joins more than 1 equation. 2 separate *pieces* of graphs are used.

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f is called a "PIECEWISE" function. Since the pieces join, and do not "JUMP", the function is CONTINUOUS for all $x \ldots -\infty, \infty$.



d. $f\ x = \sin x$

The domain for this trigonometric function is all real numbers. There are no discontinuities at all. So the interval of continuity is $-\infty, \infty$.

ONE–SIDED LIMITS:-

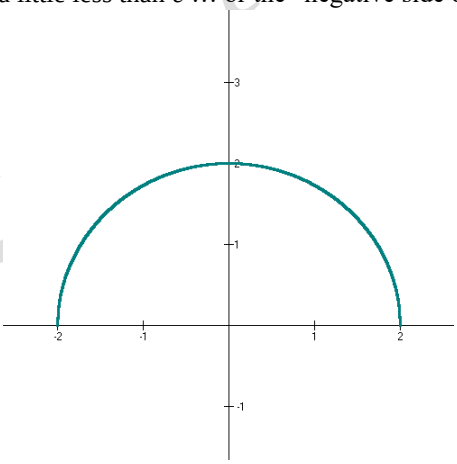
We have already suggested that it is necessary to examine "***a neighborhood***" both to the left and to the right of "*c*", when evaluating a limit. Now we will give it a formal name and formal notation.

"Limit from the right" → $\lim_{x \to c+} f\ x$

Examine the function for values just a little greater than *c* … or the "positive side of" *c*.

"Limit from the left" → $\lim_{x \to c-} f\ x$

Examine the function for values just a little less than *c* … or the "negative side of" *c*.



EXAMPLE :-

Find the limit of $f\ x = \sqrt{4 - x^2}$ as *x* approached –2 from the right.

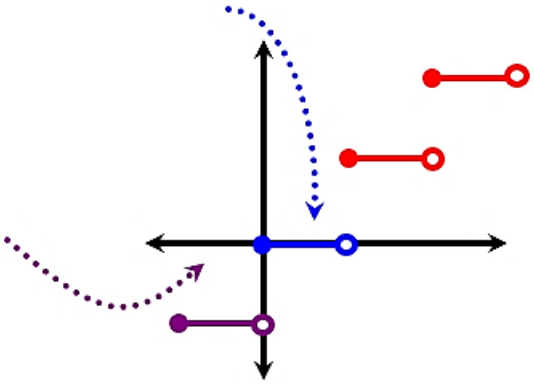
With either numerical analysis, or simple visual inspection we can see the function is approaching 0, as *x* approaches –2 from the right.

The greatest integer function is the function, f, such that $f(x) = \lfloor x \rfloor$ where $\lfloor x \rfloor$ is the greatest integer less than or equal to a real number, x. It is also written as [[x]]

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EXAMPLE :-

Find the limit of the "Greatest Integer Function"



$f\ x = \lfloor x \rfloor$ as x approaches 0 from the left and from the right. Since $\lfloor 0{,}1 \rfloor = 0$, or no matter how close x gets to 0 from the right, f is still 0 … $\lim_{x \to 0+} \lfloor x \rfloor = 0$

Yet as x gets close to 0 from the left, the value of the function remains steady at –1. Ex: $\lfloor -0{,}1 \rfloor = -1$

So then $\lim_{x \to 0-} \lfloor x \rfloor = -1$

We can conclude that the "Greatest Integer Function" is not continuous at ANY integer value. These are all JUMP discontinuities

THE EXISTENCE OF A LIMIT:-

Let f be a function and let c and L be real numbers. The limit as x approaches c exists ***if and only if***: $\lim_{x \to x-} f\ x = L$ and $\lim_{x \to x+} f\ x = L$

$\lim_{x \to c} f\ x = L$

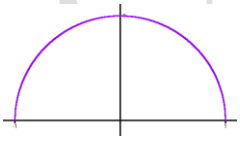
CONTINUITY ON A CLOSED INTERVAL:-

A function f is **continuous on an closed interval [a, b]**, if it is continuous on the open interval *(a, b)* and:

$$\lim_{x \to a+} f\ x = f\ a \text{ and } \lim_{x \to b-} f\ x = f\ b$$

Basically, the endpoints need to fill filled–in pieces of the graph.

EXAMPLE : - Discuss the continuity of $f\ x = \sqrt{1 - x^2}$



For this radical function, it is clearly continuous on the open interval (–1, 1).

And since

$\lim_{x \to -1+} f\ x = f\ -1 = 0$ Continuous from the right at –1

$\lim_{x \to 1-} f\ x = f\ 1 = 0$ Continuous from the left at 1

We can conclude that f is continuous on the closed interval [–1, 1], as this can be visually verified on the graph.

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PROPERTIES OF CONTINUITY:-

If *b* is a real number and *f* and *g* are continuous at $x = c$, then the following functions are also continuous at $x = c$.

1. Scalar Multiple: bf	2. Sum and Difference: $f \pm g$
3. Product: $f \bullet g$	4. Product: $f / g,\ g \neq 0$

Two continuous functions can be multiplied by a constant, added, subtracted, multiplied or divided by each other, and still be continuous!

EXAMPLE :- Describe the intervals on which each function is continuous. $f\ x = \tan x$ … The tangent function is undefined at $\frac{\pi}{2}$, and at repeated intervals $\pm n\pi$ where n is an integer. It is continuous within these discontinuous values.

So it is continuous on open intervals:

$$\ldots\left(-\frac{3\pi}{2},-\frac{\pi}{2}\right)\cup\left(-\frac{\pi}{2},\frac{\pi}{2}\right)\cup\left(\frac{\pi}{2},\frac{3\pi}{2}\right)\ldots$$

This can be more elegantly stated as follows: $\left(-\frac{\pi}{2}+n\pi,\frac{\pi}{2}+n\pi\right)$

Key Points in this Section:-

Definition of Continuity and its meaning:-

THIS is the concept that combines the value of a function's limit with the actual *function value*.

A function is **continuous at "c"** if each of these three (3) conditions is met:

1. $f(c)$ is defined
2. $\lim\limits_{x\to c} f\ x$ exists
3. $\lim\limits_{x\to c} f\ x = f\ c$

Intuitively, this says that the graph of a function is unbroken at the point where $x = a$.

When is a function NOT continuous at $x = a$?

Hole	Jump	Unbounded
$\lim\limits_{x\to a} f\ x$ exists but $\lim\limits_{x\to a} f\ x \neq f\ a$ (Removable Discontinuity)	$\lim\limits_{x\to a^-} f\ x \neq \lim\limits_{x\to a^+} f\ x$	$f(a)$ does not exist $\lim\limits_{x\to a^-} f\ x = \pm\infty$ or $\lim\limits_{x\to a^+} f\ x = \pm\infty$ (Infinite Discontinuity)

A function *f* is **continuous on an interval** if it is continuous at every point in the interval.

Formally, this is what we mean by the graph of a function being *unbroken*. You can draw the graph over the interval without lifting your pencil (no holes, jumps, or unbounded behavior).

Note:- Most functions that we are familiar with (polynomials, rational functions, trigonometric functions, logarithmic functions and exponential functions) are continuous on their *domains*.

That is, if they have a discontinuity, it is at a value of *x* NOT in the domain of the function.

Theorem:- If *f* and *g* are continuous at *a*, so are the functions

$f+g$ $\quad f-g$ $\quad f\cdot g$ $\quad \frac{f}{g}$ $\quad$ if $g(a)\neq 0$

This says if you add, subtract, multiply or divide continuous functions, the result is a continuous function.

Career Avenues

Indeterminate forms:-

Find the limit: $\lim_{x\to 1}\frac{x^3-1}{x-1}$

First, note why direct substitution will not work!!!

$\lim_{x\to 1}\frac{x^3-1}{x-1}=\frac{1^3-1}{1-1}=\frac{0}{0}=$??? … but there is a related function … identical to this one … except for at x = 1.

$f\ x=\frac{x^3-1}{x-1}=\frac{x^2+x+1}{x-1}=x^2+x+1=g\ x$

So, to evaluate the original limit for *f*, we will consider it for *g* instead:

$\lim_{x\to 1}x^2+x+1=1^2+1+1=3$

SOME IMPORTANT INDETERMINATE FORMS

1. b/0 = ∞ (be is a non zero positive constant)

   Meaning of this statement

   Note if $\lim_{x\to a} f(x)$=b and $\lim_{x\to a} g(x)$=0 then $\lim_{x\to a} f(x)/g(x)$ is said to be of the form b/0

   For any function f(x) and g(x) if $\lim_{x\to a} f(x)/g(x)$ is of the form b/0 then its value of is ∞

INDETERMINATE FORMS

1) 0/0
2) ∞/∞
3) 0 x 0
4) $\infty-\infty$
5) 0^0
6) ∞^0
7) ∞^1

2.4.1 L' HOSPITAL RULE

If $\lim_{x\to a} f\ x\ /\ g\ x$ is of the form 0/0 or ∞/∞ then

$\lim_{x\to a}\frac{f\ x}{g\ x}=\lim_{x\to a}\frac{f'\ x}{g'\ x}$

$\lim_{x\to a}\frac{f'\ x}{g'\ x}$ is of the form 0/0 or ∞/∞ then repeat the above process.

If we have any indeterminate form other than 0/0 or ∞/∞ then we convert that form to 0/0 or ∞/∞ and apply L-Hospital Rule.

USING STANDARD LIMITS AND SIMPLE TRANSFORMATION

1) $\lim_{x\to 0}\frac{\sin x}{x}=1$
2) $\lim_{x\to 0}\frac{\tan x}{x}=1$
3) $\lim_{x\to 0}\frac{\sin f'_{-}\ x_0\ x}{x}=1$
4) $\lim_{x\to 0}\frac{\tan\neq x}{x}=1$

Career Avenues

5) $\lim_{x\to a}\frac{x^{\infty}}{x-a}=na^{n-1}\ n\in R$ 6) $\lim_{x\to 0}\frac{a}{x}=\ln a$

7) $\lim_{x\to 0}\frac{e}{x}=\ln e=1$ 8) $\lim_{x\to 0}\frac{\log\ 1+x}{x}=\log_a e$

CANCELLATION AND RATIONALIZATION TECHNIQUES:-

This last example used algebraic factoring, and the canceling of common numerator and denominator factors to create a new *g(x)* function. The limit of this new *g(x)* function was easily found by direct substitution.

EXAMPLE:- Find the limit: $\lim_{x\to -3}\frac{x^2+x-6}{x+3}$

Again, note why direct substitution fails: $\frac{-3^{\ 2}+\ -3\ -6}{-3\ +3}=\frac{0}{0}=$???

Factor: $\lim_{x\to -3}\frac{x-2\quad x+3}{x+3}$ →Cancel: $\lim_{x\to -3}\ x-2$ →Substitute: -3 -2 = -5

Incidentally: We have seen the $\frac{0}{0}$ fraction appear in both of the last 2 examples. This form is **NOT** called **"UNDEFINED"** … it **IS** called an **"INDETERMINATE FORM".** The reason for this is that the limit cannot be determined while the expression is in that form.

EXAMPLE:- Find the limit $\lim_{x\to 0}\frac{\sqrt{x+1}-1}{x}$

Attempt direct substitution first: $\frac{\sqrt{0+1}-1}{0}$

To do so, we must multiply numerator and denominator by: THE CONJUGATE of the radical expression.

$$\frac{\sqrt{x+1}-1}{x}\times\frac{\sqrt{x+1}+1}{\sqrt{x+1}+1}=\frac{x+1\ -1}{x\ \sqrt{x+1}+1}=\frac{x}{x\ \sqrt{x+1}+1}=\frac{1}{\sqrt{x+1}+1}\ldots$$

Now we can again attempt the limit.

$$\lim_{x\to 0}\frac{1}{\sqrt{x+1}+1}=\frac{1}{\sqrt{0+1}+1}=\frac{1}{2}$$

L'Hospital's Rule:- If a quotient $\frac{f(x)}{g(x)}$ has the indeterminate form $\frac{0}{0}$ at $x = a$

($f(x)\to 0,\ g(x)\to 0$ as $x\to a$) or $\frac{\infty}{\infty}$ ($f(x)\to\pm\infty,\ g(x)\to\pm\infty$ as $x\to a$) and f and g are differentiable, then $\lim_{x\to a}\frac{f(x)}{g(x)}=\lim_{x\to a}\frac{f'(x)}{g'(x)}$, provided the limit on the right exists or is $\pm\infty$.Other indeterminate forms that may require L'Hospital's Rule for evaluation are $\infty-\infty, 0.\infty, 0^0, \infty^0\ and\ 1^{\infty}$ Each of these must be converted (by using algebra or logarithms) to a quotient before L'Hospital's Rule can be applied.

Career Avenues

***Solved Examples*:- How to Solve Limits Problems**

Always try direct substitution first. If the direct substitution yields an indeterminate form (i.e.: $\frac{0}{0} or \frac{\infty}{\infty}$), use factoring, rationalizing, or L'Hospital's Rule to calculate the limit.

(a) *Direct Substitution:-*

Ex. 1 – $\lim_{x\to 0}\frac{x^2}{x^3-2x^2+4}=\frac{0^2}{0^3-2\ 2^{\ 2}+4}=\frac{0}{4}=0$

Ex. 2 – $\lim_{x\to 0}\frac{1+\sin x}{\cos x} = \frac{1+\sin(0)}{\cos(0)}=\frac{1+0}{1}=1$

(b) *Factoring:-*

Ex. 3 – $\lim_{x\to 2}\frac{x^2+3x-10}{x^2-5x+6} = \frac{2^2+3(2)-10}{2^2-5(2)+6}=\frac{0}{0}$ (Indeterminate form)

$=\lim_{x\to 2}\frac{(x+5)(x-2)}{(x-3)(x-2)}$ Factor the polynomials

$=\lim_{x\to 2}\frac{(x+5)}{(x-3)}$ Divide out common factors

$=\frac{2+5}{2-3}$ Direct Substitution

$=-7$

(c) *Rationalizing*:-

Ex. 4 – $\lim_{x\to 0}\frac{\sqrt{x+1}-1}{x}=\frac{\sqrt{0+1}-1}{0}=\frac{0}{0}$ (Indeterminate form)

$=\lim_{x\to 0}\frac{\sqrt{x+1}-1}{x}\bullet\frac{\sqrt{x+1}+1}{\sqrt{x+1}+1}$ Rationalize using the conjugate

$=\lim_{x\to 0}\frac{x+1-1}{x\ \sqrt{x+1}+1}$

$=\lim_{x\to 0}\frac{x}{x\ \sqrt{x+1}+1}=\lim_{x\to 0}\frac{1}{\sqrt{x+1}+1}$ Simplify

$=\lim_{x\to 0}\frac{1}{\sqrt{x+1}+1}=\frac{1}{\sqrt{0+1}+1}=\frac{1}{2}$ Direct Substitution

(d) *L'Hospital's Rule:-*

The limit of a quotient of functions is equal to the limit of the quotient of their derivatives provided that direct substitution yields an indeterminate form.

That is to say: $\lim_{x\to a}\frac{f(x)}{g(x)}=\lim_{x\to a}\frac{f'(x)}{g'(x)}$

Ex. 5 – $\lim_{x\to 1}\frac{\ln x}{x-1}=\frac{\ln(1)}{1-1}=\frac{0}{0}$ (Indeterminate form)

$=\lim_{x\to 1}\frac{\frac{1}{x}}{1}$ (Take the derivative of the numerator and denominator)

Career Avenues

$= \frac{\frac{1}{1}}{1} = 1$ (Direct substitution and simplify)

Ex. 6 – $\lim_{x\to\infty} \frac{e^x}{x^2} = \frac{\infty}{\infty}$ (Indeterminate form)

$= \lim_{x\to\infty} \frac{e^x}{2x} = \frac{\infty}{\infty}$ (Still an Indeterminate form)

$= \lim_{x\to\infty} \frac{e^x}{2} = \frac{\infty}{2} = \infty$ (DNE)

This example shows that we had to use L'Hospital's Rule more than once until we got a non–indeterminate form. Please note that while we did indeed write that the limit was equal to infinity, it technically does not exist as we can never reach one particular number. We show it is equal to infinity to describe the behavior of the graph.

Ex. 7 – $\lim_{x\to 0} \frac{1-\cos x}{\sin x} = \frac{1-\cos(0)}{\sin(0)} = \frac{1-1}{0} = \frac{0}{0}$ (Indeterminate form)

$= \lim_{x\to 0} \frac{0-(-\sin x)}{\cos x} = \frac{\sin(0)}{\cos(0)} = \frac{0}{1} = 0$ (Using L'Hospital's Rule)

2.5 EXTREMA OF CONTINUOUS FUNCTIONS (MAXIMA & MINIMA)

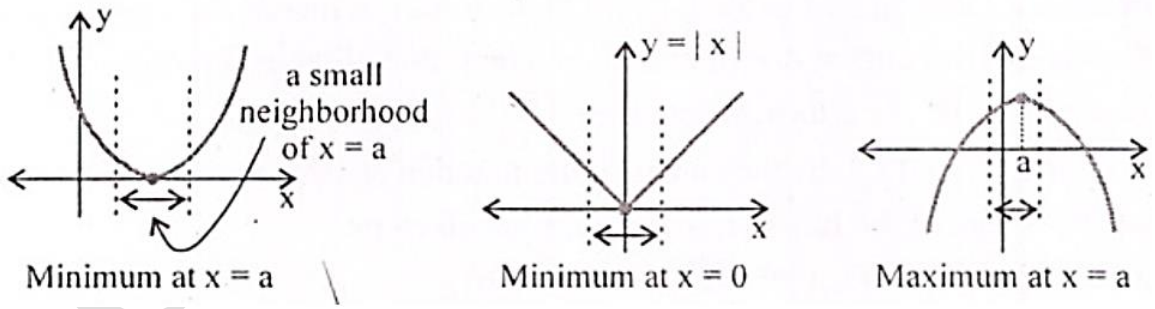
Definition:-

Maximum:-

A function f(x) is said to have a maximum or local maximum at x=a if the value of f(a) is greater than all the values of f(x) in a small neighborhood of x=a.

Minimum:-

A function f(x) is said to have a minimum at x=a if the value of f(a) is less than all the values of f(x) in small neighbourhood of x=a.



Maximum and Minimum Values of a Function:-

Testing for Maximum or minimum using change of sign of the first derivative and using sign of second derivative

First Derivative Test:-

Let c be a critical number of a function f that is continuous on the open interval I containing c. If f is differentiable on the interval, except possibly at c, then $f(c)$ can be classified as follows:

1) If $f'(x)$ changes from negative to positive at c, then $f(c)$ is a relative minimum of f

2) If $f'(x)$ changes from positive to negative at c, then $f(c)$ is a relative maximum of f

3) If $f'(x)$ does not change signs at c, then $f(c)$ is neither a relative minimum nor a relative maximum of f

Ex: $h(x) = x^3 - 6x^2 + 15$

$h(x) = 3x^2 - 12x$

$= 3x(x-4)$

Critical Values = 0,4 Increasing: $(-\infty, o) \cup (4, \infty)$

Decreasing: $(0, 4)$

Relative Max: (0,15), Relative Min: (4,–17)

***Second Derivative Test*:-**

Let f be a function such that $f'(c) = 0$ and that the second derivative of f exists on the open interval containing c.

1) If $f''(c) > 0$, then $f(c)$ is a relative minimum

2) If $f''(c) < 0$, then $f(c)$ is a relative maximum

3) If $f''(c) = 0$, then the test fails. In such cases, use the First Derivative Test.

Example: Find the minimum value of the function $f(x) = x^4 - 4x^3$.

Solution: $f(x) = x^4 - 4x^3$

$\Rightarrow f'(x) = 4x^3 - 12x^2$

$\Rightarrow f'(x) = 4x^2(x-3)$

Set $f'(x) = 0$

$\Rightarrow 4x^2(x-3) = 0$

$\Rightarrow x = 0, 3$

Since $f(0)$ and $f(3)$ exist, therefore $x = 0, 3$ are the critical numbers of the function.

These points divide the domain of the function into three intervals:

$(-\infty, 0) \cup (0, 3) \cup (3, \infty)$.

INTERVALS	TEST POINT	$f'(x)$	CONCLUSION
$(-\infty, 0)$	$x = -1$	$-ive$	
$(0, 3)$	$x = 1$	$-ive$	
$(3, \infty)$	$x = 4$	$+ive$	Since $f'(x)$ changes from $-ive$ to $+ive$ at 3, so the function has a local minimum at x = 3.

Minimum value is

$f(3) = (3)^4 - 4(3)^3 = 81 - 108 = -27$

Real World Application

Use of First and second derivative in optimization problems:

Ex: A fast–food restaurant has determined that the monthly demand for its hamburgers is . $P = \frac{6000 - x}{20000}$

Find the increase in revenue per hamburger (marginal revenue) for monthly sales of 20,000 hamburgers.

Career Avenues

(Total revenue is given by R=xp)

Solution: $R = xp = x\left(\frac{60000 - x}{20000}\right) = \frac{1}{20000}\ 60000x - x^2$

Marginal Revenue is: $\frac{dR}{dx} = \frac{1}{20000}\ 60000 - 2x$

When x=20000 the marginal revenue is:

$$\frac{dR}{dx} = \frac{1}{20000}\left[60000 - 2\ 20000\ \right] = \frac{20000}{20000} = \$1/unit$$

Absolute Minimum and Absolute Maximum Values :-

Closed Interval Method:-

To find the absolute maximum and minimum values of a continuous function *f* on a closed interval [*a*, *b*]:

1. Find the values of *f* at the critical numbers of *f* in (*a*, *b*).
2. Find the values of *f* at the end points of the interval.
3. The largest of the values from step 1 and 2 is the absolute maximum value; the smallest of the values from step 1 and 2 is the absolute minimum value.

Example:- Find the absolute maximum and minimum values of the function

$f(x) = 3x^2 - 12x + 5,\ [0, 3]$

Solution: - $f(x) = 3x^2 - 12x + 5$

$f'(x) = 6x - 12$

$f'(x) = 6(x - 2)$

Set $f'(x) = 0$

$\Rightarrow 6(x - 2) = 0 \ \Rightarrow x = 2$

$x = 2$ is the only critical number.

Now, $f(2) = 3(2)^2 - 12(2) + 5 = 12 - 24 + 5 = -7$

$f(0) = 3(0)^2 - 12(0) + 5 = 0 - 0 + 5 = 5$

$f(3) = 3(3)^2 - 12(3) + 5 = 27 - 36 + 5 = -4$

Absolute maximum value = 5

Absolute minimum value = –7

2.5.1 CONDITIONS FOR MAXIMA AND MINIMA:-

At a maximum point, the function y = f(x) changes from an increasing to a decreasing state. $\therefore$ $\frac{dy}{dx}$ changes from positive to negative value. In changing from positive to negative values $\frac{dy}{dx}$ must pass through the value zero. Hence, $\frac{dy}{dx}$ = 0 at a maximum point. Similarly, at a minimum point the function y = f(x) changes from a decreasing to an increasing state.

$\therefore$ $\frac{dy}{dx}$ changes from negative to positive value. Hence $\frac{dy}{dx}$ = 0 at a minimum point.

Thus $\frac{dy}{dx}$ = 0 for both maximum and minimum points on the curve y = f (x).

Career Avenues

Note:- $\frac{dy}{dx}$ **= 0 is only a necessary but not sufficient condition for maximum and minimum points. For a maximum point,** $\frac{dy}{dx}$ **changes from positive to negative. This indicates that** $\frac{dy}{dx}$ **is a decreasing function of x,** □ **its differential co–efficient, i.e** $\frac{d^2y}{dx^2}$ **< 0. Similarly for a minimum point,** $\frac{dy}{dx}$ **changes from negative to positive. This indicates that** $\frac{dy}{dx}$ **is an increasing function of x.** □ **its differential co–efficient,** $\frac{d^2y}{dx^2}$ **> 0.**

Hence, for a maximum point:- (i) $\frac{dy}{dx} = 0$ (ii) $\frac{d^2y}{dx^2} < 0$ at that Point

for a minimum point :- (i) $\frac{dy}{dx} = 0$ (ii) $\frac{d^2y}{dx^2} > 0$ at that point

2.5.2 GEOMETRICAL ILLUSTRATIONS

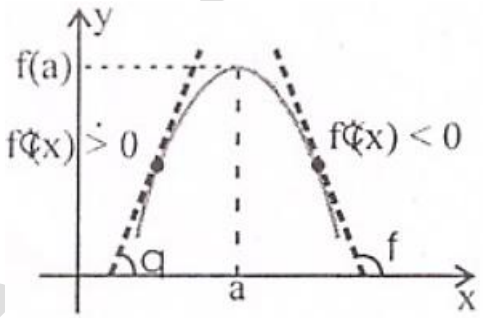
1. ***Maximum:-***

At x=a the nature of the function f(x) changes from (Positive) to (Negative).

i.e. f'(x) changes from (Positive) to (Negative)

Thus at x=a it must take the value zero.

Since the function f'(x) deceases from positive to zero to negative, therefore its derivatives is negative at x=a.

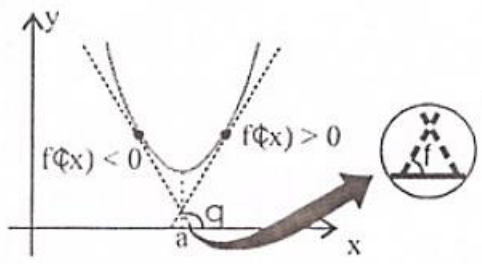


2. ***Minimum:-***

At x=a the nature of the function f(x) changes from decreasing to increasing.

i.e. f'(x) changes from negative to positive at x=a therefore f'(a) is 0.

Since the function f'(x) increases from negative to zero to positive therefore its derivatives is positive at x=a.

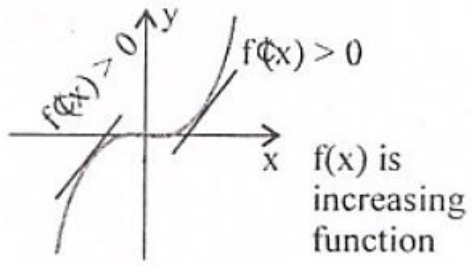


3. ***Neither minimum nor maximum:-***

In the figure f'(x) changes from positive to zero to positive. Therefore f(x) has neither

Career Avenues

minimum nor maximum at x=0 (point of inflexion)

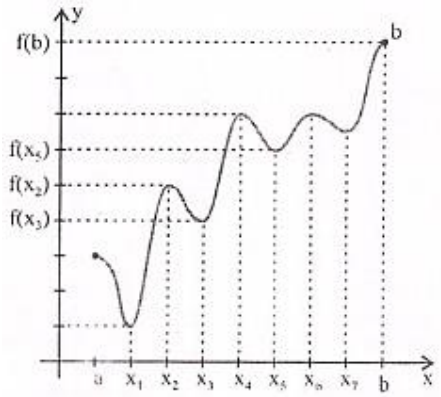


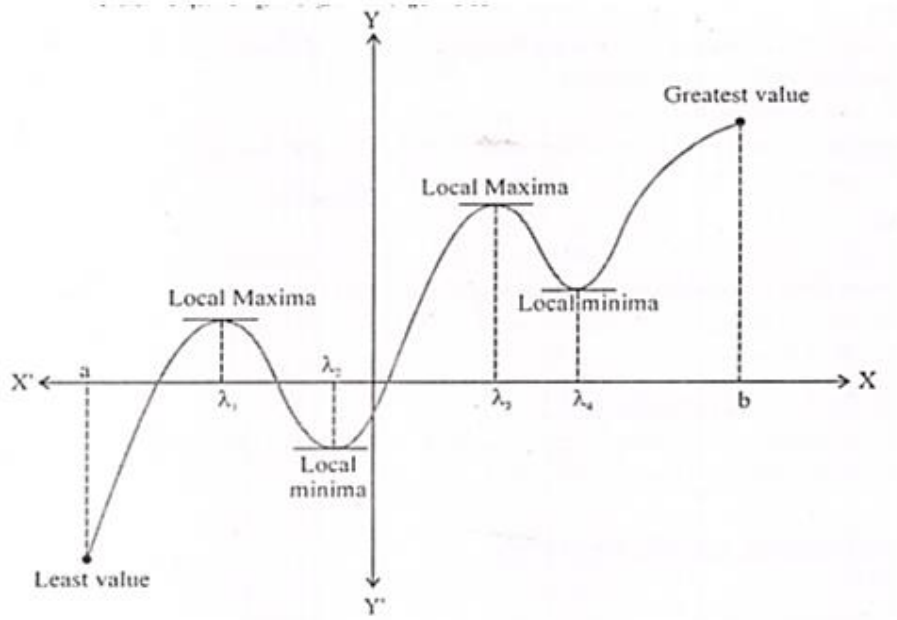
4. ***Minima and maxima occur alternately:-***

In the graph below

Minima occur at x_1, x_3, x_5, x_7

Maxima occur at x_2, x_4, x_6

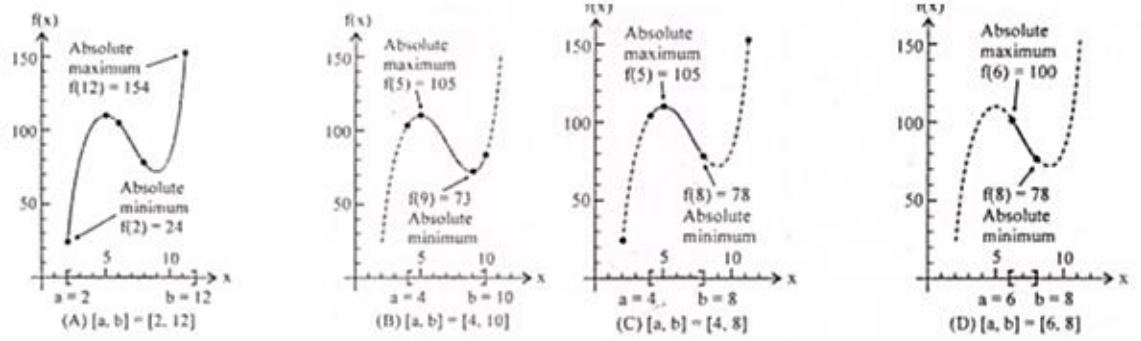




Career Avenues

Illustration

$f(x) = x^3-21x^2+135x-170$



Example :-

(i) Find the greatest and the least values of the function $f(x) = x^3-6x^2+9x+1$ on [0,2].

Solution:-

Given $f(x) = x^3-6x^2+9x+1$

$f'(x) = 3x^2-12x+9 = 3(x^2-4x+3)=3(x-1)(x-3)$

For maxima or minima f(x) =0

x=1,3 but 3[0,2]

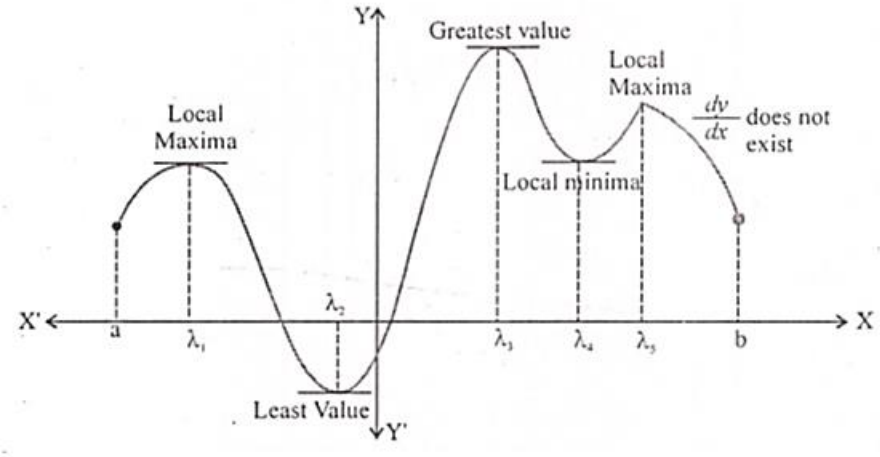
Only one critical points in [0,2] is x=1

Greatest value = max {f(0),f(1),f(2)}=max{1,5,3}=5

And least value = min{f(0),f(1),f(2)}=min{1,5,3}=1

Hence greatest and least values of f'(x) in [0,2] are 5 and 1 respectively.

(ii)**Greatest or least values of a function in open interval (a,b)**.



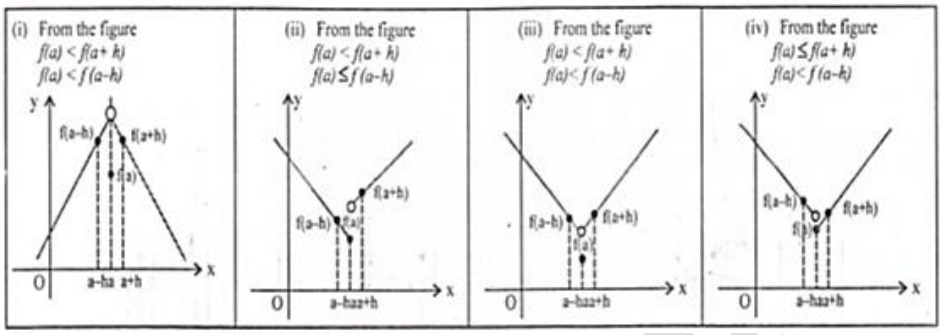
2.5.3 EXTREMA OF DISCONTINUOUS FUNCTION

Maxima & minima of discontinuous functions

In discontinuous functions we don't apply any of the methods discussed in previous section but we observe certain conditions and their graph which would give you more clear concept.

1. ***Minimum of discontinuous functions:-***

   The following four cases arise for minimum at x=a



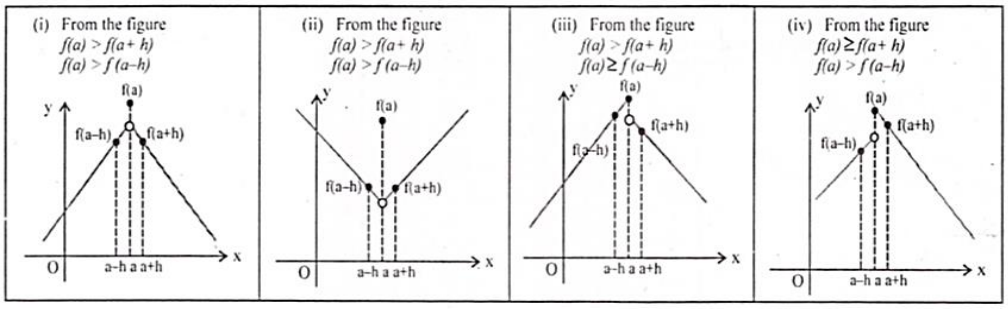
   From all the four cases discussed above, for minimum discontinuous functions, we have

   f(a) f(a+h)

   f(a) f(a-h)

2. ***Maximum of discontinuous functions:-***

   The following four cases arises for maximum at x=a



   From all the four cases discussed above, for maximum discontinuous functions, we have

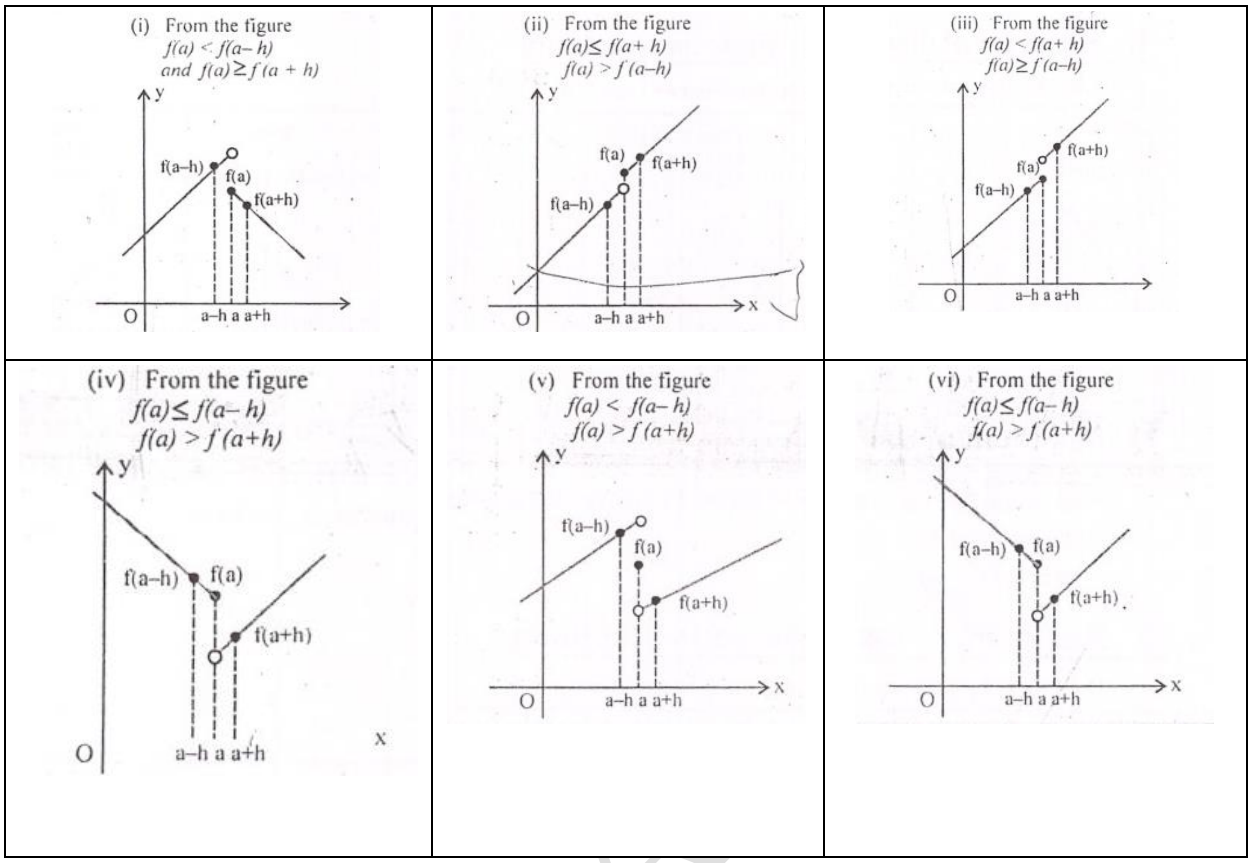
   f(a) f(a+h)

   f(x) f(a-h)

3. ***Neither maximum nor minimum of discontinuous functions:-***

   The following cases arise for neither maximum nor minimum at x=a

Career Avenues



In above all cases no extremum exists.

2.6 Derivatives:

The ***derivative*** is a measure of how a function changes as its input changes. Loosely speaking, a derivative can be thought of as how much a quantity is changing at a given point. For example, the derivative of the position (or distance) of a vehicle with respect to time is the instantaneous velocity (respectively, instantaneous speed) at which the vehicle is travelling. Conversely, the integral of the velocity over time is the vehicle's position.

The derivative of a function at a chosen input value describes the best linear approximation of the function near that input value. For a real–valued function of a single real variable, the derivative at a point equals the slope of the tangent line to the graph of the function at that point. In higher dimensions, the derivative of a function at a point is a linear transformation called the linearization. A closely related notion is the differential of a function.

The process of finding a derivative is called ***differentiation***. The fundamental theorem of calculus states that differentiation is the reverse process to integration.

2.6.1 Differentiation and the derivative

Differentiation is a method to compute the rate at which a dependent output y, changes with respect to the change in the independent input x. This rate of change is called the ***derivative*** of y with respect to x. In more precise language, the dependence of y upon x means that y is a function of x. If x and y are real numbers, and if the graph of y is plotted against x, the derivative measures the slope of this graph at each point. This functional relationship is often denoted $y = f(x)$, where f denotes the function.

Career A

The simplest case is when y is a linear function of x, meaning that the graph of y against x is a straight line. In this case, $y = f(x) = m x + c$, for real numbers m and c, and the slope m is given by

$$m = \frac{change\ in\ y}{change\ in\ x} = \frac{\Delta y}{\Delta x}$$

where the symbol Δ (the uppercase form of the Greek letter Delta) is an abbreviation for "change in." This formula is true because

$y + \Delta y = f(x + \Delta x) = m (x + \Delta x) + c = m x + c + m \Delta x = y + m\Delta x$.

It follows that $\Delta y = m \Delta x$.

This gives an exact value for the slope of a straight line. If the function f is not linear (i.e. its graph is not a straight line), however, then the change in y divided by the change in x varies: differentiation is a method to find an exact value for this rate of change at any given value of x.

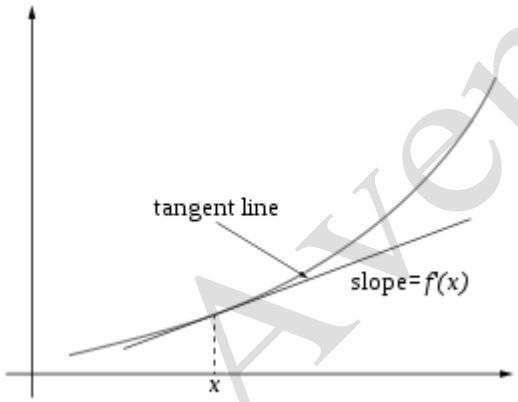


Figure 2.7 The tangent line at (x, f(x))

The idea, is to compute the rate of change as the limiting value of the ratio of the differences Δy /Δx as Δx becomes infinitely small.

In Leibniz's notation, such an infinitesimal change in x is denoted by dx, and the derivative of y with respect to x is written as dy / dx

suggesting the ratio of two infinitesimal quantities. (The above expression is read as "the derivative of y with respect to x", "d y by d x", or "d y over d x".

The derivative as a function

Let f be a function that has a derivative at every point a in the domain of f. Because every point a has a derivative, there is a function which sends the point a to the derivative of f at a. This function is written f'(x) and is called the derivative function or the derivative of f. The derivative of f collects all the derivatives of f at all the points in the domain of f.

Sometimes f has a derivative at most, but not all, points of its domain. Using this idea, differentiation becomes a function of functions: The derivative is an operator whose domain is the set of all functions which have derivatives at every point of their domain and whose range is a set of functions. If we denote this operator by D, then D(f) is the

Career Avenues

function f'(x). Since D(f) is a function, it can be evaluated at a point a. By the definition of the derivative function, D(f)(a) = f'(a).

The derivative of a function can, in principle, be computed from the definition by considering the difference quotient, and computing its limit. In practice, once the derivatives of a few simple functions are known, the derivatives of other functions are more easily computed using rules for obtaining derivatives of more complicated functions from simpler ones.

Derivative of a function

Let g be any general function that is defined over an open interval that contains c. The general form of the derivative for any x value throughout g would be $\lim_{\Delta x \to 0} \frac{g(x+\Delta x)-g(x)}{\Delta x} = g'(x)$. This is read as "g prime of x."

***Example*:-** Find the derivative of g(x)= $3x^2 - 2$ where x = 2.

$\lim_{\Delta x \to 0} \frac{g(x+\Delta x)-g(x)}{\Delta x} = g'(x)$.

$\lim_{\Delta x \to 0} \frac{3(x+\Delta x)^2 - 2 - (3x^2 - 2)}{\Delta x}$ =g'(x).

$\lim_{\Delta x \to 0} \frac{3(x^2 + 2x \cdot \Delta x + \Delta x^2) - 2 - 3x^2 + 2}{\Delta x}$ =g'(x).

$\lim_{\Delta x \to 0} \frac{3x^2 + 6x \cdot \Delta x + 3\Delta x^2 - 2 - 3x^2 + 2}{\Delta x}$ = g'(x).

$\lim_{\Delta x \to 0} \frac{6x \cdot \Delta x + 3\Delta x^2}{\Delta x}$ = g'(x).

$\lim_{\Delta x \to 0} \frac{\Delta x(6x + 3\Delta x)}{\Delta x}$ = g'(x).

$\lim_{\Delta x \to 0} 6x + 3\Delta x$ =g'(x).

6x + 0 =g'(x).

6x = g'(x).

Now for x = 2, we find that the derivative is 6(2) = 12. The advantage to using this formula is that once we find the simplified derivative, all we have to do is plug in specific domain values to find their individual derivative.

Let's look at this for the function f(x) = sin(x).

$\lim_{\Delta x \to 0} \frac{f(x+\Delta x)-f(x)}{\Delta x} = f'(x)$.

$\lim_{\Delta x \to 0} \frac{\sin(x+\Delta x)-\sin(x)}{\Delta x} = f'(x)$.

$\lim_{\Delta x \to 0} \frac{\sin(x)\cos(\Delta x) + \sin(\Delta x)\cos(x) - \sin(x)}{\Delta x} = f'(x)$.

$\lim_{\Delta x \to 0} \frac{\sin(x)\cos(\Delta x) - \sin x}{\Delta x} + \frac{\cos(x)\sin(\Delta x)}{\Delta x}$ = f'(x).

$\lim_{\Delta x \to 0} \frac{\sin(x)\cos(\Delta x) - \sin x}{\Delta x}$]+ $\lim_{\Delta x \to 0}$ [$\frac{\cos(x)\sin(\Delta x)}{\Delta x}$ = f'(x).

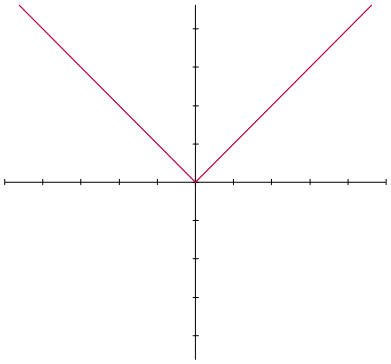
$\lim_{\Delta x \to 0} \sin(x)\frac{\cos(\Delta x) - 1}{\Delta x}$ + $\lim_{\Delta x \to 0} \cos(x)\frac{\sin(\Delta x)}{\Delta x}$ = f'(x).

sin(x)*0 + cos(x)*1= f'(x).

cos(x) = f'(x).

Career Avenues

If a function has a derivative at c then we say that the function, or any equation for that matter, is ***differentiable*** at c. Two common ways for a function not to have a derivative at c are if the function is not continuous at c, and if the function has a corner at c. For the latter of the two ways consider the graph of $g(x) = |x|$ for c = 0.



A corner exists at c = 0; therefore f'(0) does not exist. Also consider that you could not draw just one tangent line at that particular value. A third way for a function not to be continuous at a point is if the tangent line at that particular x–value is vertical.

A function must be continuous to be differentiable but a function does not have to be differentiable to be continuous.

***Definition*:-** Let f: I→R be a function defined on an interval I and let $a \in I$. We say that f is differentiable at a if $\lim_{x\to a}\frac{f(x)-f(a)}{x-a}$ exists and is finite, and we write

$$f'(a) = \lim_{x\to a}\frac{f(x)-f(a)}{x-a}$$

where $f'(a)$ is the derivative of f at a.

Example: Let $g(x) = x^2$.

$$\lim_{x\to a}\frac{g(x)-g(a)}{x-a} = \lim_{x\to a}\frac{x^2-a^2}{x-a} = \lim_{x\to a}\frac{x-a \quad x+a}{x-a} = \lim_{x\to a} x+a = 2a \Rightarrow g'(a) = 2a .$$

Example: Let $f(x) = x^n$.

$$\lim_{x\to a}\frac{f(x)-f(a)}{x-a} = \lim_{x\to a}\frac{x^n-a^n}{x-a} = \lim_{x\to a}\frac{x-a \quad x^{n-1}+ax^{n-2}+.....+a^{n-2}x+a^{n-1}}{x-a}$$

$$= \lim_{x\to a}(x^{n-1}+ax^{n-2}+.....+a^{n-2}x+a^{n-1}) = a^{n-1}+a^{n-1}+.....+a^{n-1}+a^{n-1} = na^{n-1}$$

$$\Rightarrow f'(a) = na^{n-1}$$

2.7 Differentiability and Continuity:-

What is the relationship between continuity and differentiability?

Career Avenues

Example:-
The function f(x) = | x | is continuous everywhere. Is it also differentiable everywhere ?
Solution:-

We know that f is continuous. To check for differentiability, $\lim_{x\to a}\frac{f(x)-f(a)}{x-a}=\lim_{x\to a}\frac{|x|-|a|}{x-a}$

1) If a > 0 then x > 0 eventually. Then there is no need for the absolute value. The limit become +1, and therefore f' (a) = 1.
2) If a < 0 then x < 0 eventually. The absolute values are resolved by an additional negative sign. The limit becomes –1, and therefore f' (a) = –1.
3) If a = 0, then the left and the right–handed limits will be different (–1 and +1). Therefore, the function is not differentiable at 0.

Example:-The function f(x) = $\begin{cases} x^2\sin\frac{1}{x} & x\neq 0 \\ 0 & x=0 \end{cases}$ is continuous at x = 0.

Is the function differentiable at x = 0 ?
Solution:-

$\lim_{x\to 0}\frac{f(x)-f(0)}{x-0}=\lim_{x\to 0}\frac{x^2\sin\frac{1}{x}-0}{x-0}=\lim_{x\to 0}(x\sin\frac{1}{x})=0$.

Therefore, f is differentiable at 0 and f' (0) = 0.
Hence, every differentiable function is continuous (theorem).
There are continuous functions that are not differentiable.

Theorem :-

If f is differentiable at a point a, then f is continuous at a.

Proof: f is differentiable at a $\Rightarrow f'(a)=\lim_{x\to a}\frac{f(x)-f(a)}{x-a}$. We will prove that $\lim_{x\to a} f(x)=f(a)$.

For $x\neq a$ we have $f(x)=\frac{f(x)-f(a)}{x-a}(x-a)+f(a)$

$\Rightarrow \lim_{x\to a} f(x)=\lim_{x\to a}\left[\frac{f(x)-f(a)}{x-a}(x-a)+f(a)\right]=f'(a).0+f(a)=f(a)$.

Hence, f is continuous at a.

Theorem:-

Let f and g be functions that are differentiable at the point a and let c be a real number, then each of the functions cf, f + g, fg, and ($\frac{f}{g}, g(a)\neq 0$) is also differentiable at a with derivatives:

(i) $(cf)'(a)=cf'(a)$ (ii) $(f+g)'(a)=f'(a)+g'(a)$

(iii) $(fg)'(a)=f(a)g'(a)+f'(a)g(a)$ (iv) $(\frac{f}{g})'(a)=\frac{f'(a)g(a)-f(a)g'(a)}{g^2(a)}$

Right hand derivative (RHD):-

For right hand derivative of f(x) at $x=x_0$, we have

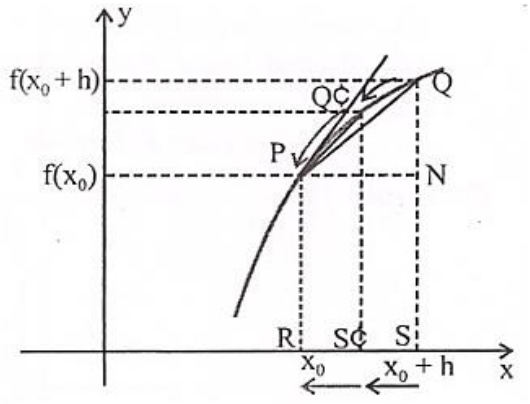
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Symbol

$f'_+(x_0)$ =(+ indicates right)

$$= \lim_{h \to 0} \frac{f\ x\infty + h\ - f\ x\infty}{h}$$

Definition:



Geometrical meaning:

Slope of secant PQ=QN/PN $= \frac{f\ x\infty + h\ - f\ x\infty}{x\infty + h\ - \ x\infty}$

As h approaches 0, S approaches through S' to R

And Q approaches through Q' to P

& the secant PQ approaches through PQ' to tangent at P from right.

RHD is the limit of slope of the secant PQ as Q approaches towards P. In other words it is the slope of tangent (at $x=x_0$) from right on the graph of $y=f(x)$.

Left hand derivatives (LHD):

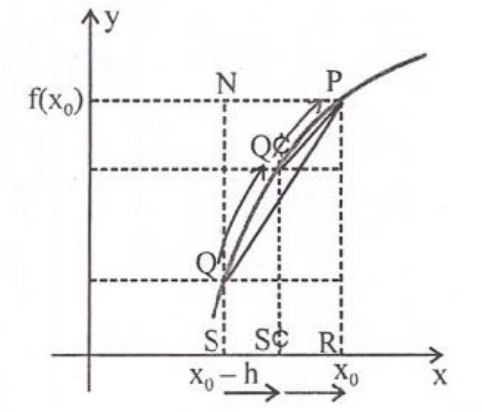
For left hand derivative of f(x) at $x=x_0$, we have

$f'_-(x_0)$ =(- indicates left)

$$= \lim_{h \to 0} \frac{f\ x\infty - h\ - f\ x^{-1}}{h}$$

Career Avenues

Definition:



Geometrical meaning:

It is the limit of slope of the secant PQ as Q approaches towards P. In other words, it is the slope of the tangent (at $x=x_0$) on the graph $y=f(x)$ from left

Slope of secant PQ=QN/PN $= \dfrac{f\ x^{-1} - h\ - f\ x^n - a^n}{x \in -h\ -\ x^x - 1}$

As h approaches 0, S approaches through S' to R

And Q approaches through Q' to P

& the secant PQ approaches through PQ' to tangent at P from left.

RHD is the limit of slope of the secant PQ as Q approaches towards P. In other words it is the slope of tangent (at $x=x_0$) from right on the graph of $y=f(x)$.

Definition of differentiability:

A function f(x) is called differentiable at $x=x_0$, if its right hand derivative at $x=x_0$ is equal to its left hand derivatives at $x=x_0$.

Symbolically,

$f_+\ x_0\ = f_-\ x_0$

OR

(RHD at $x=x_0$) = (LHD at $x=x_0$)

OR

$= \lim_{h\to 0} \dfrac{f\ x\infty + h\ - f\ x\infty}{h} = \lim_{h\to 0} \dfrac{f\ x\infty - h\ - f\ x^{-1}}{h}$

Career Avenues

Continuity of a differentiable function

Consider a function f(x) differentiable at $x=x_0$ i.e. $f_+ x_0$ & $f_- x_0$ exist and are equal. A differential function is always continuous.

Now

$f_+ x_0$ x 0 = 0

$\lim_{h\to 0} x\lambda_2 + h - f\ x\lambda_3 = 0$

i.e. $\lim_{h\to 0} x\lambda_n + h = f\ x\lambda_1$

i.e. RHL = Value

Similarly, LHL = value of function at $x=x_0$

Hence f(x) is continuous at $x=x_0$

Conclusion:

Differentiability $\longrightarrow$ Continuity

Differentiability of a continuous function

A function f(x) continuous at $x=x_0$ may not be differentiable at $x=x_0$

i.e. $f_+ x_0 \neq f_- x_0$

(RHD) $\neq$ (LHD)

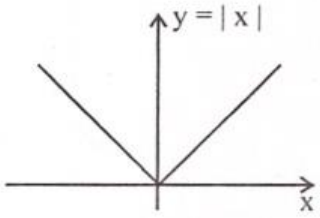
i.e. slope of tangent from right $\neq$ slope of tangent from left

i.e. slope of tangent changes abruptly at $x=x_0$

i.e. f(x) has a corner at $x=x_0$

i.e. f(x) is not a smooth function

Example :



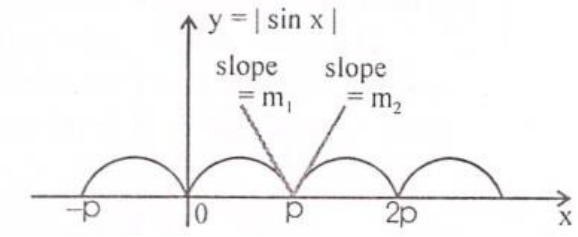
At x = 0: / x / is continuous

But ∵ RHD = 1 & LHD = 1

∴ non differentiable

Example :

Career Avenues



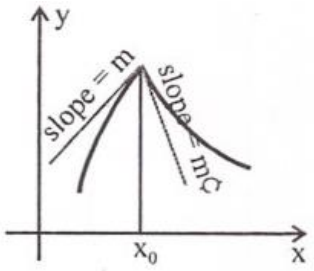
At x = π(*in* general nπ,n $\in$ I)

/sin x / is continuous

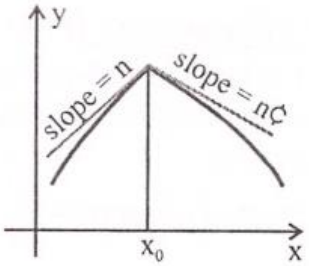
But RHD $= m_2 \neq LHD = m_1$

$\therefore$ Non differentiable

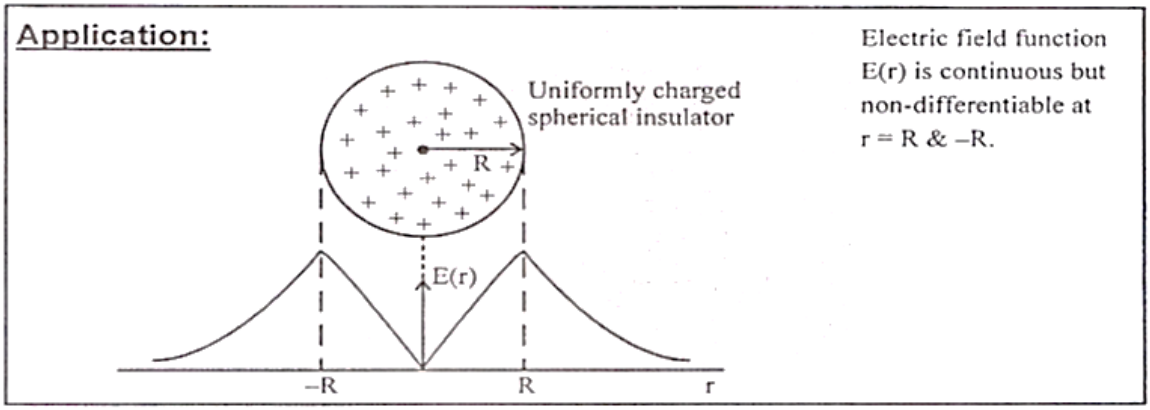
Example :



Example :



Electric field function E(r) is continuous but non differentiable at r=R & -R.



2.7.1 Some standard results on differentiability

Previously we saw the differentiability of some of the functions, now we, give an overview of the standard result.

Sl. No.	Function f(x)	Intervals in which f(x) is differentiable
1.	Polynomial	$-\infty, \infty$
2.	Exponential (a^x, a>0)	∞, ∞
3.	Constant	∞, ∞
4.	Logarithmic	Each point in its domain
5.	Trigonometric	Each point in its domain
6.	Inverse trigonometric	Each point in its domain

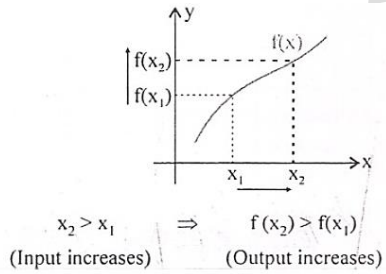
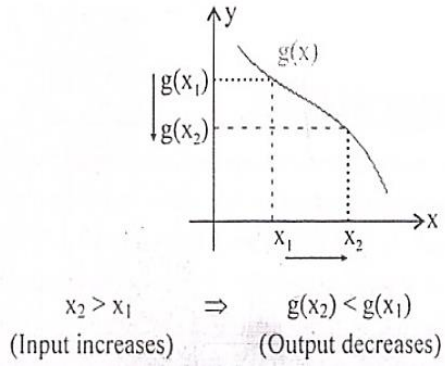
1. Sum, difference, product & quotient of two differentiable functions is differentiable.
2. The composition of differentiable function is also differentiable.

   If f ang g are continuous then

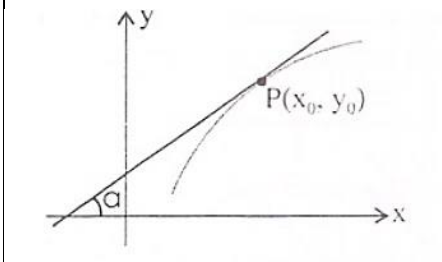
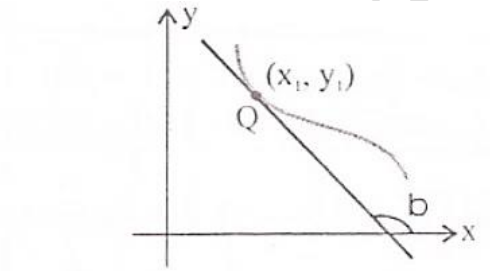
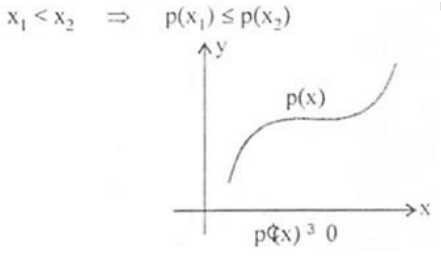
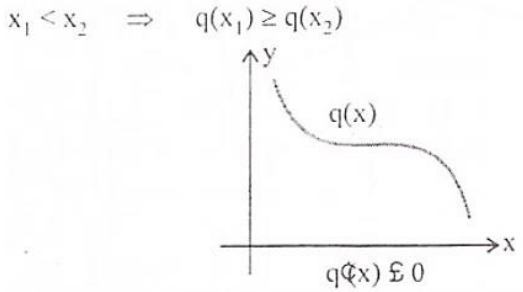
   fog = gof

   f [g(x)]=g [(x)]

Comparison of strictly increasing & decreasing functions

Strictly increasing function	**Strictly decreasing function**
1. Definition A function whose value of output (image) increases on increasing the value of input (pre-image) is called strictly increasing function	**1. Definition** A function whose value of output (image) decreases on increasing the value of input (pre-image) is called strictly decreasing function
2. Graphical illustration  	2. Graphical illustration 
3. Effect on Inequality An inequality remains unchanged if we apply a strictly increasing function on both sides of it	3. Effect on Inequality An inequality gets reversed if we apply a strictly decreasing function on both sides of it i.e.

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i.e. $\alpha < \beta \Rightarrow f\ \alpha\ < f\ \beta$	$\alpha < \beta \Rightarrow g\ \alpha\ > g\ \beta$
4. Derivative: RHD = $f_+\ x_0\ =$ $\lim_{h\to 0} \frac{f\ x\lambda_2 + h\ - f\ x\lambda_3}{h} = \frac{+ve}{+ve} = +ve$ LHD $f_-'\ x_0\ =$ $\lim_{h\to 0} \frac{f\ x\lambda_1 - h\ - f\ x\lambda_2}{-h} = \frac{-ve}{-ve} = +ve$	Derivative: RHD= $f_+'\ x_0$ $\lim_{h\to 0} \frac{f\ x\lambda_n + h\ - f\left(x \frac{a+x\ \ b+x}{c+x} in\ c, \infty\right)}{h} = \frac{-ve}{+ve} = -ve$ LHD= $f_-'\ x_0\ =$ $\lim_{h\to 0} \frac{f\left(x \frac{c+x\ \ a+b+2x\ -\ a+x\ \ b+x}{c+x^{\ 2}} = h\right) - f\ \ x \pm \sqrt{a-c\ \ b-c}}{-h} \frac{+ve}{-ve} = -ve$
5. Slop of tangent:  m(Tp) i.e. f'(x0) = tan α = positive	5. Slope of tangent  m(T1) i.e. f'(x1) = tan β = negative
6. A function p(x) is an increasing function if $x_1 < x_2$ ⇨ $P(x_1) \le P(x_2)$  	6. A function q(x) is an decreasing function if $x_1 < x_2$ $q(x_1) \ge q(x_2)$ 

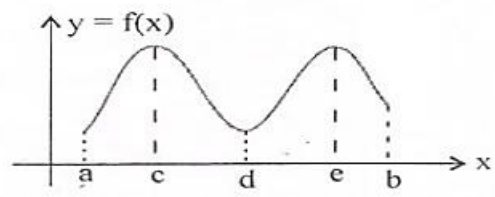
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Monotonic function:-

A function, which is either strictly increasing or strictly decreasing in its domain is called a monotonic function.

Non monotonic function:-

A function, which is increasing in some intervals & decreasing in some other intervals is called a non monotonic function e.g. Y=f(x). Hence, f(x) is a strictly increasing function in the intervals (a,c) & (d,e), while it is a strictly decreasing function in the intervals (c,d) & (e,b)



Increasing and Decreasing Functions:-

A function is *increasing* on an interval if for any x_1 and x_2 in the interval then

$$x_1 < x_2 \quad \Rightarrow \quad f(x_1) < f(x_2)$$

A function is *decreasing* on an interval if for any x_1 and x_2 in the interval then

$$x_1 < x_2 \quad \Rightarrow \quad f(x_1) > f(x_2)$$

How does this relate to derivatives? Recall that the derivative is the limit

$$\frac{f(x_2) - f(x_1)}{x_2 - x_1}$$

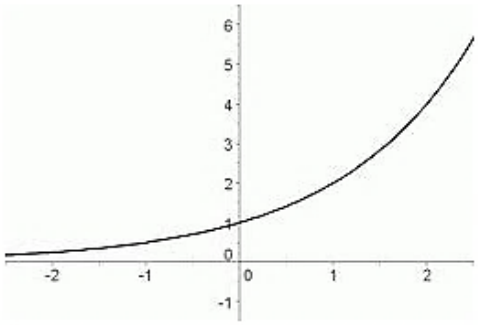
If $x_1 < x_2$, then the denominator will be positive. Now if

1) $f(x_1) < f(x_2)$, then the numerator will be positive, hence the derivative will be positive.
2) $f(x_1) > f(x_2)$, then the numerator will be negative and the derivative will be negative.

A function $f(x)$ is called an ***increasing function*** in an interval (a,b) if

$f(x)$ is differentiable on the interval (a,b).

$f'(x) > 0$ for all values of $x \in (a,b)$.

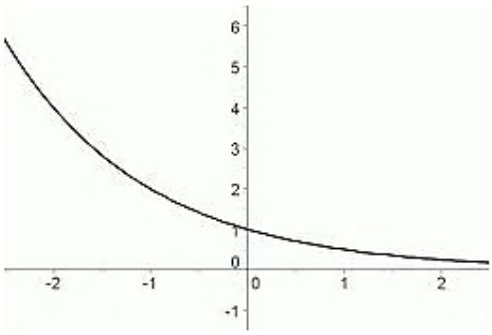


A function $f(x)$ is called a ***decreasing function*** in an interval (a,b) if

$f(x)$ is differentiable on the interval (a,b).

$f'(x) < 0$ for all values of $x \in (a,b)$.

Career Avenues



Working Rule:-

Step 1: Find $f'(x)$ and factor it.

Step 2: Set $f'(x)=0$ and solve for *x*.

These *x* values will divide real number line into intervals I.

Step 3: If $f'(x)>0$ for all $x\in I$, then the function $f(x)$ is increasing on I.

If $f'(x)<0$ for all $x\in I$, then the function $f(x)$ is decreasing on I.

Example:-

Determine the intervals on which the function $f(x)=x^2-8x+4$ is increasing or decreasing.

Solution:-

$f(x)=x^2-8x+4$

$f'(x)=2x-8$

$f'(x)=2(x-4)$

Set $f'(x)=0 \Rightarrow 2(x-4)=0 \Rightarrow x=4$

Here the intervals are $(-\infty, 4)\cup(4, \infty)$.

INTERVALS	TEST POINT	$f'(x)$	CONCLUSION
$(-\infty, 4)$	$x=0$	$-ive$	Function is decreasing
$(4, \infty)$	$x=5$	$+ive$	Function is increasing

Critical Number :-

A number $x=c$ is known as a critical number of a function $f(x)$ if

1) $f(c)$ exists 2) $f'(c)=0$ or $f'(c)$ does not exist

Working Rule:-

Step 1: Find $f'(x)$ and factor it.

Step 2: Set $f'(x)=0$ and solve for *x*.

These *x* values are the candidates for critical numbers.

Step 3: The values from step 2 will be the critical numbers if $f(x)$ exist at these points.

Example:-

Find the critical numbers of the function $f(x)=5x^2+4x$

Solution:-

We have $f(x)=5x^2+4x$

Career Avenues

Step 1: Find $f'(x)$ and factor it.

$f(x) = 5x^2 + 4x$

$f'(x) = 10x + 4$

$f'(x) = 2(5x + 2)$

Step 2: Set $f'(x) = 0 \Rightarrow 2(5x+2) = 0 \Rightarrow x = -\frac{2}{5}$

Step 3: $f(-\frac{2}{5})$ exists.

Thus $x = -\frac{2}{5}$ is the critical number of the function

Example:-

Find the critical numbers of the function $f(x) = x^3 + x^2 - x$

Solution:-

Step 1: Find $f'(x)$ and factor it.

$f(x) = x^3 + x^2 - x$

$f'(x) = 3x^2 + 2x - 1$

$f'(x) = 3x^2 + 3x - x - 1$

$f'(x) = 3x(x+1) - 1(x+1)$

$f'(x) = (3x-1)(x+1)$

Step 2: Set $f'(x) = 0$

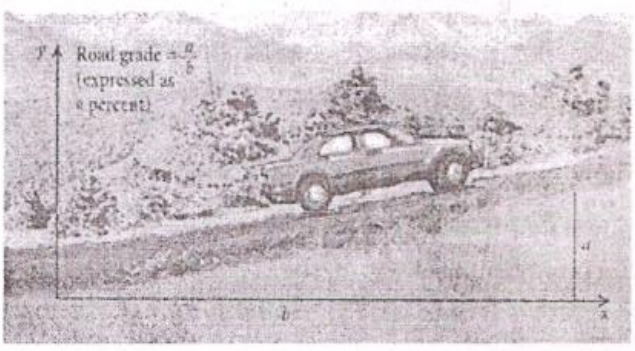
$\Rightarrow f'(x) = (3x-1)(x+1) \Rightarrow x = \frac{1}{3}, \quad -1$

Step 3: $f(-1)$ and $f(\frac{1}{3})$ both exist.

Thus $x = -1, \frac{1}{3}$ are the critical numbers of the function.

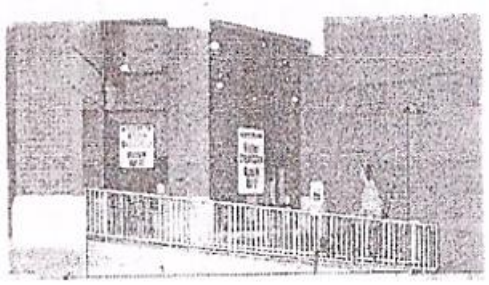
Application of slope:-

Slop has many real world applications. % road grade tell how steep a road is on hill or mountain. For example a 4% grade means that the road rises 4 feet for every horizontal distance of 100 feet if a vehicle is going up and vice versa.

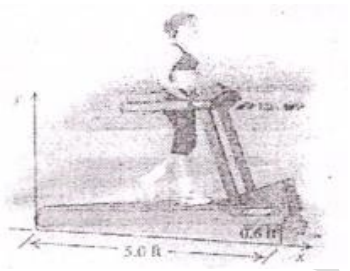


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It is also used in construction laws regard ramps for the handicapped state that very rise of 1 feet required much horizontal run.



The concept of slop is also used in cardiology when a person runs on a trade mill. A physician may change the slope or grade, of a treadmill to measure its effect on heart pulse rate.



Another example occur in hydrology when a river flows, the strength or force of the river depends on how for the river falls vertically compared to how far it flows horizontally.

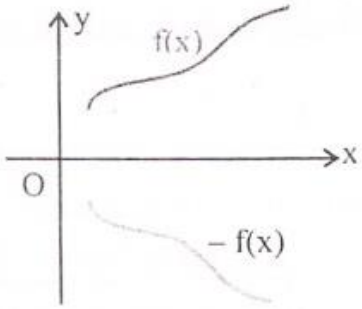
Properties of increasing & decreasing function:-

(i) If f is an increasing function then its negative is a decreasing function and vice versa i.e. f(x) is increasing function then – f(x) is decreasing function Or Let F(x)=-f(x)

F'(x)=-f'(x)

F'(x)<0 (since f'(x)>0)

⇨ F(x) is a decreasing function (se fig. 1))



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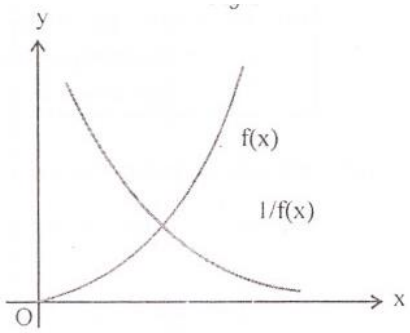
(ii) If f is an increasing function then its reciprocal is a decreasing function and vice versa

i.e. f(x) is increasing function then 1/f(x) is decreasing function

or Let $F(x)=1/f(x)$ $F'(x)=-1/f(x)^2 . F'(x)$

$F'(x)<0$

⇨ F(x) is a decreasing function (see fig. (2)



(iii) If f is an increasing function and g is also an increasing function then their sum is an increasing function

i.e. if f(x) and g(x) are increasing functions then f(x)+g(x) is also increasing function.

Or Let $F(x)=f(x)+g(x)$ $F'(x) = f'(x)+g'(x)$

$F'(x)>0$ (since $f'(x)>0$ and $g'(x)>0$)

⇨ F(x) is an increasing function (see fig. 3))

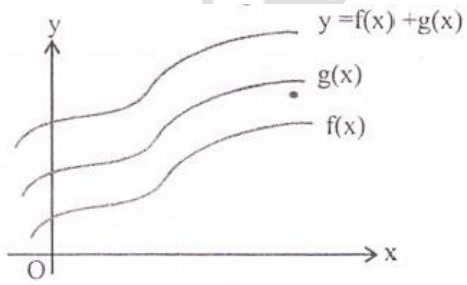


Fig 3

Note:-

Decreasing function + decreasing function = decreasing function
Increasing function + decreasing function = increasing or decreasing or neither increasing nor decreasing

(iv) If f is an increasing function and g is decreasing function (i.e. –g is increasing function and g is increasing then f-g is decreasing function)

According to graph f(x) – g(x) is decreasing for x (see fig. (4))

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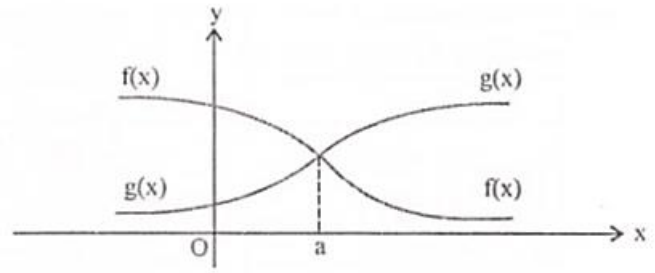


Fig 4

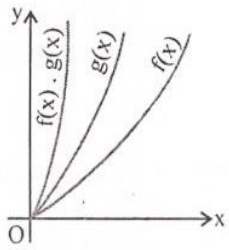
Note:-

(a) Increasing – increasing = increasing + decreasing = increasing or decreasing or neither increasing nor decreasing
(b) Decreasing – decreasing = decreasing + increasing = increasing or decreasing or neither increasing nor decreasing

if f is an increasing function and g is also an increasing function then f(x). g(x) is also an increasing function (see fig. (5)).

Note:

(a) Increasing X decreasing = decreasing
(b) Increasing X decreasing = Decreasing X increasing = increasing or decreasing or neither increasing nor decreasing



(vi) if f is an increasing function and g is a decreasing function then f(x)/g(x) is an increasing function (e.g. 1/g increasing function) or g/f is a decreasing function

(i.e., 1/f decreasing function) (see fig. (6)).

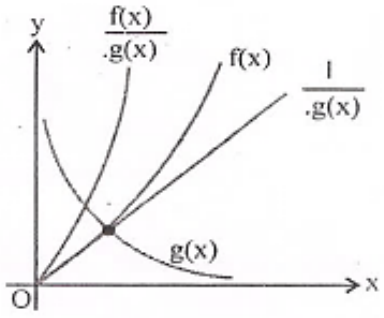


Fig 6

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Note:

(vii) If f is an increasing function and g is also an increasing then composite function fog is an increasing function (se Fig. (7))

F	G	Fog
Increasing	Decreasing	Decreasing
Decreasing	Increasing	Decreasing
Decreasing	Decreasing	Increasing

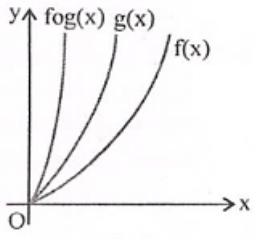


Fig 7

(viii) If f(x) is a monotonic increasing function in the interval [a,b] then greatest value and least of f(x) in the interval [a,b] are f(b) and f(a) respectively. (see fig (8)).

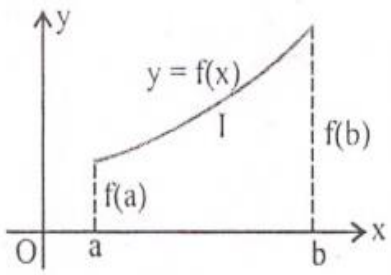


Fig 8

(ix) If f(x) is a monotonic decreasing function in the interval [a,b] then greatest and least values of f(x) in the intervals [a,b] are f(a) and f(b) respectively (see fig. (9))

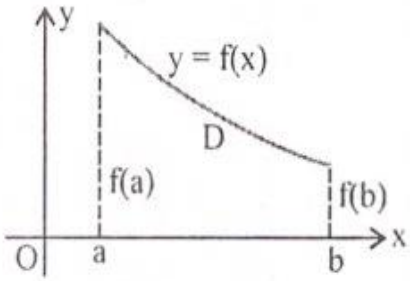


Fig 9

Career Avenues

(x) If f(x) is monotonic increasing in [a,b] and monotonic decreasing in [b,c] then greatest value f(x)=f(b) and least value of f(x)= minimum {f(a). f(c)} (see fig. (10))

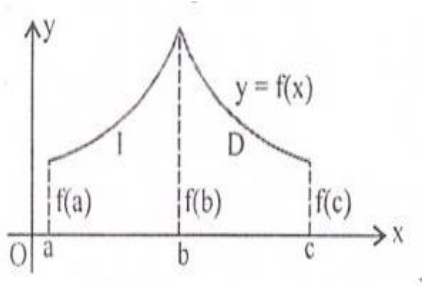


Fig 10

(xi) if f(x) is monotonic decreasing in [a,b] and monotonic increasing in [b,c] then the greatest value of f(x) = maximum {f(a), f(c)} and least value of f(x)=f(b) (see fig. (11))

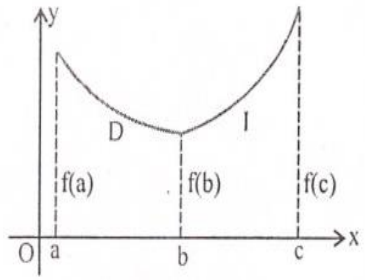


Fig 11

(xii) If f(x) is strictly increasing function on an interval [a,b] then $f^{-1}(x)$ exists and it is also a strictly increasing function (see fig (12))

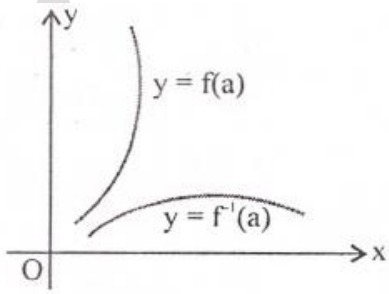


Fig 12

Computation of derivatives of different functions is described as following:-

Derivatives of elementary functions:-

Most derivative computations eventually require taking the derivative of some common functions. The following list gives some of the most frequently used functions of a single real variable and their derivatives.

If, $f(x) = x^r$, where *r* is any real number, then
$f'(x) = r\,x^{r-1}$,

Career Avenues

wherever this function is defined. For example, if $r = 1/2$, then

$f'(x) = \frac{1}{2}x^{-1/2}$

and the function is defined only for non–negative x. When r = 0, this rule recovers the constant rule.

- ***Exponential and logarithmic functions***:-

  $\frac{d}{dx}a^x = a^x \ln a$

  $\frac{d}{dx}\ln x = \frac{1}{x}$, $x>0$

  $\frac{d}{dx}\log_a x = \frac{d}{dx}\frac{\ln x}{\ln a} = \frac{1}{x \ln a}$

- ***Trigonometric functions***:-

  $\frac{d}{dx}\sin x = \cos x$ $\quad$ $\frac{d}{dx}\cos x = -\sin x$

  $\frac{d}{dx}\tan x = \sec^2 x$ $\quad$ $\frac{d}{dx}\cot x = -\operatorname{cosec}^2 x$

  $\frac{d}{dx}\sec x = \sec x \tan x$ $\quad$ $\frac{d}{dx}\operatorname{cosec} x = -\operatorname{cosec} x \cot x$

- ***Inverse trigonometric functions***:-

  $\frac{d}{dx}\sin^{-1} x = \frac{1}{\sqrt{1x2}}$ $\quad$ $\frac{d}{dx}\cos^{-1} x = -\frac{1}{\sqrt{1x2}}$ $\quad$ $\frac{d}{dx}\tan^{-1} x = \frac{1}{1+x2}$ $\quad$ $\frac{d}{dx}\cot^{-1} x = -\frac{1}{1+x^2}$

  $\frac{d}{dx}\sec^{-1} x = \frac{1}{|x|\sqrt{(x^2-1)}}$

Other Useful formulae:-

$\frac{d}{dx}\sinh(x) = \cosh(x)$ $\quad$ $\frac{d}{dx}\cosh(x) = \sinh(x)$

$\frac{d}{dx}\tanh(x) = \operatorname{sech}^2(x)$ $\quad$ $\frac{d}{dx}\coth(x) = -\operatorname{csch}^2(x)$

$\frac{d}{dx}\operatorname{sech}(x) = -\operatorname{sech}(x)\tanh(x)$ $\quad$ $\frac{d}{dx}\operatorname{csch}(x) = -\operatorname{csch}(x)\coth(x)$

Rules for finding the derivative:-

In many cases, complicated limit calculations by direct application of Newton's difference quotient can be avoided using differentiation rules. Some of the most basic rules are the following.

- ***Constant rule***:- if 'c' is constant, then $\frac{d}{dx}(c) = 0$
- ***Sum rule***:- $\frac{d}{dx}(u+v) = \frac{d}{dx}u + \frac{d}{dx}v$
- ***Product rule***:- $\frac{d}{dx}(uv) = u\frac{dv}{dx} + v\frac{du}{dx}$

Career Avenues

- ***Quotient rule***:- $\frac{d}{dx}\frac{u}{v} = \frac{v\frac{du}{dx}u\frac{dv}{dx}}{v2}$
- ***Chain rule***:- If y = f(u) and u = g(x), then $\frac{dy}{dx} = \frac{dy}{du}\frac{du}{dx}$

Chain Rule:-
The Chain Rule is used when we want to take the derivative of a function that has been formed by taking the *composition* $f \circ g$ of two other functions. An example would be:

$f(x) = \sqrt{x}, g(x) = x^2 + 1,$

$(f \circ g)(x) = f(g(x)) = \sqrt{x^2 + 1}$

Note, at the end, what we have is an 'inside' function, $x^2 + 1$, and an 'outside function', $\sqrt{x}$. Usually, we will have to recognize when a function has been formed via composition, i.e. we will have to recognize that there is, indeed, and inside and outside function.

2.8 Rolle's & Lagrange's Theorem:-

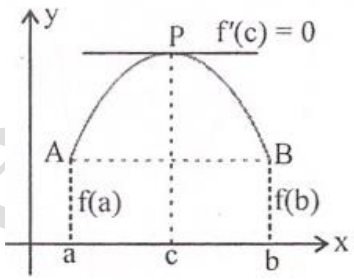
2.8.1 Rolle's Theorem:-

If a function f(x) is

(i) Continuous in a closed interval [a,b]

(ii) derivable in the open interval (a,b)

(iii) f(a) = f(b), then there exists at least one real number c in (a,b) such that f'(c)=0

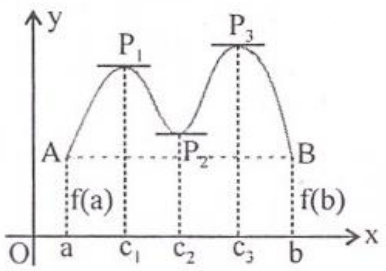


Geometrical significance of Rolle's Theorem:-

The existence of a real number c(a,b) such that f'(c)=0 shows that there is at least one point P on the curve between A & B, the tangent at which is parallel to x axis (see above figure.

There may exist more than one point between A & B, the tangents at which are parallel to x-axis i.e. there exists more than one real number c in (a,b) such that f'(c)=0.

Career Avenues



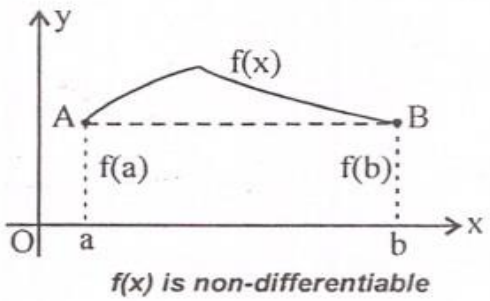
Proof:-

Since, f is continuous on [a,b] it has a maximum M and a minimum m on that interval. If M=m then f is a constant and in that case f'(x)=0 at all points in (a,b)

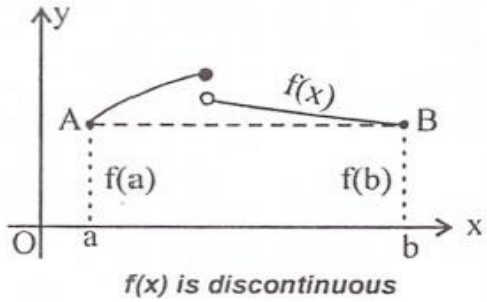
Suppose Mm. If f attains maxima at the end points, then it has a minima at some interior point and if f attains minima at the end points, then it has a maxima at some interior point. In either case f'(x) =0 at that interior point where f attains its maxima or minima.

Remarks:-

(1) Rolle's theorem fails for the function which does not satisfy even one of the three conditions.

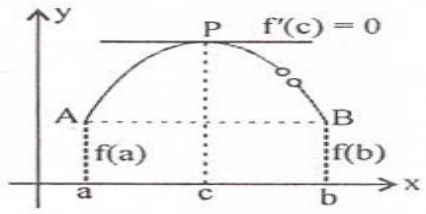


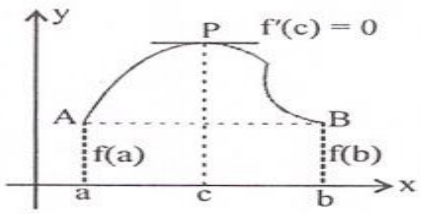
f(x) is non-differentiable



f(x) is discontinuous

(2) The converse of Rolle's Theorem may not be true i.e. f'(c) may be zero at a point in (a,b) without satisfying all the three conditions of Rolle's Theorem.

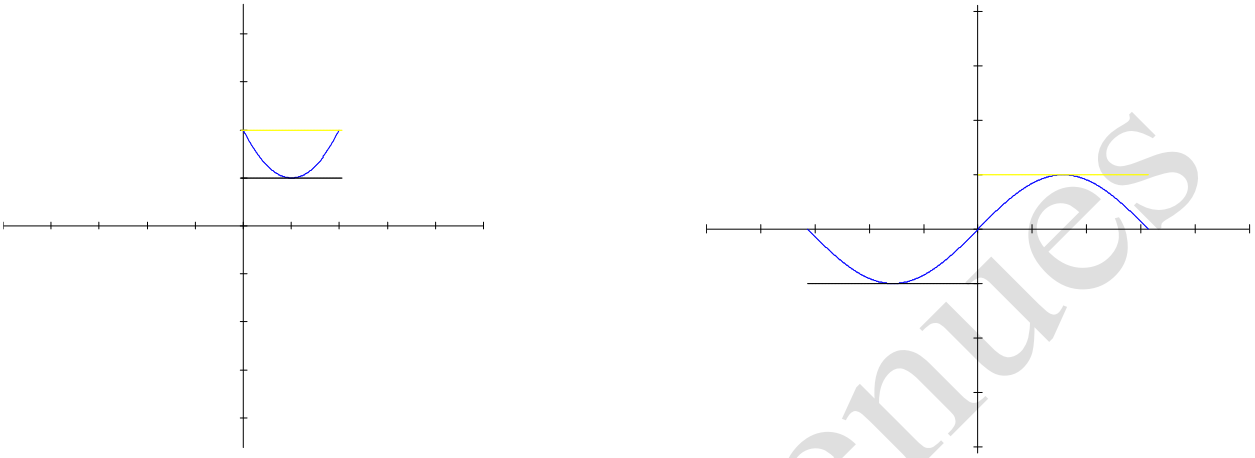




Rolle's Theorem:-

Rolle's Theorem is named for the French mathematician Michel Rolle. He noted that if you take a function *f* that is continuous over [*a*,*b*] and differentiable over (*a*,*b*) and *f*(*a*) = *f*(*b*), then there must exist at least one horizontal tangent line to *f* over [*a*,*b*].

Let's look at two sample graphs that will illustrate this theorem.



In the left–hand example, [0,2], for $f(x) = x^2 - 2x + 2$, $f'(1) = 0$. In the right–hand example $[-\pi, \pi]$ for y=*sin x*, $f'(-\pi/2) = 0$ and $f'(\pi/2) = 0$.

Rolle's Theorem :- Let f be a continuous function over the closed interval [*a*,*b*], and let f be differentiable over the open interval (*a*,*b*). If $f(a) = f(b)$, then there exists at least one *c* in (*a*,*b*) such that $f'(c) = 0$.

2.8.2 Lagrange's Mean value theorem:-

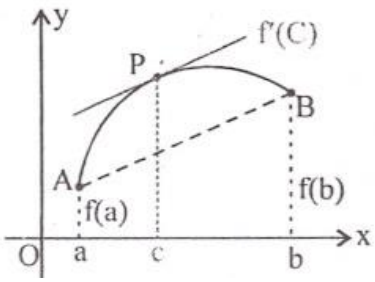
If a function f(x) is

(i) Continuous in a closed interval [a,b] &

(ii) derivable in the open interval (a,b)

then there exists at least one real number c in (a,b) such that f'(c)=

Geometrical IIlustration:-



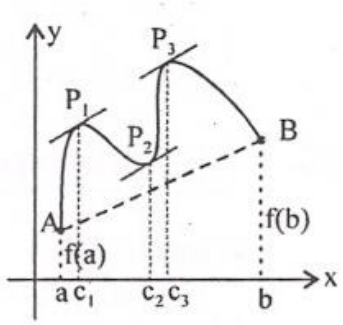
Let P be a point on the curve f(x) such that

=f'(c) …(i)

The slope of chord AB is & that of the tangent at point P is f'(c). These being equal by (i), it follows that there exists a point P on the curve, the tangent at which is parallel to the chord AB.

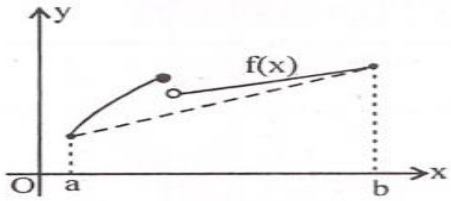
There may exist more than one point between A & B, the tangents at which are parallel to the chord AB, Lagrange's mean value theorem ensures the existence of at least one real number c in (a,b) such that f'(c)= .

Career Avenues

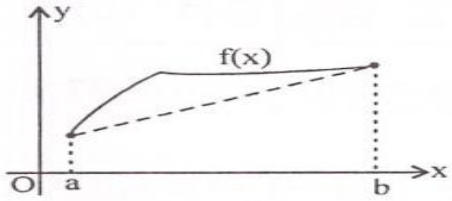


Remarks:-

1. Rolle's theorem is a special case of Lagrange's mean value theorem in which f(a)=f(b), thereby f'(c)=0
2. Lagrange's mean value theorem fails for the function which does not satisfy even one of the two conditions.

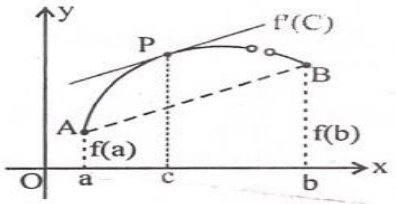


f(x) is discontinuous

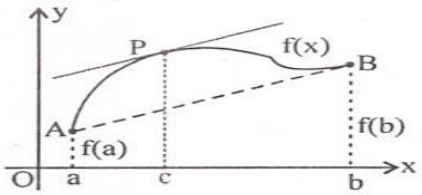


f(x) is non differentiable

3. The converse of Lagrange's mean value theorem may not be true i.e. f'(c) may be equal to at a point c in (a,b) without satisfying both the condition of Lagrange's mean value theorem.

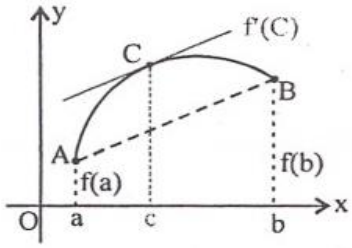


f(x) is discontinuous



f(x) is non differentiable

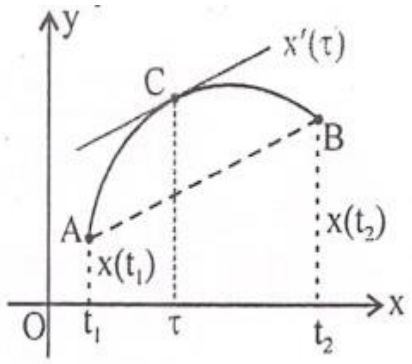
Physical IIIustration:-



[f(b)-f(a)] is the change in the function f as x changes from a to b so that is the average rate of change of the function over the interval [a,b].

Also f'(c) is the actual rate of change of the function for x=c. Thus the theorem states that the average rate of change of a function over an interval is also the actual rate of change of the function at some point of the interval. In particular, for instance, the average velocity of a particle over an interval of time is equal to the velocity at some instant belonging to the interval. i.e. there exists a time for which x'()=

i.e. instantaneous velocity = average velocity

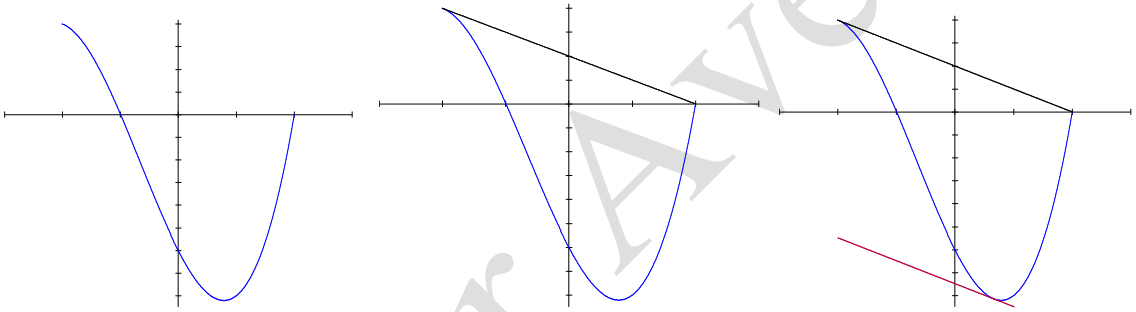


2.9 The Mean Value Theorem:-

The Mean Value Theorem is key component of Calculus.

Mean Value Theorem (MVT): Let f be a continuous function over the closed interval $[a,b]$, and let f be differentiable over the open interval (a,b). There exists at least one c in (a,b) such that $f'(c) = \dfrac{f(b)-f(a)}{b-a}$

Example :-



The left–hand graph is the graph of $f(x) = x^3 + 2x^2 - 5x - 6$ from (-2,2). We then determine $\dfrac{f(b)-f(a)}{b-a}$. The resulting answer leads to a slope of –1. This is shown in the middle illustration. We then find the derivative of f and determine it to be $f'(x) = 3x^2 + 4x - 5$. Now according to the MVT we can conclude that there is some value from (–2,2) where the slope at that point will be –1. Therefore we will set $-1 = 3x^2 + 4x - 5$. We find that the results are –2 and 2/3. Because –2 does not fit for the open interval restriction, we will look at 2/3. The right–hand graph shows the illustration of this.

You can see the way that Rolle's Theorem is an example of the Mean Value Theorem. Only it is for the specific case where $f(a) = f(b)$. There are other theorems that can be derived by playing around with the MVT.

Career Avenues

Example:-

Show that $|\cos 2a - \cos 2b| \le 2|a-b|$ for all real numbers *a* and *b*.

Solution:-

Set f(x) = cos 2x. Then we have $f'(x) = -2\sin 2x$. The inequality clearly holds when a = b.

For any numbers *a* and *b*, with $a \neq b$, the Mean Value Theorem implies that $f'(c) = \dfrac{f(b)-f(a)}{b-a}$

$\Rightarrow -2\sin 2c = \dfrac{\cos 2b - \cos 2a}{b-a}$ for some $c \in (a,b)$.

Since $|\sin 2c| \le 1 \Rightarrow |-2\sin 2c| \le 2 \Rightarrow \left|\dfrac{\cos 2b - \cos 2a}{b-a}\right| \le 2$

$\Rightarrow |\cos 2a - \cos 2b| \le 2|a-b|$.

2.10 PARTIAL DERIVATIVES:

If a derivative of a function of several independent variables be found with respect to any one of them, keeping the other as constants, it is said to be a partial derivatives. The operations of finding the partial derivative of a function or more than one independent variables is called partial differentiation.

The symbols etc. are used to denote such differentiations and the expressions , etc are respectively called partial differential coefficients of u with respect to x and y.

If u=f(x,y,z) the partial differential coefficient of u with respect to x i.e. is obtained by differentiating u with respect to x keeping y and z as constants.

The rate of change of f with respect to x (holding y constant) is called the **partial derivative of f with respect to x** and is denoted by $f_x(x,y)$. Similarly, the rate of change of f with respect to y is called the **partial derivative of f with respect to y** and is denoted by $f_y(x,y)$.

To calculate $f_x(x,y)$ **,** you view y as a constant and differentiate f(x,y) with respect to x:

$f_x(x,y) = y^2$ as expected since $\dfrac{dx}{dx} = 1$, and $f_y(x,y)=2xy$ since $\dfrac{d}{dy}[y^2]=2y$.

Let z=f(x,y).The partial derivative $f_x(x,y)$ can also be written as: $\dfrac{\partial z}{\partial x}, z_x, \dfrac{\partial f(x,y)}{\partial x}$

Similarly, $f_y(x,y)$ can also be written as $\dfrac{\partial z}{\partial y}, z_y, \dfrac{\partial f(x,y)}{\partial y}$

The partial derivative fy(x y) evaluated at the point (x0 y0) can be expressed in several ways: $f_x(x_0,y_0)$, $\dfrac{\partial f(x_o,y_o)}{\partial x}$

Higher partial derivative:-

To get the higher derivatives it is found that we must use the first derivative to get the second one if we differentiate $\dfrac{\partial Z}{\partial x}$ we have two options we can differentiate it with respect to x or differentiate it with respect to y. if we differentiate it with respect to x we get $\dfrac{\partial}{\partial x}\left(\dfrac{\partial Z}{\partial x}\right)$ or what we call it $\dfrac{\partial^2 Z}{\partial x^2}$ or Zxx and it is defined by

$$\frac{\partial^2 Z}{\partial x^2} = \frac{\partial}{\partial x}\left(\frac{\partial Z}{\partial x}\right) = \lim_{\Delta x \to 0} \frac{f_x(x+\Delta x, y) - f_x(x,y)}{\Delta x}$$

and with respect to y we get $\dfrac{\partial}{\partial y}\left(\dfrac{\partial Z}{\partial x}\right)$ or Zxy and it is defined by

Career Avenues

$$\frac{\partial^2 Z}{\partial y \partial x} = \frac{\partial}{\partial y}\left(\frac{\partial Z}{\partial x}\right) = \lim_{\Delta y \to 0} \frac{f_x(x, y+\Delta y) - f_x(x, y)}{\Delta y}$$

Similarly, we can get the first derivate with respect to y to get the second derivative with respect to x or y so we get

$$\frac{\partial^2 Z}{\partial y^2} = \frac{\partial}{\partial y}\left(\frac{\partial Z}{\partial y}\right) = \lim_{\Delta y \to 0} \frac{f_y(x, y+\Delta y) - f_y(x, y)}{\Delta y}$$

And $\frac{\partial^2 Z}{\partial x \partial y} = \frac{\partial}{\partial x}\left(\frac{\partial Z}{\partial y}\right) = \lim_{\Delta x \to 0} \frac{f_y(x+\Delta x, y) - f_y(x, y)}{\Delta x}$

Example:-

Determine ∂z/∂x, ∂z/∂y in each case

i) $z = x^2 + y^2$ ii) $z = \frac{x}{y}$

iii) $z = \frac{1}{\sqrt{x^2 + y^2}}$ iv) $z = \ln(1 + xy)$

v) $z = e^{-xy}(2 + 3xy)$ vi) $z = x^3 \tan^{-1}\left(\frac{x}{y}\right)$

Solution:-

i) $z = x^2 + y^2$

$\frac{\partial z}{\partial x} = 2x$ (y is constant) $\frac{\partial z}{\partial y} = 2y$ (x is constant)

ii) $z = \frac{x}{y}$

$z_x = \frac{1}{y}$ $z_y = -\frac{x}{y^2}$

iii) $z = \frac{1}{\sqrt{x^2 + y^2}} = (x^2 + y^2)^{-1/2}$

so $z_x = -\frac{1}{2}(x^2 + y^2)^{-3/2}(2x) = -\frac{x}{(x^2 + y^2)^{3/2}}$

and by symmetry $z_y = -\frac{y}{(x^2 + y^2)^{3/2}}$

iv) $z = \ln(1 + xy)$

$z_x = \frac{y}{1 + xy}$ and by symmetry $z_y = \frac{x}{1 + xy}$

v) $z = e^{-xy}(2 + 3xy)$

$z_x = (-y)e^{-xy}(2 + 3xy) + e^{-xy}(3y) = ye^{-xy}(1 - 3xy)$

$z_y = (-x)e^{-xy}(2 + 3xy) + e^{-xy}(3x) = xe^{-xy}(1 - 3xy)$

vi) $z = x^3 \tan^{-1}\left(\frac{x}{y}\right)$

$z_x = 3x^2 \tan^{-1}\left(\frac{x}{y}\right) + x^3 \frac{1}{1 + (x/y)^2}\left(\frac{1}{y}\right)$

Career Avenues

$= 3x^2 \tan^{-1}\left(\frac{x}{y}\right) + \frac{yx^3}{x^2+y^2}$

$z_y = x^3 \frac{1}{1+(x/y)^2}\left(\frac{x}{y^2}\right) = -\frac{x^4}{x^2+y^2}$

Example:-

1. Given $y = f(x_1, x_2) = 3x_1^2 + x_1x_2 + 4x_2^2$. Find Partial derivatives of x_1 and x_2.

Solution:-

$\frac{\partial y}{\partial x_1} \equiv f_1 = 6x_1 + x_2$

$\frac{\partial y}{\partial x_2} \equiv f_2 = x_1 + 8x_2$

Example:-

Determine $\partial z/\partial x$, $\partial z/\partial y$ in each case

i) $z = x^3 + x^2y + y^4$ ii) $z = e^{xy}\cos(3y^2)$

iii) $z = \frac{x^2+y^2}{\sqrt{1+y}}$

Solution:-

i) $z = x^3 + x^2y + y^4$

$z_x = 3x^2 + 2xy$

$z_y = x^2 + 4y^3$

ii) $z = e^{xy}\cos(3y^2)$

$z_x = ye^{xy}\cos(3y^2)$

$z_y = xe^{xy}\cos(3y^2) + e^{xy}(-\sin(3y^2))\,6y$

$= e^{xy}(x\cos(3y^2) - 6y\sin(3y^2))$

iii) $z = \frac{x^2+y^2}{\sqrt{1+y}}$

$z_x = \frac{2x}{\sqrt{1+y}}$ (*remember y and therefore* $\sqrt{1+y}$ *is a constant*)

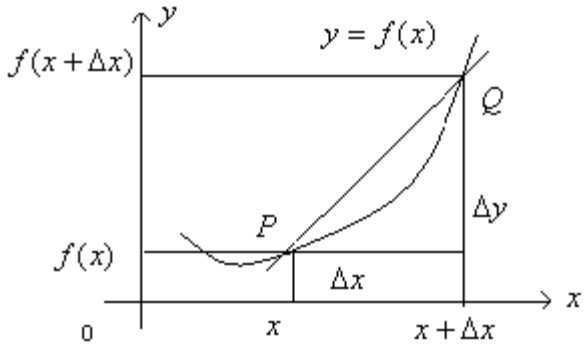
$z_y = \frac{2y}{\sqrt{1+y}} - \frac{1}{2}\frac{x^2+y^2}{(1+y)^{3/2}} = \frac{4y(1+y)x^2y^2}{2(1+y)^{3/2}} = \frac{3y^2+4yx^2}{2(1+y)^{3/2}}$

SUCCESSIVE DIFFERENTIATION:-

Differential calculus (DC) deals with problem of calculating rates of change. When we have a formula for the distance that a moving body covers as a function of time, DC gives us the formulas for calculating the body's **velocity** and **acceleration** at any instant.

Definition of derivative of a function y = f(x) :–

Career Avenues



Slope of the line PQ is $\frac{f(x+\Delta x)-f(x)}{\Delta x}$

The derivative of a function y = f(x) is the function $f'(x)$ whose value at each x is defined as

$\frac{dy}{dx}$ = $f'(x)$ = Slope of the line PQ

$= \lim_{\Delta x \to 0} \frac{f(x+\Delta x)-f(x)}{\Delta x}$ ———— (1)

$= \lim_{\Delta x \to 0}$ (Average rate change)

= Instantaneous rate of change of *f* at *x* provided the limit exists.

The instantaneous velocity and acceleration of a body (moving along a line) at any instant *x* is the derivative of its position co–ordinate *y = f(x)* w.r.t x, i.e.,

Velocity = $\frac{dy}{dx} = f'(x)$ ———— (2)

And the corresponding acceleration is given by

Acceleration $= \frac{d^2y}{dx^2} = f''(x)$ ———— (3)

***Successive Differentiation*:–**

The process of differentiating a given function again and again is called as ***Successive differentiation*** and the results of such differentiation are called ***successive derivatives.***

The higher order differential coefficients will occur more frequently in spreading a function all fields of scientific and engineering applications.

Notations:-

i. $\frac{dy}{dx}, \frac{d^2y}{dx^2}, \frac{d^3y}{dx^3}$,……., n^{th} order derivative: $\frac{d^ny}{dx^n}$

ii. $f'(x)$, $f''(x)$, $f'''(x)$,….., n^{th} order derivative: $f^n(x)$

iii. Dy, D^2y, D^3y ,……..., n^{th} order derivative: D^ny

iv. y' , y'' , y''' ,……, n^{th} order derivative: $y^{(n)}$

v. $y_1, y_2, y_3 \ldots$, n^{th} order derivative: y_n

Career Avenues

Example:-

If $y = \sin(\sin x)$, prove that $\frac{d^2y}{dx^2} + \tan x \frac{dy}{dx} + y\cos^2 x = 0$

Solution:-

Differentiating $y = \sin(\sin x)$ ———— (1) with respect to (w.r.t) x, we get

$y_1 = \frac{dy}{dx} = \cos(\sin x).\cos x$ ———————— (2)

Again differentiating $y_1 = \frac{dy}{dx}$ w.r.t. x gives

$y_2 = \frac{d^2y}{dx^2} = \cos(\sin x)(-\sin x) + \cos x(-\sin(\sin x)\cos x$ Using product rule

$y_2 = \frac{d^2y}{dx^2} = -\left[\sin x \cos(\sin x) + \cos^2 x \sin(\sin x)\right]$

i.e. $y_2 = -\left[\frac{\sin x}{\cos x}\cos x \cos(\sin x) + \cos^2 x \sin(\sin x)\right]$

$y_2 = -\left[\tan x y_1 + \cos^2 xy\right]$, using Eqs. (1) and (2)

or $y_2 + \tan x y_1 + \cos^2 xy = 0$

or $\frac{d^2y}{dx^2} + \tan x \frac{dy}{dx} + y\cos^2 x = 0$

Calculation of nth derivatives of some standard functions:-

1. Consider e^{mx} . Let $y = e^{mx}$. Differentiating w.r.t x, we get

$y_1 = me^{mx}$. Again differentiating w.r.t x, we get

$y_2 = m\ me^{mx} = m^2e^{mx}$

Similarly, we get

$y_3 = m^3e^{mx}$

$y_4 = m^4e^{mx}$

…….

And hence we get $y_n = m^ne^{mx}$ $\therefore \frac{d^n}{dx^n}\left[e^{mx}\right] = m^ne^{mx}$.

2. $(ax+b)^m$

Let $y = (ax+b)^m$ Differentiating w.r.t x,

$y_1 = m(ax+b)^{m-1}a$. Again differentiating w.r.t x, we get

$y_2 = m(m-1)(ax+b)^{m-2}a^2$

Similarly, we get

$y_3 = m(m-1)(m-2)(ax+b)^{m-3}a^3$

…….

And hence we get

$y_n = m(m-1)(m-2)\ldots\ldots(m-n+1)(ax+b)^{m-n}a^n$ for all m.

Case (i) If $m = n$ (m–positive integer),then the above expression becomes

$y_n = n(n-1)(n-2)\ldots\ldots.3.2.1(ax+b)^{n-n}a^n$

Career Avenues

i.e. $y_n = n!\ a^n$

Case (ii) If m<n, (i.e. if n > m) which means if we further differentiate the above expression, the right hand site yields zero. Thus $D^n\left[\ ax+b\ ^m\right]=0 \quad if \quad m<n$

Case (iii) If m>n, then $y_n = m\ \ m-1\ \ m-2\ \ldots\ m-n+1\ \ ax+b\ ^{m-n}\ a^n$ becomes

$$= \frac{m\ \ m-1\ \ m-2\ \ldots\ m-n+1\ \ m-n\ !}{m-n\ !}\ ax+b\ ^{m-n}\ a^n$$

i.e. $y_n = \frac{m!}{m-n\ !}\ ax+b\ ^{m-n}\ a^n$

3. $\frac{1}{ax+b\ ^m}$

$Let \quad y = \frac{1}{ax+b\ ^m} = \ ax+b\ ^{-m}$

Differentiating w.r.t x

$y_1 = -m\ \ ax+b\ ^{-m-1}\ a = \ -1\ m\ \ ax+b\ ^{-m+1}\ a$

$y_2 = \ -1\ \ m\left[-\ m+1\ \ ax+b\ ^{-m+1-1}\ a\right] = \ -1\ ^2\ m\ \ m+1\ \ ax+b\ ^{-m+2}\ \ a^2$

Similarly, we get $y_3 = \ -1\ ^3\ m\ \ m+1\ \ m+2\ \ ax+b\ ^{-m+3}\ \ a^3$

$y_4 = \ -1\ ^4\ m\ \ m+1\ \ m+2\ \ m+3\ \ ax+b\ ^{-m+4}\ \ a^4$

……

$y_n = \ -1\ ^n\ m\ \ m+1\ \ m+2\ \ldots\ m+n-1\ \ ax+b\ ^{-m+n}\ \ a^n$

This may be rewritten as

$$y_n = \frac{-1\ ^n\ \ m+n-1\ \ m+n-2\ \ldots\ m+1\ m\ \ m-1\ !}{m-1\ !}\ ax+b\ ^{-m+n}\ \ a^n$$

Or $y_n = \frac{-1\ ^n\ \ m+n-1\ !}{m-1\ !\ ax+b\ ^{m+n}} a^n$

4. $\frac{1}{ax+b}$

Putting $m=1$, in the result

$$D^n\left[\frac{1}{(ax+b)^m}\right] = \frac{(-1)^n(m+n-1)!}{(m-1)!(ax+b)^{m+n}}\,a^n$$

we get $D^n\left[\frac{1}{(ax+b)}\right] = \frac{(-1)^n(1+n-1)!}{(1-1)!(ax+b)^{1+n}}\,a^n$

or $D^n\left[\frac{1}{(ax+b)}\right] = \frac{(-1)^n n!}{(ax+b)^{1+n}}\,a^n$

Career Avenues

***Table of nth order derivatives of some standard functions*:-**

	y = f(x)	$y_n = \frac{d^n y}{dx^n} = D^n y$
1.	e^{mx}	$m^n e^{mx}$
2.	a^{mx}	$m^n \log a^{\,n}\, a^{mx}$
3.	$(ax+b)^{m}$	i) $m\ (m-1)\ (m-2)\ \ldots\ (m-n+1)\ a^n\ (ax+b)^{m-n}$ for all m. ii) 0 if $m<n$ iii) $n!\ a^n$ if $m=n$ iv) $\frac{m!}{(m-n)!}x^{m-n}$ if $m<n$
4.	$\frac{1}{ax+b}$	$\frac{(-1)^n n!}{(ax+b)^{n+1}}a^n$
5.	$\frac{1}{(ax+b)^{m}}$	$\frac{(-1)^n (m+n-1)!}{(m-1)!(ax+b)^{m+n}}a^n$
6.	$\log(ax+b)$	$\frac{(-1)^{n-1}(n-1)!}{(ax+b)^n}a^n$
7.	$\sin(ax+b)$	$a^n \sin(ax+b+n\pi/2)$
8.	$\cos(ax+b)$	$a^n \cos(ax+b+n\pi/2)$
9.	$e^{ax}\sin(bx+c)$	$r^n e^{ax}\sin(bx+c+n\theta)$, $r=\sqrt{a^2+b^2}$ $\quad\theta=\tan^{-1}(b/a)$
10.	$e^{ax}\cos(bx+c)$	$r^n e^{ax}\cos(bx+c+n\theta)$, $r=\sqrt{a^2+b^2}$ $\quad\theta=\tan^{-1}(b/a)$

Total Derivative:-

a) *Total differential and Total derivative*:–

For a function $z=f(x,y)$ of two variables, x and y the ***total differential (or exact differential) dz*** is defined by:

$$dz=\frac{\partial f}{\partial x}dx+\frac{\partial f}{\partial x}dy \qquad (1)$$

Further, if $z=f(x,y)$ where $x=x(t)$, $y=y(t)$ i.e. x and y are themselves functions of an independent variable t, then total ***derivative of z*** is given by

$$\frac{dz}{dt}=\frac{\partial f}{\partial x}\frac{dx}{dt}+\frac{\partial f}{\partial y}\frac{dy}{dt} \qquad (2)$$

Similarly, the total differential of a function $u=f(x,y,z)$ is defined by

$$du=\frac{\partial f}{\partial x}dx+\frac{\partial f}{\partial y}dy+\frac{\partial f}{\partial z}dz \qquad (3)$$

Further, if $u=f(x,y,z)$ and if $x=x(t)$, $y=y(t)$, $z=z(t)$, then the total derivative of u is given by

$$\frac{du}{dt}=\frac{\partial f}{\partial x}\frac{\partial x}{dt}+\frac{\partial f}{\partial y}\frac{dy}{dt}+\frac{\partial f}{\partial z}\frac{dz}{dt} \qquad (4)$$

Career Avenues

(b) Differentiation of implicit functions :–

An implicit function with x as an independent variable and y as the dependent variable is generally of the form $z = f(x, y) = 0$. This gives $\left(\frac{dz}{dx}\right) = \left(\frac{df}{dx}\right) = 0$ .Then, by virtue of expression (2) above, we get

$\frac{dz}{dx} = \frac{\partial f}{\partial x} + \frac{\partial f}{\partial y}\frac{dy}{dx}$ or $\frac{df}{dx} = \frac{\partial f}{\partial x} + \frac{\partial f}{\partial y}\frac{dy}{dx}$, and hence

$$0 = \frac{\partial f}{\partial x} + \frac{\partial f}{\partial y}\frac{dy}{dx} \text{ so that we get } \frac{dy}{dx} = -\frac{\partial f / \partial x}{\partial f / \partial y} \tag{5}$$

(c) *Differentiation of composite functions*:–

Let z be an function of x and y and that $x = \varphi(u, v)$ and $y = \phi(u, v)$ are functions of u and v then.

$$\frac{\partial z}{\partial u} = \frac{\partial f}{\partial x}\frac{\partial x}{\partial u} + \frac{\partial f}{\partial y}\frac{\partial y}{\partial u}$$

$$\& \quad \frac{\partial z}{\partial v} = \frac{\partial f}{\partial x}\frac{\partial x}{\partial v} + \frac{\partial f}{\partial y}\frac{\partial y}{\partial v} \tag{6}$$

Similarly, if $z = f(u, v)$ are functions of u and v and if $u = \varphi(x, y)$ and $v = \phi(x, y)$ are functions of x and y then,

$$\frac{\partial z}{\partial x} = \frac{\partial f}{\partial u}\frac{\partial u}{\partial x} + \frac{\partial f}{\partial v}\frac{\partial v}{\partial x}$$

$$\& \frac{\partial z}{\partial y} = \frac{\partial f}{\partial u}\frac{\partial u}{\partial y} + \frac{\partial f}{\partial v}\frac{\partial v}{\partial y} \tag{7}$$

Note*:–**1) The above formulae can be extended to functions of three are more variables and formulas (6) and(7) are called ***Chain rule for partial differentiation.

2) The second and higher order partial derivatives of $z = f(x, y)$ can be obtained by repeated applications of the above formulas

***Examples*:–**

1. Find the total differential of

(i) $e^x \; x\sin y + y\cos y$

(ii) e^{xyz}

***Solution*:-**

Ans:– (i) Let $z = f\;(x, y) = e^x \; x\sin y + y\cos y$ Then

$$\frac{\partial z}{\partial x} = e^x\left[\,1 + x\;\sin y + y\cos y\right]$$

and $\frac{\partial z}{\partial y} = e^x\left[\,1 + x\;\cos y - y\sin y\right]$ Hence, using formula (1), we get

$$dz = \frac{\partial z}{\partial x}dx + \frac{\partial z}{\partial y}dy$$

i.e $dz = e^x\left[\,1 + x\;\sin y + y\cos y\right]dx + e^x\left[\,1 + x\;\cos y - y\sin y\right]dy$

(ii) Let $z = f(x, y, z) = e^{xyz}$ Then

Career Avenues

$$\frac{\partial u}{\partial x} = yz\ e^{xyz}; \frac{\partial u}{\partial y} = xz\ e^{xyz}; \frac{\partial u}{\partial z} = xy\ e^{xyz}$$

$\therefore$ Total differential of $z = f(x, y, z)$ is (see formula (3) above)

$$du = \frac{\partial u}{\partial x}dx + \frac{\partial u}{\partial y}dy + \frac{\partial u}{\partial z}dz$$

$$= e^{xyz}\ \ yzdx + zxdy + xydz$$

Example:-

Find $\frac{dz}{dt}$ if (i) $z = xy^2 + x^2y$,where $x = at^2$, $y = 2at$

(ii) $u = \tan^{-1}\ y/x$,where $x = e^t - e^{-t}$, $y = e^t + e^{-t}$

Solution:– ***(i) Consider*** $z = xy^2 + x^2y$

$$\frac{\partial z}{\partial x} = y^2 + 2xy \ \ \& \frac{\partial z}{\partial y} = 2xy + x^2$$

Since $x = at^2$ & $y = 2at$, We have $\frac{dx}{dt} = 2at$, $\frac{dy}{dt} = 2a$

Hence, using formula (2), we get

$$\frac{dz}{dt} = \frac{\partial z}{\partial x}\frac{dx}{dt} + \frac{\partial z}{\partial y}\frac{dy}{dt}$$

$$= \ y^2 + 2xy\ \ 2at\ +\ 2xy + x^2\ \ 2a$$

$$= \ y^2 + 2xy\ \ y + 2a\ \ 2xy + x^2\ \ , \text{Using}\ \ y = 2at$$

$$\frac{dz}{dt} = y^3 + 2xy^2 + 4axy + 2ax^2$$

To get $\left(\frac{dz}{dt}\right)$ explicitly in terms of t, we substitute

$x = at^2$ & $y = 2at$, to get

$$\left(\frac{dz}{dt}\right) = 2a^3\left(8t^3 + 5t^4\right)$$

(ii) Consider $u = \tan^{-1}\ y/x$

$$\frac{\partial u}{\partial x} = \frac{-y}{x^2 + y^2}, \frac{\partial u}{\partial y} = \frac{x}{x^2 + y^2}$$

Since $x = e^t - e^{-t}$ & $y = e^t + e^{-t}$,we have

$$\frac{dx}{dt} = e^t + e^{-t} = y \ \ \frac{dy}{dt} = e^t - e^{-t} = x$$

Hence $\frac{du}{dt} = \frac{\partial u}{\partial x}\frac{dx}{dt} + \frac{\partial u}{\partial y}\frac{dy}{dt}$ (see Eqn (2))

$$= \left(\frac{-y}{x^2 + y^2}\right)y + \left(\frac{x}{x^2 + y^2}\right)x = \left(\frac{x^2 - y^2}{x^2 + y^2}\right)$$

Career Avenues

Substituting $x = e^t - e^{-t}$ & $y = e^t + e^{-t}$, we get $\frac{du}{dt} = \frac{-2}{e^{2t} + e^{-2t}}$

Jacobians:-

Jacobians were invented by German mathematician C.G. Jacob Jacobi (1804–1851), who made significant contributions to mechanics, Partial differential equations and calculus of variations.

Definition:– Let u and v are functions of x and y, then Jacobian of u and v w.r.t x and y, denoted by

J or $J\left(\frac{u,v}{x,y}\right)$ or $\frac{\partial\ u,v}{\partial\ x,y}$

is defined by

$$J\left(\frac{u,v}{x,y}\right) = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix}$$

Similarly, if u, v, w are functions of three independent variables of x, y, z, then

$$J = J\left(\frac{u,v,w}{x,y,z}\right) = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} & \frac{\partial u}{\partial z} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} & \frac{\partial u}{\partial z} \\ \frac{\partial w}{\partial x} & \frac{\partial w}{\partial y} & \frac{\partial w}{\partial z} \end{vmatrix}$$

Remark:– In a similar way, Jacobian of n functions in n–variables can be defined

Properties of Jacobians **:–**

Property **1:–** If $J = \frac{\partial\ u,v}{\partial\ x,y}$ and $J' = \frac{\partial\ x,y}{\partial\ u,v}$ then $JJ' = 1$

Proof:– Consider

$$JJ^1 = \frac{\partial\ u,v}{\partial\ x,y} \times \frac{\partial\ x,y}{\partial\ u,v} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix} \times \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix}$$

$$= \begin{vmatrix} \frac{\partial u}{\partial x}\frac{\partial x}{\partial u} + \frac{\partial u}{\partial y}\frac{\partial y}{\partial u} & \frac{\partial u}{\partial x}\frac{\partial x}{\partial v} + \frac{\partial u}{\partial y}\frac{\partial y}{\partial v} \\ \frac{\partial v}{\partial x}\frac{\partial x}{\partial u} + \frac{\partial v}{\partial y}\frac{\partial y}{\partial u} & \frac{\partial v}{\partial x}\frac{\partial x}{\partial v} + \frac{\partial v}{\partial y}\frac{\partial y}{\partial v} \end{vmatrix} = \begin{vmatrix} 1 & 0 \\ 1 & 0 \end{vmatrix} = 1$$

Property **2:–** (Chain rule for Jacobians):– If u and v are functions of r & s and r, s are functions x&y,then

$$J = \left(\frac{u,v}{x,y}\right) = J\left(\frac{u,v}{r,s}\right) \times J\left(\frac{r,s}{x,y}\right)$$

Proof **:–** Consider

$$J\left(\frac{u,v}{r,s}\right) \times J\left(\frac{r,s}{x,y}\right) = \begin{vmatrix} \frac{\partial u}{\partial r} & \frac{\partial u}{\partial s} \\ \frac{\partial v}{\partial r} & \frac{\partial v}{\partial s} \end{vmatrix} \times \begin{vmatrix} \frac{\partial r}{\partial x} & \frac{\partial r}{\partial y} \\ \frac{\partial s}{\partial x} & \frac{\partial s}{\partial y} \end{vmatrix}$$

Career Avenues

$$= \begin{vmatrix} \frac{\partial u}{\partial r}\frac{\partial r}{\partial x} + \frac{\partial u}{\partial s}\frac{\partial s}{\partial x} & \frac{\partial u}{\partial r}\frac{\partial r}{\partial y} + \frac{\partial u}{\partial s}\frac{\partial s}{\partial y} \\ \frac{\partial v}{\partial r}\frac{\partial r}{\partial x} + \frac{\partial v}{\partial s}\frac{\partial s}{\partial x} & \frac{\partial v}{\partial r}\frac{\partial r}{\partial y} + \frac{\partial v}{\partial s}\frac{\partial s}{\partial y} \end{vmatrix}$$

$$= \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix} = J\left(\frac{u,v}{x,y}\right)$$

***Jacobians in various co–ordinate systems*:–**

***Example*:**

(1) Show that in Polar co–ordinates, $x = r\cos\theta$, $y = r\sin\theta$

we have $\frac{\partial\ u,v}{\partial\ r,\theta} = r$

(2) Show that in spherical coordinates, $x = \rho\cos\varphi, y = \rho\sin\varphi, z = z,$ we have

$$\frac{\partial\ x,y,z}{\partial\ \rho,\varphi,z} = \rho$$

(3) Show that in spherical polar co–ordinates, $x = r\sin\theta\cos\varphi$, $y = r\sin\theta\sin\varphi$, $z = r\cos\theta$

Solutions:-

(1) We have, $\frac{\partial x}{\partial r} = \cos\theta$ and $\frac{\partial y}{\partial r} = \sin\theta$

$$\frac{\partial x}{\partial\theta} = -r\sin\theta \text{ and } \frac{\partial y}{\partial\theta} = r\cos\theta$$

$$\therefore \frac{\partial\ x,y}{\partial\ r,\theta} = \begin{vmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial\theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial\theta} \end{vmatrix} = \begin{vmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{vmatrix}$$

$$= r\cos^2\theta + r\sin^2\theta = r\ \ \cos^2\theta + \sin^2\theta\ \ = r$$

(2) We have $\frac{\partial x}{\partial\rho} = \cos\varphi$, $\frac{\partial y}{\partial\rho} = \sin\varphi$, $\frac{\partial z}{\partial\rho} = 0$

$$\frac{\partial x}{\partial\varphi} = -\rho\sin\varphi, \frac{\partial y}{\partial\varphi} = \rho\cos\varphi \text{ , } \frac{\partial z}{\partial\varphi} = 0$$

$$\frac{\partial z}{\partial\rho} = 0, \frac{\partial z}{\partial\varphi} = 0, \frac{\partial z}{\partial z} = 0$$

$$\therefore \frac{\partial\ x,y,z}{\partial\ \rho,\varphi,z} = \begin{vmatrix} \frac{\partial x}{\partial\rho} & \frac{\partial x}{\partial\varphi} & \frac{\partial x}{\partial z} \\ \frac{\partial y}{\partial\rho} & \frac{\partial y}{\partial\varphi} & \frac{\partial y}{\partial z} \\ \frac{\partial z}{\partial\rho} & \frac{\partial z}{\partial\varphi} & \frac{\partial z}{\partial z} \end{vmatrix} = \begin{vmatrix} \cos\varphi & -\rho\sin\varphi & 0 \\ \sin\varphi & \rho\cos\varphi & 0 \\ 0 & 0 & 1 \end{vmatrix} = \rho$$

(3) We have $\frac{\partial x}{\partial r} = \sin\theta\cos\theta, \frac{\partial x}{\partial\theta} = r\cos\theta\cos\varphi, \frac{\partial x}{\partial\varphi} = -r\sin\theta\sin\varphi$

Career Avenues

$$\frac{\partial y}{\partial r}=\sin\theta\sin\varphi,\ \frac{\partial y}{\partial \theta}=r\cos\theta\sin\varphi,\ \frac{\partial y}{\partial \varphi}=r\sin\theta\cos\varphi$$

$$\frac{\partial z}{\partial r}=\cos\theta,\ \frac{\partial z}{\partial \theta}=-r\sin\theta,\ \frac{\partial z}{\partial \varphi}=0$$

$$\therefore \frac{\partial\ x,y,z}{\partial\ r,\theta,\varphi}=\begin{vmatrix}\sin\theta\cos\theta & r\cos\theta\cos\varphi & -r\sin\theta\sin\varphi\\ \sin\theta\sin\varphi & r\cos\theta\sin\varphi & r\sin\theta\cos\varphi\\ \cos\theta & -r\sin\theta & 0\end{vmatrix}$$

$$=r^2\sin\theta$$

Leibnitz's theorem :-

Leibnitz's theorem is useful in the calculation of nth derivatives of product of two functions.

***Statement of the theorem* :-**

If u and v are functions of x, then $D^n(uv)=D^nuv+{}^nC_1D^{n-1}uDv+{}^nC_2D^{n-2}uD^2v+....+{}^nC_rD^{n-r}uD^rv+...uD^nv$,

where $D=\frac{d}{dx}$, ${}^nC_1=n$, ${}^nC_2=\frac{n\ n-1}{2}$,....., ${}^nC_r=\frac{n!}{r!\ n-r\ !}$

Example:-

If $x=\sin t, y=\sin pt$ *prove that*

$$1-x^2\ \ y_{n+2}-\ 2n+1\ xy_{n+1}+\ p^2-n^2\ \ y_n=0$$

Solution:-

Note that the function $y=f(x)$ is given in the parametric form with a parameter t.

So, we consider

$$\frac{dy}{dx}=\frac{dy/dt}{dx/dt}=\frac{p\cos pt}{\cos t}\qquad (p-\text{constant})$$

or $$\left(\frac{dy}{dx}\right)^2=\frac{p^2\cos^2 pt}{\cos^2 t}=\frac{p^2(1-\sin^2 pt)}{1-\sin^2 t}=\frac{p^2(1-y^2)}{1-x^2}$$

or $$1-x^2\ \ y_1^2=p^2\ \ 1-y^2$$

So that $$1-x^2\ \ y_1^2-p^2\ \ 1-y^2\ \ =0$$ Differentiating w.r.t. x,

$$\left[\ 1-x^2\ \ \ 2y_1y_2\ \ +y_1^2\ \ -2x\ \right]-p^2\ \ -2yy_1\ \ =0$$

$$1-x^2\ \ y_2-xy_1+p^2y=0 \qquad (1)$$

[$\div\ 2y_1$, throughout]

Equation (1) has second order derivative y_2 in it. We differentiate (1), n times, term wise, using Leibnitz's theorem as follows.

$$D^n\left[\ 1-x^2\ \ y_2-xy_1-p^2y\right]=0$$

i.e $$D^n\ \ (1-x^2)y_2\ \ -D^n\ \ xy_1\ \ -D^n\ \ p^2y\ \ =0 \qquad (2)$$

(a) (b) (c)

Career Avenues

Consider the term (a) :-

$D^n\left[\ 1-x^2\ \ y_2\right]$. Taking $u = y_2$ and $v = (1-x^2)$ and applying Leibnitz's theorem we get

$D^n\ uv = D^n uv + {}^nC_1 D^{n-1} uDv + {}^nC_2 D^{n-2} uD^2 v + {}^nC_3 D^{n-3} uD^3 v + ...$

i.e

$D^n\left[y_2(1-x^2)\right] = D^n(y_2).(1-x^2) + {}^nC_1 D^{n-1}(y_2).D(1-x^2) + {}^nC_2 D^{n-2}(y_2)D^2(1-x^2) + {}^nC_3 D^{n-3}(y_2)D^3(1-x^2) + ...$

$= y_{(n)+2}.(1-x^2) + ny_{(n-1)+2}.(-2x) + \frac{n(n-1)}{2!} y_{(n-2)+2}.(-2) + \frac{n(n-1)(n-2)}{3!}.y_{(n-3)+2}.(0) + ...$

$D^n\left[\ 1-x^2\ \ y_2\right] = \ 1-x^2\ \ y_{n+2} - 2nxy_{n+1} - n(n-1)y_n$ (3)

Consider the term (b) :-

$D^n\ xy_1$. Taking $u = y_1$ and $v = x$ and applying Leibnitz's theorem, we get

$D^n\ y_1(x) = D^n(y_1).(x) + {}^nC_1 D^{n-1} y_1.D(x) + {}^nC_2 D^{n-2}(y_1).D^2(x) + ...$

$= y_{(n)+1}.x + ny_{(n-1)+1} + \frac{n(n-1)}{2!} y_{(n-2)+2}(0) + ...$

$D^n\ xy_1 = xy_{n+1} + ny_n$ (4)

Consider the term (c) :-

$D^n(p^2 y) = p^2 D^n(y) = p^2 y_n$ (5)

Substituting these values (3), (4) and (5) in Eq (2) we get

$1-x^2\ \ y_{n+2} - 2nxy_{n+1} - n(n-1)y_n\ \ -\ \ xy_{n+1} + ny_n\ \ +\ \ p^2 y_n\ \ = 0$

i.e. $1-x^2\ \ y_{n+2} - (2n+1)xy_{n+1} - n^2 y_n + ny_n - ny_n + p^2 y_n = 0$

$\therefore\ 1-x^2\ \ y_{n+2} - (2n+1)xy_{n+1} +\ \ p^2 - n^2\ \ y_n = 0$ as desired.

***Example*:-** If $\sin^{-1} y = 2\log(x+1)$ or

or $y = \sin\left[\log(x+1)^2\right]$

or $y = \sin\log(x^2+2x+1)$, show that

$x+1\ ^2\ y_{n+2} +\ 2n+1\ \ x+1\ y_{n+1} +\ n^2+4\ \ y_n = 0$

***Solution*:-**

Out of the above four versions, we consider the function as

$\sin^{-1}(y) = 2\log(x+1)$

Differentiating w.r.t x, we get

$\frac{1}{\sqrt{1-y^2}}(y_1) = \left(\frac{2}{x+1}\right)$ ie $(x+1)y_1 = 2\sqrt{1-y^2}$

Squaring on both sides

$x+1\ ^2\ y_1^2 = 4(1-y^2)$

Again differentiating w.r.t x,

$x+1\ ^2\ \ 2y_1 y_2\ \ + y_1^2\ \ 2(x+1)\ \ = 4(-2yy_1)$

or $\quad x+1\ ^2\ y_2 + (x+1)y_1 = -4y \quad (\div 2y_1)$

or $\quad x+1\ ^2\ y_2 + (x+1)y_1 + 4y = 0$ (1)

Career Avenues

Differentiating (1) w.r.t x, n–times, using Leibnitz's theorem,

$$\left\{D^n\ y_2\ \ x+1\ ^2+nD^{n-1}\ \ y_2\ \ 2\ \ x-1\ +\frac{n\ \ n-1}{2!}D^{n-2}\ \ y_2\ \ \ 2\ \right\}+\ \ D^n\ \ g_1\ \ \ x+1\ +nD^{n-1}\ y_1\ \ 1\ \ \ +4D^n y=0$$

On simplification, we get

$$x+1\ ^2\ y_{n+2}+\ \ 2n+1\ \ \ x+1\ \ y_{n+1}+\ \ n^2+4\ \ y_n=0$$

2.10.1 Homogenous function:

An expression in which every term is of the same degree is called Homogenous function.

$=x^n$

Or $x^n f(y/x)$ where f(y/x) is same function of y/x

To test whether is a given function f(x,y) is Homogenous or not we put for x and for y in it.

If we get $f= t^n f(x,y)$, the function f(x,y) is Homogenous of degree in otherwise f(x,y) is not a Homogenous function.

2.10.2 Euler's Theorem:

The following theorem, named in honour of the great Swiss mathematician Euler (1707-1783) gives a very useful formula for a particular combination of partial derivatives of homogenous functions.

Statement of Euler's theorem:- If u is a homogenous function of x and y with degree n, then $x\frac{\partial u}{\partial x}+y\frac{\partial u}{\partial y}=nu$

Proof: - Since u is a *homogenous function of degree n*, we can put u is of the form $u=x^n\varphi\ \ y/x\ \ .$

Differentiating this w.r.t x partially, we get

$$\frac{\partial u}{\partial x}=x^n\ \phi^1\ \ y/x\ \left(-y/x^2\right)+\phi\ \ y/x\ \ \ nx^{n-1}$$

$$\text{or}\ \frac{\partial u}{\partial x}=-x^{n-2}y\varphi^1\ \ y/x\ \ +nx^{n-1}\varphi\ \ \ y/x\ \ \text{----------(1)}$$

Again differentiating $u=x^n\varphi\ \ y/x\ \ $ partially w.r.t y, we get

$$\frac{\partial u}{\partial y}=x^n\varphi^1\ \ y/x\ \ \ 1/x\ \ +0$$

$$\frac{\partial u}{\partial y}=x^{n-1}\varphi^1\ \ y/x\ \ \text{…….. (2)}$$

Consider

$$x\frac{\partial u}{\partial x}+y\frac{\partial u}{\partial y}=x\ \ -x^{n-2}y\varphi^1\ \ y/x\ \ +nx^{n-1}\varphi\ \ y/x\ \ \ +y\ \ x^{n-1}\varphi^1\ \ y/x$$

$$=-x^{n-1}\ y\phi^1\ \ y/x\ \ +nx^n\phi\ \ y/x\ \ +x^{n-1}\ y\phi^1\ \ y/x$$

$$=nx^n\phi\ \ y/x$$

= nu, Since $u=x^n\varphi\ \ y/x$

Career Avenues

Note: **-** In general, if u is a homogenous function of order x, y, z, t… then

$$x\frac{\partial u}{\partial x}+y\frac{\partial u}{\partial y}+z\frac{\partial u}{\partial z}+t\frac{\partial u}{\partial t}+\ldots\ldots\ldots=nu$$

Examples :-

Verify Euler's theorem for

1) $u=\left(\frac{xy}{x+y}\right)$

2) $u=ax^2+2hxy+by^2$

Solution:- (i) Consider $u=\frac{xy}{x+y}=\frac{x^2\ y/x}{x\ 1+y/x}=x\left(\frac{y/x}{1+y/x}\right)=x\varphi\ {}^{y}\!/_{x}\ =x^1\varphi\ {}^{y}\!/_{x}$

This shows that $u=\frac{xy}{x+y}$ is a homogenous function of degree1 (1=n). Hence, Euler's theorem when applied to u becomes

$x\frac{\partial u}{\partial x}+y\frac{\partial u}{\partial y}=(1)u$ ……(*) (n=1)

we verify the equation (*) by showing actually LHS=RHS

now as $u=\left(\frac{xy}{x+y}\right), \frac{\partial u}{\partial x}=\frac{y^2}{x+y^{\ 2}}$ & $\frac{\partial u}{\partial y}=\frac{x^2}{x+y^{\ 2}}$

$\therefore$ LHS of (*) $=x\frac{\partial u}{\partial x}+y\frac{\partial u}{\partial y}$

$=x\left(\frac{y^2}{x+y^{\ 2}}\right)+y\left(\frac{x^2}{x+y^{\ 2}}\right)=\frac{xy\ \ x+y}{x+y^{\ 2}}=\frac{xy}{x+y}$ =RHS

This verifies the result(*)

(ii) Consider $u=ax^2+2hxy+by^2=x^2\left({}^{a}\!/_{x^2}+2h\ {}^{y}\!/_{x}\ +b\,{}^{y^2}\!/_{x^2}\right)$

$=x^2\varphi\ {}^{y}\!/_{x}$,which means u is a homogenous function of degree 2. Hence, on applying Euler's theorem to u, we get

$x\frac{\partial u}{\partial x}+y\frac{\partial u}{\partial y}=2u$ ……(*)

Now, as $u=ax^2+2hxy+by^2$, we see that $\frac{\partial u}{\partial x}=2\ \ ax+hy$ & $\frac{\partial u}{\partial y}=2\ \ ay+hx$

Consider, LHS of (*) $x\frac{\partial u}{\partial x}+y\frac{\partial u}{\partial y}$

$=x\ \ 2\ \ ax+hy\ \ +y\ \ 2\ \ ay+hx$

$=2ax^2+2hxy+2hxy+2ay^2=2\ \ ax^2+2hxy+by^2$ $=2u$ = RHS

This verifies the result (*)

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Example:-

If $\sin u = \left(\frac{x^2y^2}{x+y}\right)$, show that $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = 3\tan u$

Solution:- Consider $\sin u = \left(\frac{x^2y^2}{x+y}\right) = f$,say We note that $f = \left(\frac{x^2y^2}{x+y}\right) = \frac{x^4\ {y}/{x}^{\,2}}{x\ 1+{y}/{x}} = x^3\varphi\ {y}/{x}$ .Thus, `f` is a homogenous function of degree 3. Applying Euler's theorem to f, we get

$x\frac{\partial f}{\partial x} + y\frac{\partial f}{\partial y} = 3f$ As $f = \sin u$,we see that

$x\frac{\partial}{\partial x}\ \sin u\ + y\frac{\partial}{\partial y}\ \sin u\ = 3\ \sin u$

i.e. $x\left\{\ \cos u\ \frac{\partial u}{\partial x}\right\} + y\left\{\ \cos u\ \frac{\partial u}{\partial y}\right\} = 3\sin u$

or we get $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = 3\tan u$ as desired

2.11 Taylor series :-

For a function $f\ x$

$$f\ x+h\ = f\ x\ + f'\ x\ h + \frac{f''\ x}{2!}h^2 + \frac{f'''\ x}{3!}h^3 + \cdots$$

provided all derivatives of $f\ x$ exist and are continuous between x and $x+h$.

It is very important to note that the Taylor series is not asking for the expression of the function and its derivatives, just the value of the function and its derivatives at a single point.

All the derivatives have to exist and be continuous between x (the point where you are) to the point, $x+h$ where you are wanting to calculate the function at. However, if you want to calculate the function approximately by using the nth order Taylor polynomial, then $1^{st}, 2^{nd}, ..., n^{th}$ derivatives need to exist and be continuous in the closed interval $[x, x+h]$, while the $(n+1)^{th}$ derivative needs to exist and be continuous in the open interval $(x, x+h)$.

Example:-

Take $f\ x\ = \sin\ x$. We all know the value of $\sin\left(\frac{\pi}{2}\right) = 1$. We also know the $f'\ x\ = \cos\ x$ and $\cos\left(\frac{\pi}{2}\right) = 0$.

Similarly $f''\ x\ = -\sin(x)$ and $\sin\left(\frac{\pi}{2}\right) = 1$. In a way, we know the value of $\sin\ x$ and all its derivatives at $x = \frac{\pi}{2}$

. Can you use Taylor series and this information to find the value of $\sin\ 2$?

Solution:-

$x = \frac{\pi}{2}$

$x + h = 2$

$h = 2 - x$

$= 2 - \frac{\pi}{2} = 0.42920$

So

$$f\ x+h\ = f\ x\ + f'\ x\ h + f''\ x\ \frac{h^2}{2!} + f'''\ x\ \frac{h^3}{3!} + f''''(x)\frac{h^4}{4!} + \cdots$$

Career Avenues

$x=\frac{\pi}{2}$

$h=0.42920$

$f\ x\ =\sin\ x\ ,\ f\left(\frac{\pi}{2}\right)=\sin\left(\frac{\pi}{2}\right)=1$

$f'\ x\ =\cos\ x\ ,\ f'\left(\frac{\pi}{2}\right)=0$

$f''\ x\ =-\sin\ x\ \ f''\left(\frac{\pi}{2}\right)=-1$

$f'''\ x\ =-\cos(x)\ ,\ f'''\left(\frac{\pi}{2}\right)=0$

$f''''\ x\ =\sin(x)\ ,\ f''''\left(\frac{\pi}{2}\right)=1$

Hence

$$f\left(\frac{\pi}{2}+h\right)=f\left(\frac{\pi}{2}\right)+f'\left(\frac{\pi}{2}\right)h+f''\left(\frac{\pi}{2}\right)\frac{h^2}{2!}+f'''\left(\frac{\pi}{2}\right)\frac{h^3}{3!}+f''''\left(\frac{\pi}{2}\right)\frac{h^4}{4!}+\cdots$$

$$f\left(\frac{\pi}{2}+0.42920\right)=1+0\ \ 0.42920\ -1\frac{0.42920^{\ 2}}{2!}+0\frac{0.42920^{\ 3}}{3!}+1\frac{0.42920^{\ 4}}{4!}+\cdots$$

$$=1+0-0.092106+0+0.00141393+\cdots\ \ \cong 0.90931$$

2.12.Maclaurin series:-

Maclaurin series is simply a Taylor series for the point $x=0$.

Example:-

Derive the Maclaurin series of $\sin\ x\ =x-\frac{x^3}{3!}+\frac{x^5}{5!}-\frac{x^7}{7!}+\cdots$

Solution:- In the previous example, we wrote the Taylor series for $\sin\ x$ around the point $x=\frac{\pi}{2}$.

$f\ x=\sin\ x\ ,\ f\ 0\ =0$

$f'\ x\ =\cos\ x\ ,f'\ 0\ =1$

$f''\ x\ =-\sin\ x\ ,f''\ 0\ =0$

$f'''\ x\ =-\cos\ x\ ,f'''\ 0\ =-1$

$f''''\ x\ =\sin\ x\ ,f''''\ 0\ =0$

$f'''''\ x\ =\cos\ x\ ,f'''''\ 0\ =1$

Using the Taylor series now,

$$f\ x+h\ =f\ x\ +f'\ x\ h+f''\ x\ \frac{h^2}{2!}+f'''\ x\ \frac{h3}{3!}+f''''\ x\ \frac{h^4}{4}+f'''''\ x\ \frac{h^5}{5}+...$$

$$f\ 0+h\ =f\ 0\ +f'\ 0\ h+f''\ 0\ \frac{h^2}{2!}+f'''\ 0\ \frac{h^3}{3!}+f''''\ 0\ \frac{h^4}{4}+f'''''\ 0\ \frac{h^5}{5}+...$$

$$f\ h\ =f\ 0\ +f'\ 0\ h+f''\ 0\ \frac{h^2}{2!}+f'''\ 0\ \frac{h^3}{3!}+f''''\ 0\ \frac{h^4}{4}+f'''''\ 0\ \frac{h^5}{5}+...$$

$$=0+1\ h\ -0\frac{h^2}{2!}-1\frac{h^3}{3!}+0\frac{h^4}{4}+1\frac{h^5}{5}+...=h-\frac{h^3}{3!}+\frac{h^5}{5!}+...$$

Career Avenues

$So, f\ x = x - \frac{x^3}{3!} + \frac{x^5}{5!}$

$\sin\ x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \ldots$

Example :-

Find the value of $f\ 6$ given that $f\ 4 = 125$, $f'\ 4 = 74$, $f''\ 4 = 30$, $f'''\ 4 = 6$ and all other higher derivatives of $f\ x$ at $x = 4$ are zero.

Solution:-

$$f\ x+h = f\ x + f'\ x\ h + f''\ x\ \frac{h^2}{2!} + f'''\ x\ \frac{h^3}{3!} + \cdots$$

$x = 4$

$h = 6 - 4 = 2$

Since fourth and higher derivatives of $f\ x$ are zero at $x = 4$.

$$f\ 4+2 = f\ 4 + f'\ 4\ 2 + f''\ 4\ \frac{2^2}{2!} + f'''\ 4\ \frac{2^3}{3!}$$

$$f\ 6 = 125 + 74\ 2 + 30\left(\frac{2^2}{2!}\right) + 6\left(\frac{2^3}{3!}\right)$$

$$= 125 + 148 + 60 + 8 = 341$$

2.13 INTEGRAL CALCULUS:-

2.13.1 Introduction to Integration:-

The dictionary definition of *integration* is combining parts so that they work together or form a whole. Mathematically, integration stands for finding the area under a curve from one point to another. It is represented by

$$\int_a^b f(x)dx$$

where the symbol $\int$ is an integral sign, and a and b are the lower and upper limits of integration, respectively, the function f is the integrand of the integral, and x is the variable of integration. Figure 1 represents a graphical demonstration of the concept.

Riemann Sum:-

Let f be defined on the closed interval $[a,b]$, and let Δ be an arbitrary partition of $[a,b]$ such as: $a = x_0 < x_1 < x_2 < \ldots.. < x_{n-1} < x_n = b$, where Δx_i is the length of the ith subinterval (Figure 2).

If c_i is any point in the ith subinterval, then the sum

$$\sum_{i=1}^{n} f\ c_i\ \Delta x_i, x_{i-1} \le c_i \le x_i$$

is called a Riemann sum of the function f for the partition Δ on the interval $[a,b]$. For a given partition Δ, the length of the longest subinterval is called the norm of the partition. It is denoted by $\|\Delta\|$ (the norm of Δ). The following limit is used to define the definite integral.

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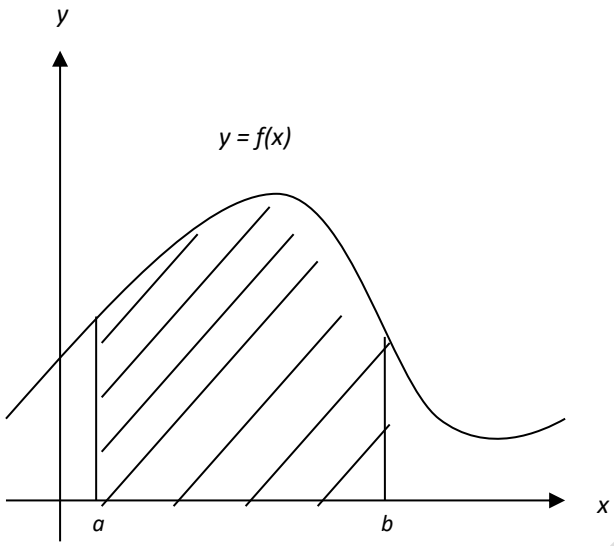


***Figure 1*:-** The definite integral as the area of a region under the curve, $Area = \int_a^b f(x)dx$.

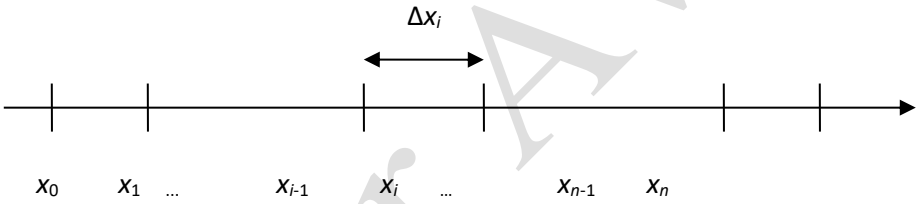


Figure 2 :- Division of interval into n segments.

$$\lim_{|\Delta|\to 0}\sum_{i=1}^{n} f\ c_i\ \Delta x_i = I$$

This limit exists if and only if for any positive number ε, there exists a positive number δ such that for every partition Δ of $[a,b]$ with $\|\Delta\| < \delta$, it follows that

$$\left| I - \sum_{i=1}^{n} f\ c_i\ \Delta x_i \right| < \varepsilon$$

for any choice of c_i in the ith subinterval of Δ.

If the limit of a Riemann sum of f exists, then the function f is said to be integrable over $[a,b]$ and the Riemann sum of f on $[a,b]$ approaches the number I.

$$\lim_{|\Delta|\to 0}\sum_{i=1}^{n} f\ c_i\ \Delta x_i = I$$

where

$$I = \int_a^b f\ x\ dx$$

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Example 1:-

Find the area of the region between the parabola $y = x^2$ and the x-axis on the interval $[0, 4.5]$. Use Riemann's sum with four partitions.

Solution:-

We evaluate the integral for the area as a limit of Riemann sums. We sketch the region (Figure 3), and partition $[0, 4.5]$ into four subintervals of length

$$\Delta x = \frac{4.5 - 0}{4} = 1.125.$$

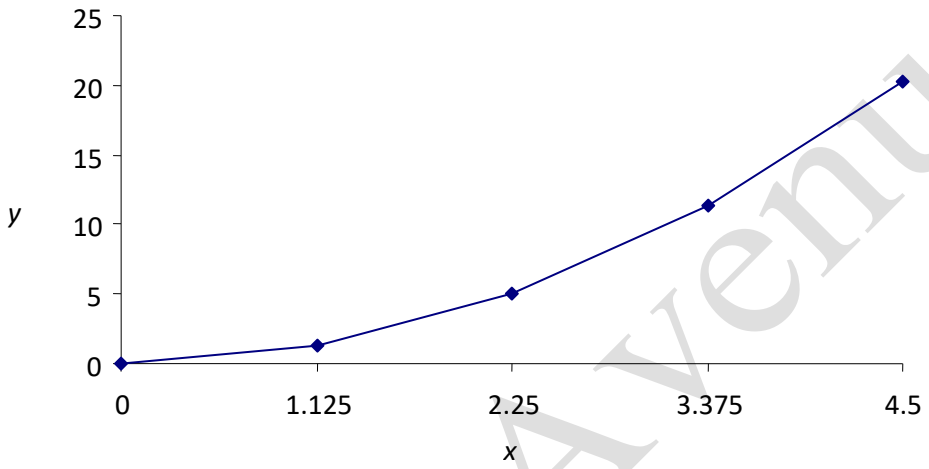


Figure 3 Graph of the function $y = x^2$.

The points of partition are

$$x_0 = 0,\ x_1 = 1.125,\ x_2 = 2.25,\ x_3 = 3.375,\ x_4 = 4.5$$

Let's choose c_i's to be right hand endpoint of its subinterval. Thus,

$$c_1 = x_1 = 1.125,\ c_2 = x_2 = 2.25,\ c_3 = x_3 = 3.375,\ c_4 = x_4 = 4.5$$

The rectangles defined by these choices have the following areas:

$$f(c_1)\Delta x = f(1.125) \times (1.125) = (1.125)^2 (1.125) = 1.4238$$

$$f(c_2)\Delta x = f(2.25) \times (1.125) = (2.25)^2 (1.125) = 5.6953$$

$$f(c_3)\Delta x = f(3.375) \times (1.125) = (3.375)^2 (1.125) = 12.814$$

$$f(c_4)\Delta x = f(4.5) \times (1.125) = (4.5)^2 (1.125) = 22.781$$

The sum of the areas then is

$$\int_0^{4.5} x^2 dx \approx \sum_{i=1}^{4} f(c_i)\Delta x,$$

$$= 1.4238 + 5.6953 + 12.814 + 22.781$$

$$= 42.715$$

How does this compare with the exact value of the integral $\int_0^{4.5} x^2 dx$?

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Example 2:-

Find the exact area of the region between the parabola $y = x^2$ and the $x-$axis on the interval $[0, b]$. Use Riemann's sum.

Solution:-

Note that in Example 1 for $y = x^2$ that

$$f\ c_i\ \Delta x = i^2\ \Delta x^{\ 3}$$

As $f\ c_i\ = i.\Delta x^{\ 2}$

Thus, the sum of these areas, if the interval is divided into n equal segments is

$$S_n = \sum_{i=1}^{n} f(c_i)\Delta x$$

$$= \sum_{i=1}^{n} i^2(\Delta x)^3$$

$$= (\Delta x)^3 \sum_{i=1}^{n} i^2$$

Since

$$\Delta x = \frac{b}{n}, \text{ and}$$

$$\sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6}$$

then

$$S_n = \frac{b^3}{n^3}\frac{n(n+1)(2n+1)}{6}$$

$$= \frac{b^3}{6}\frac{2n^2+n+2n+1}{n^2}$$

$$= \frac{b^3}{6}\left(2+\frac{3}{n}+\frac{1}{n^2}\right)$$

The definition of a definite integral can now be used

$$\int_a^b f\ x\ dx = \lim_{|\Delta x|\to 0} \sum_{i=1}^{n} f\ c_i\ \Delta x$$

To find the area under the parabola from $x = 0$ to $x = b$, we have

$$\int_0^b x^2 dx = \lim_{|\Delta|\to 0} \sum_{i=1}^{n} f(c_i)\Delta x$$

$$= \lim_{n\to\infty} S_n$$

$$= \lim_{n\to\infty} \frac{b^3}{6}\left(2+\frac{3}{n}+\frac{1}{n^2}\right)$$

$$= \frac{b^3}{6}\ 2+0+0$$

$$= \frac{b^3}{3}$$

For the value of $b = 4.5$ as given in Example 1,

Career Avenues

$$\int_0^{4.5} x^2 dx = \frac{4.5^3}{3}$$

= 30.375

The Mean Value Theorem for Integrals:-

The area of a region under a curve is usually greater than the area of an inscribed rectangle and less than the area of a circumscribed rectangle. The mean value theorem for integrals states that somewhere between these two rectangles, there exists a rectangle whose area is exactly equal to the area of the region under the curve, as shown in Figure 4. Another variation states that if a function f is continuous between a and b, then there is at least one point in $[a,b]$ where the function equals the average value of the function f over $[a,b]$.

***Theorem*:-** If the function f is continuous on the closed interval $[a,b]$, then there exists a number c in $[a,b]$ such that:

$$f(c) = \frac{1}{b-a}\int_a^b f(x)dx$$

Example 3:-

Graph the function $f(x) = (x-1)^2$, and find its average value over the interval $[0,3]$. At what point in the given interval does the function assume its average value?

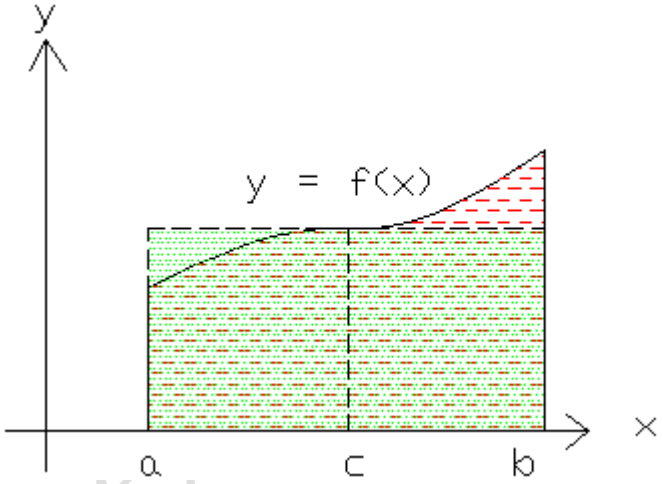


Figure 4 Mean value rectangle.

Solution:-

$$\text{Average } f = \frac{1}{b-a}\int_a^b f\ x\ dx$$

$$= \frac{1}{3-0}\int_0^3 x-1\ ^2\ dx$$

$$= \frac{1}{3}\int_0^3 x^2 - 2x + 1\ dx$$

$$= \frac{1}{3}\left[\left(\frac{1}{3}\times 27 - 9 + 3\right) - 0\right]$$

$$= 1$$

Career Avenues

The average value of the function f over the interval $[0,3]$ is 1. Thus, the function assumes its average value at

$$f(c)=1$$
$$(c-1)^2=1$$
$$c=0,\ 2$$

The connection between integrals and area can be exploited in two ways. When a formula for the area of the region between the x-axis and the graph of a continuous function is known, it can be used to evaluate the integral of the function. However, if the area of region is not known, the integral of the function can be used to define and calculate the area. Table 1 lists a number of standard indefinite integral forms.

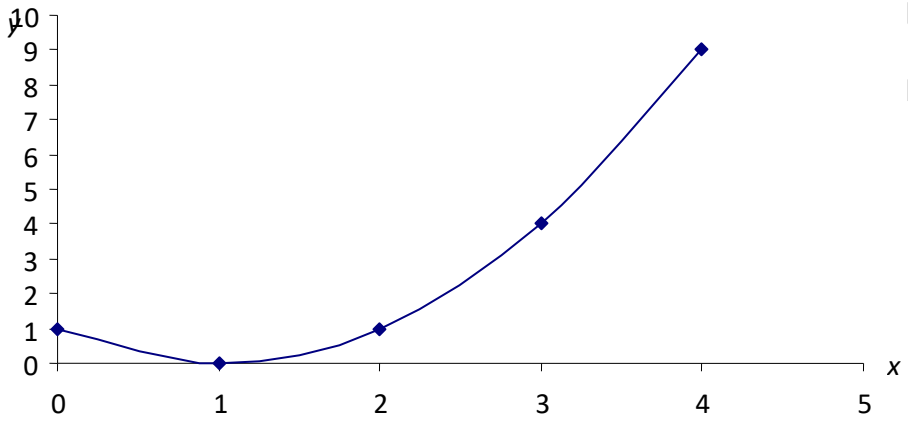


Figure 5:- The function $f(x)=(x-1)^2$.

Example 4:-

Find the area of the region between the circle $x^2+y^2=1$ and the x-axis on the interval $[0,1]$ (the shaded region) in two different ways.

Solution:-

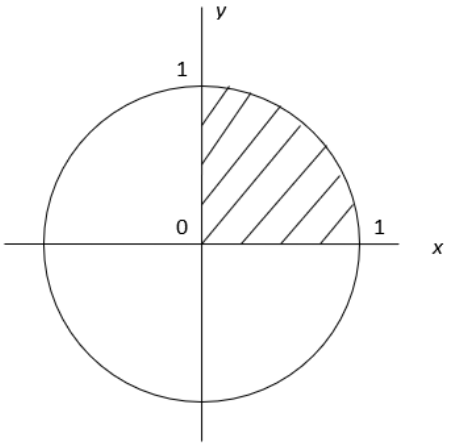


Figure 6:- Graph of the function $x^2+y^2=1$.

The first and easy way to solve this problem is by recognizing that it is a quarter circle. Hence the area of the shaded area is

$$A = \frac{1}{4}\pi r^2$$
$$= \frac{1}{4}\pi(1)^2 \quad = \frac{\pi}{4}$$

The second way is to use the integrals and the trigonometric functions. First, let's simplify the function $x^2 + y^2 = 1$.

$$x^2 + y^2 = 1$$
$$y^2 = 1 - x^2$$
$$y = \sqrt{1-x^2}$$

The area of the shaded region is the equal to

$$A = \int_0^1 \sqrt{1-x^2}\,dx$$

We set $x = \sin\theta$, $dx = \cos\theta\, d\theta$

$$A = \int_0^1 \sqrt{1-x^2}\,dx$$
$$= \int_0^{\pi/2} \sqrt{1-\sin^2\theta}\ \cos\theta\ d\theta$$
$$= \int_0^{\pi/2} \sqrt{\cos^2\theta}\ \cos\theta\ d\theta$$
$$= \int_0^{\pi/2} \cos^2\theta\ d\theta$$

By using the following formula

$$\cos^2\theta = \frac{1+\cos 2\theta}{2},$$

we have

$$A = \int_0^{\pi/2} \frac{1+\cos 2\theta}{2} d\theta$$
$$= \int_0^{\pi/2} \left(\frac{1}{2} + \frac{\cos 2\theta}{2}\right) d\theta$$
$$= \left[\frac{1}{2}\theta + \frac{\sin 2\theta}{4}\right]_0^{\pi/2}$$
$$= \left(\frac{\pi}{4} + 0\right) - \ 0 + 0$$
$$= \frac{\pi}{4}$$

Career Avenues

You can use the brief table of integrals given in Table 1.

Table 1:- ***A brief table of integrals:-***

$\int dx = x + C$	$\int \sin x dx = -\cos x + C$
$\int a\, f(x)\, dx = a\int f(x)\, dx + C$	$\int \cos x dx = \sin x + C$
$\int [u\ x \pm v\ x\ ] dx = \int u\ x\ dx \pm \int v\ x\ dx + C$	$\int \tan x dx = -\ln\lvert\cos x\rvert + C = \ln\lvert\sec x\rvert + C$
$\int x^n\, dx = \frac{x^{n+1}}{n+1} + C$	$\int \sec\ ax\ dx = \frac{1}{a}\ln\lvert\sec\ ax\ + \tan\ ax\ \rvert + C$
$\int u\, dv = uv - \int v\, du + C$	$\int \cot x dx = -\ln\lvert\csc x\rvert + C = \ln\lvert\sin x\rvert + C$
$\int \frac{dx}{ax+b} = \frac{1}{a}\ln\lvert ax+b\rvert + C$	$\int \sec^2 ax dx = \frac{1}{a}\tan(ax) + C$
$\int a^x\, dx = \frac{a^x}{\ln a} + C$	$\int \sec\ x\ \tan\ x\ dx = \sec\ x\ + C$
$\int e^{ax}\, dx = \frac{e^{ax}}{a} + C$	$\int \csc\ x\ \cot\ x\ dx = -\csc\ x\ + C$

2.13.2 ***Integrating discrete functions:-***

Two methods of integrating discrete functions are shown below using an example.

Example:-

The upward velocity of a rocket is given as a function of time in Table 1.

Table 1 Velocity as a function of time.

t (s)	$v(t)$ (m/s)
0	0
10	227.04
15	362.78
20	517.35
22.5	602.97
30	901.67

Determine the distance, s, covered by the rocket from $t = 11$ to $t = 16$ using the velocity data provided and use any applicable numerical technique.

Solution:-

Career Avenues

Method 1:- Average Velocity Method

The velocity of the rocket is not provided at $t = 11$ and $t = 16$, so we will have to use an interval that includes $11, 16$ to find the average velocity of the rocket within that range. In this case, the interval $10, 20$ will suffice.

$v(10) = 227.04$

$v(15) = 362.78$

$v(20) = 517.35$

$$\text{Average Velocity} = \frac{v(10)+v(15)+v(20)}{3} = \frac{227.04+362.78+517.35}{3} = 369.06 \text{ m/s}$$

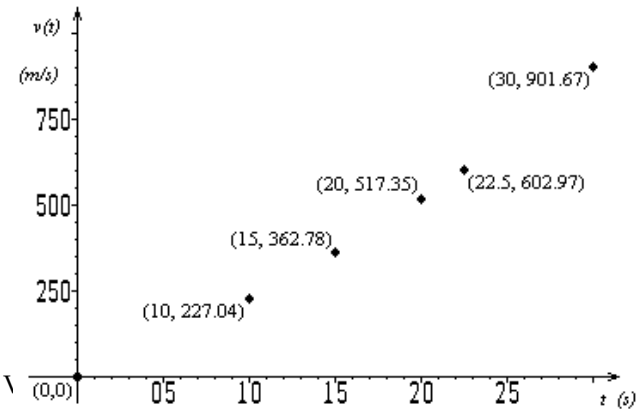


Figure **:-** Velocity V

Using

$s = \bar{v}\Delta t,$

we get

$s = (369.06)(16-11) = 1845.3$ m

Method 2:- Trapezoidal Rule

If we were finding the distance traveled between times in the data table, we would simply find the area of the trapezoids with the corner points given as the velocity and time data points. For example

$$\int_{10}^{20} v(t)dt = \int_{10}^{15} v(t)dt + \int_{15}^{20} v(t)dt$$

and applying the trapezoidal rule over each of the above integrals gives

$$\int_{10}^{20} v(t)dt \approx \frac{15-10}{2}[v(10)+v(15)] + \frac{20-15}{2}[v(15)+v(20)]$$

The values of $v(10)$, $v(15)$ and $v(20)$ are given in Table 1.

However, we are interested in finding

$$\int_{11}^{16} v(t)dt = \int_{11}^{15} v(t)dt + \int_{15}^{16} v(t)dt$$

and applying the trapezoidal rule over each of the above integrals gives

$$\int_{11}^{16} v(t)dt \approx \frac{15-11}{2}[v(11)+v(15)] + \frac{16-15}{2}[v(15)+v(16)]$$

$$= \frac{15-11}{2}(v(11)+362.78) + \frac{16-15}{2}(362.78+v(16))$$

How do we find $v(11)$ and $v(16)$? We use linear interpolation. To find $v(11)$,

$v(t) = 227.04 + 27.148\ t-10\ ,\ 10 \le t \le 15$

Career Avenues

$$v(11) = 227.04 + 27.148\ 11-10$$
$$= 254.19 \text{ m/s}$$

and to find $v(16)$

$$v(t) = 362.78 + 30.913\ t-15\ ,\ 15 \le t \le 20$$
$$v(16) = 362.78 + 30.913\ 16-15$$
$$= 393.69 \text{ m/s}$$

Then

$$\int_{11}^{16} v(t)dt \approx \frac{15-11}{2}(v(11)+362.78) + \frac{16-15}{2}(362.78+v(16))$$
$$= \frac{15-11}{2}(254.19+362.78) + \frac{16-15}{2}(362.78+393.69)$$
$$= 1612.2 \text{ m}$$

Trapezoidal Rule for Discrete Functions with Unequal Segments:-

For a general case of a function given at n data points $x_1, f\ x_1\ ,\ x_2, f\ x_2\ ,\ x_3, f\ x_3\ , \ldots..,\ x_n, f\ x_n\ $, where, $x_1, .x_2, ..., x_n$ are in an ascending order, the approximate value of the integral $\int_{x_1}^{x_n} f\ x\ dx$ is given by

$$\int_{x_1}^{x_n} f\ x\ dx = \int_{x_1}^{x_2} f\ x\ dx + \int_{x_2}^{x_3} f\ x\ dx + ... + \int_{x_{n-1}}^{x_n} f\ x\ dx$$
$$\approx\ x_2 - x_1\ \frac{f\ x_1\ + f\ x_2}{2} + x_3 - x_2\ \frac{f\ x_2\ + f\ x_3}{2} + ...$$
$$.....+\ x_n - x_{n-1}\ \frac{f\ x_{n-1}\ + f\ x_n}{2}$$

This approach uses the trapezoidal rule in the intervals $x_1, x_2\ ,\ x_2, x_3\ , \ldots..,\ x_{n-1}, x_n$ and then adds the obtained values.

2.14 Definite Integral:-

Let f be a continuous function defined on a closed interval [a,b]. For a positive number n, let $\Delta x = \frac{b-a}{n}$ and partition [a,b] into n subintervals of length Δx .

Let $x_0(=a), x_1, x_2, ..., x_n(=b)$ be the endpoints of the subintervals and choose a point x_i* from each of the subintervals $[x_{i-1}, x_i]$, $i = 1, 2, ..., n$. Finally, form the Riemann sum

$$f(x_1^*)\Delta x + f(x_2^*)\Delta x + ... + f(x_n^*)\Delta x = \sum_{i=1}^{n} f(x_i^*)\Delta x\ .$$

Then the definite integral of f from a to b is defined by $\int_a^b f(x)dx = \lim_{n\to\infty} \sum_{i=1}^{n} f(x_i^*)\Delta x$

***Area under a curve*:-** If f is nonnegative on the interval [a,b], then the definite integral gives the area under the curve $y = f(x)$ from $x = a$ to $x = b$.

$$A = \int_a^b f(x)dx$$

Career Avenues

Fundamental Theorem of Calculus, Part 1:- If f is continuous on $[a,b]$ and $g(x)=\int_a^x f(t)dt$, then $g'(x)=f(x)$

Anti-derivative:- A function F is an antiderivative for a function f if $F'(x)=f(x)$.

Fundamental Theorem of Calculus, Part 2:- If f is continuous on $[a,b]$ and F is any antiderivative for f, then $\int_a^b f(x)dx=F(b)-F(a)$.

The Fundamental Theorem, Part 2, is used to evaluate definite integrals

2.15 *Indefinite integral:-*

An antiderivative F is also called an indefinite integral for f. Since any two antiderivatives for f differ by a constant, we write

$$\int f(x)dx=F(x)+C$$

Table of Indefinite Integrals:-

$$\int cf(x)dx=c\int f(x)dx \qquad \int [f(x)+g(x)]dx=\int f(x)dx+\int g(x)dx$$

$$\int kdx=kx+C \qquad \int x^n dx=\frac{x^{n+1}}{n+1}+C \quad (n\neq -1)$$

$$\int \sin x dx=-\cos x+C \qquad \int \cos x dx=\sin x+C$$

$$\int \sec^2 x dx=\tan x+C \qquad \int \csc^2 x dx=-\cot x+C$$

$$\int \sec x\tan x dx=\sec x+C \qquad \int \csc x\cot x dx=-\csc x+C$$

Substitution Rule:- If an integral has the form $\int f(g(x))g'(x)dx$, the substitution $u=g(x)$ and $du=g'(x)dx$ reduces the integral to the form $\int f(u)du$.

Applications of the definite integral:-

1. ***Area between two curves*:-** If f and g are continuous functions on the closed interval $[a,b]$ and $f(x)\geq g(x)$ for all x in $[a,b]$, then the area bounded by the curves $y=f(x)$ and $y=g(x)$ and the vertical lines $x=a$ and $x=b$ is given by the formula

$$A=\int_a^b f(x)-g(x)\,dx.$$

2. ***Volume of a solid by slicing:-*** If a solid S lies between the lines $x=a$ and $x=b$, let $A(x)$ represent the area of a vertical cross section of S at the point x. Then the volume of S is given by the integral

$$V=\int_a^b A(x)dx \qquad (1)$$

If the solid lies between the lines $y=c$ and $y=d$ and the area of a cross section perpendicular to the y-axis is given by $A(y)$, then the volume of the solid is given by

$$V=\int_c^d A(y)dy \qquad (2)$$

3. ***Volume of a solid using the disk method:-*** A cross section of a solid of revolution obtained by revolving a region in the xy-plane about a horizontal or vertical axis has the shape of either a disk or a washer. If the cross section is a disk (this happens when the axis of rotation forms one boundary of the region), find the radius of the disk (in terms of x or y) to get the cross sectional area $A=\pi(\text{radius})^2$. Then use either formula (1) or formula (2) to find the volume. If the cross section is a washer, find the inner radius and outer radius of the washer to get the area

Career Avenues

$A = \pi(\text{outer radius})^2 - \pi(\text{inner radius})^2$ in terms of *x* or *y*. Again formula (1) or formula (2) may be used to obtain the volume.

4. ***Volume of a solid by cylindrical shells:-*** The volume of a solid of revolution may also be found by the method of cylindrical shells, where a portion of the generating region parallel to the axis of rotation may be thought of as generating a hollow shell with volume given by the generic formula

$$2\pi(\text{radius})(\text{height})(\text{thickness}).$$

If the region under the curve $y = f(x)$ from $x = a \geq 0$ to $x = b$ is revolved around the *y*-axis, the volume generated is given by the formula $V = \int_a^b 2\pi x f(x)dx$. If the region between the curve $x = g(y)$ and the *y*-axis from $y = c \geq 0$ to $y = d$ is revolved around the x-axis, the volume generated is given by the formula

$V = \int_c^d 2\pi y g(y)dy$

Disk Method:- $\int_a^b \pi f(x)^2 dx$ ***Shell Method:-*** $\int_a^b 2\pi x f(x)dx$

5. ***Work:-*** If a variable force $f(x)$ moves an object along the *x*-axis from $x = a$ to $x = b$, the work done is given by the integral $W = \int_a^b f(x)dx$.

Exponential and logarithmic functions:- A function *f* is one-to-one (1-1) on an interval if $f(x_1) = f(x_2) \Rightarrow x_1 = x_2$ for all values x_1 and x_2 in the interval. The graph of a 1-1 function must pass the Horizontal Line Test: No horizontal line can intersect the graph in more than one point. If a function *f* is 1-1, it has an inverse function defined by

$$f^{-1}(y) = x \text{ if and only if } f(x) = y$$

This means that if *f* takes *x* to *y*, then f^{-1} takes *y* to *x*. The domain of *f* is the range of f^{-1}, and the range of *f* is the domain of f^{-1}. A function and its inverse have the following cancellation relations:

$$f^{-1}(f(x)) = x \quad \text{and} \quad f(f^{-1}(y)) = y$$

Exponential functions:- If *n* is a positive integer and *a* is a positive real number, then we know that $a^n = \underbrace{a \cdot a \cdot \ldots \cdot a}_{n \text{ times}}$, $a^{-n} = \frac{1}{a^n}$, and $a^0 = 1$. For a rational number $r = \frac{p}{q}$ with $q > 0$, $a^r = a^{p/q} = \sqrt[q]{a^p}$. If *x* is a real number, define $a^x = \lim_{x \to r} a^r$ for rational numbers *r*.

Properties of exponential functions: **-** The exponential function $f(x) = a^x$ is a continuous function with domain $(-\infty, \infty)$ and range $(0, \infty)$. If *x* and *y* are real numbers,

$$a^x \cdot a^y = a^{x+y} \qquad \frac{a^x}{a^y} = a^{x-y} \qquad (a^x)^y = a^{xy} \qquad (ab)^x = a^x b^x$$

The number e :- The number *e* is defined to be that unique positive number with the property that the slope of the tangent line to the graph of $y = e^x$ is 1 when $x = 0$. The value of *e* is approximately 2.718. The function $f(x) = e^x$ is called the natural exponential function. This function is differentiable with

Career Avenues

$$\frac{d}{dx}(e^x) = e^x \qquad \text{and} \qquad \text{(if u is a differentiable function of x)} \quad \frac{d}{dx}(e^u) = e^u \frac{du}{dx}$$

The associated indefinite integral is $\int e^x dx = e^x + C$

Logarithm functions: - The general exponential function $f(x) = a^x$ is increasing if $a > 1$ and decreasing if $0 < a < 1$. This means the function is 1-1 and has an inverse. The inverse of the exponential function $f(x) = a^x$ is called the logarithm function to the base *a*, which is defined by

$y = \log_a x$ if and only if $x = a^y$

Since the *range* of the exponential function is $(0, \infty)$, the *domain* of the logarithm function is $(0, \infty)$, which means that the logarithm function is defined only for *positive* values of *x*.

Cancellation properties: -

$a^{\log_a y} = y$ and $\log_a a^x = x$

Properties of logarithm functions: - If $a > 1$, the logarithm function $y = \log_a x$ is a continuous, increasing function with domain $(0, \infty)$ and range $(-\infty, \infty)$.

***Natural logarithm*:-** The logarithm function to the base *e* is called the natural logarithm function: $\ln x = \log_e x$. The natural logarithm function is related to the general logarithm function by the formula

$$\log_a x = \frac{\ln x}{\ln a}.$$

If *x* and *y* are positive numbers and *r* is any real number, then

$$\ln(xy) = \ln x + \ln y \qquad \ln\left(\frac{x}{y}\right) = \ln x - \ln y \qquad \ln x^r = r \ln x$$

The natural logarithm function and the natural exponential functions are inverses of each other, so $\ln(e^x) = x$ for all *x* and $e^{\ln x} = x$ for $x > 0$. In particular, $\ln e = 1$.

The natural logarithm function is differentiable with $\frac{d}{dx}(\ln x) = \frac{1}{x}$ and $\frac{d}{dx}(\ln u) = \frac{1}{u}\frac{du}{dx}$ if *u* is a differentiable function of *x*.

The corresponding integration formula is $\int \frac{1}{x} dx = \ln|x| + C$.

This formula leads to four other useful integrals:

$$\int \tan x dx = \ln|\sec x| + C \qquad \int \cot x dx = \ln|\sin x| + C$$

$$\int \sec x dx = \ln|\sec x + \tan x| + C \qquad \int \csc x dx = \ln|\csc x - \cot x| + C$$

Career Avenues

Inverse trigonometric functions: **-** If the sine function is restricted so that its domain is $\left[-\frac{\pi}{2},\frac{\pi}{2}\right]$, the resulting function is 1-1 and has an inverse. The remaining trigonometric functions also have inverses when their domains are restricted. The definitions for the three inverse trigonometric functions most frequently used are given below:

$$y=\sin^{-1}x \Leftrightarrow \sin x=y \text{ and } -\frac{\pi}{2}\le x\le\frac{\pi}{2} \qquad y=\tan^{-1}x \Leftrightarrow \tan x=y \text{ and } -\frac{\pi}{2}<x<\frac{\pi}{2}$$

$$y=\sec^{-1}x \Leftrightarrow \sec x=y \text{ and } 0\le x<\frac{\pi}{2} \text{ or } \pi\le x<\frac{3\pi}{2}$$

Each inverse trigonometric function is differentiable, with the following derivatives:-

$$\frac{d}{dx}(\sin^{-1}x)=\frac{1}{\sqrt{1-x^2}} \qquad \frac{d}{dx}(\tan^{-1}x)=\frac{1}{1+x^2} \qquad \frac{d}{dx}(\sec^{-1}x)=\frac{1}{x\sqrt{x^2-1}}$$

These can be used to derive the following useful indefinite integral formulas:

$$\int\frac{1}{\sqrt{a^2-x^2}}\,dx=\sin^{-1}\left(\frac{x}{a}\right)+C \qquad \int\frac{1}{x^2+a^2}\,dx=\frac{1}{a}\tan^{-1}\left(\frac{x}{a}\right)+C$$

Hyperbolic functions:- The hyperbolic functions are special combinations of the exponential functions e^x and e^{-x} that occur frequently in applications.

$$\sinh x=\frac{e^x-e^{-x}}{2} \qquad \cosh x=\frac{e^x+e^{-x}}{2} \qquad \tanh x=\frac{\sinh x}{\cosh x}=\frac{e^x-e^{-x}}{e^x+e^{-x}}$$

$$\operatorname{csch} x=\frac{1}{\sinh x} \quad \operatorname{sech} x=\frac{1}{\cosh x} \quad \coth x=\frac{1}{\tanh x}$$

The hyperbolic functions satisfy identities, differentiation formulas, and integration formulas that are similar to those satisfied by the trigonometric functions.

Integration techniques:-

Powers and products of trigonometric functions:- For the most part, the idea behind this integration technique is to use trigonometric identities to reduce the integrand to a power of a trigonometric function times the derivative of the function. The integral can then be evaluated by u-substitution. One exception is odd powers of the secant function, which must be integrated by parts.

Substitutions involving :- $\sqrt{a^2-x^2}$, $\sqrt{x^2+a^2}$, or $\sqrt{x^2-a^2}$:

1. If the integrand involves $\sqrt{a^2-x^2}$, use the substitution $x=a\sin\theta,\ dx=a\cos\theta$.
2. If the integrand involves $\sqrt{x^2+a^2}$, use the substitution $x=a\tan\theta,\ dx=a\sec^2\theta$.
3. If the integrand involves $\sqrt{x^2-a^2}$, use the substitution $x=a\sec\theta,\ dx=a\sec\theta\tan\theta$.

Partial Fractions:- A rational function is the quotient of two polynomials. The method of partial fractions can be used to reduce a rational function to a sum of fractions (called the *partial fraction decomposition*) with denominators that involve linear (first degree) or quadratic (second degree) polynomials. To apply the method of partial fractions to integrate a rational function, follow these steps:

1. If the degree of the numerator is greater than or equal to the degree of the denominator, perform a long division to produce a fraction of the form $\frac{P(x)}{Q(x)}$, where $\deg(P(x))<\deg(Q(x))$.

Career Avenues

2. Factor the denominator *Q(x)* and set up the partial fraction decomposition using these rules:
   (a)If *Q(x)* contains distinct, linear factors a_1x+b_1, a_2x+b_2,..., a_kx+b_k, the partial fraction decomposition contains the terms $\frac{A_1}{a_1x+b_1}+\frac{A_2}{a_2x+b_2}+...+\frac{A_k}{a_kx+b_k}$.
   (b)If *Q(x)* contains a repeated linear factor $(ax+b)^r$, then the partial fraction decomposition contains the terms $\frac{A_1}{ax+b}+\frac{A_2}{(ax+b)^2}+...+\frac{A_r}{(ax+b)^r}$.
   (c)If *Q(x)* contains a quadratic term ax^2+bx+c, then the partial fraction decomposition contains the term $\frac{Ax+B}{ax^2+bx+c}$.
   (d)If *Q(x)* contains a repeated quadratic term $(ax^2+bx+c)^r$, then the partial fraction decomposition contains the terms $\frac{A_1x+B_1}{ax^2+bx+c}+\frac{A_2x+B_2}{(ax^2+bx+c)^2}+...+\frac{A_rx+B_r}{(ax^2+bx+c)^r}$
3. In the equation with the rational function $\frac{P(x)}{Q(x)}$ equal to the partial fraction decomposition, multiply *both sides* by the denominator *Q(x)*.
4. Group like powers of *x* in the resulting equation and equate coefficients on the two sides. Then solve these equations for the unknowns *A*, *B*, … from the partial fraction decomposition.
5. Integrate the partial fraction decomposition of the rational function.

Approximate Integration:- Certain integrals cannot be evaluated by any known techniques. In this case, the value of a definite integral can be approximated by various numerical methods. These involve partitioning the interval of integration into subintervals and using a formula such as the Midpoint Rule, the Trapezoidal Rule, or Simpson's Rule to estimate the value of the integral.

Improper Integrals:- The definition of a definite integral requires that the function *f* have no infinite discontinuities and that the interval [*a*,*b*] be of finite length. If either of these conditions fails, the integral is improper. Two types of improper integrals (infinite interval and infinite discontinuity) can be defined, provided the given limits exist:

1. $\int_a^{\infty} f(x)\,dx=\lim_{t\to\infty}\int_a^t f(x)\,dx$ $\int_{-\infty}^b f(x)\,dx=\lim_{t\to-\infty}\int_t^b f(x)\,dx$
   $\int_{-\infty}^{\infty} f(x)\,dx=\int_{-\infty}^a f(x)\,dx+\int_a^{\infty} f(x)\,dx$, where *a* is any real number
2. If *f* is continuous on [*a*,*b*) and discontinuous at *b*, $\int_a^b f(x)\,dx=\lim_{t\to b-}\int_a^t f(x)\,dx$. If *f* is continuous on (*a*,*b*] and discontinuous at *a*, $\int_a^b f(x)\,dx=\lim_{t\to a+}\int_t^b f(x)\,dx$ If *f* has an infinite discontinuity at *c*, where $a<c<b$, then $\int_a^b f(x)\,dx=\int_a^c f(x)\,dx+\int_c^b f(x)\,dx$.

Example:-
Give some examples of improper integrals
Solution:-
The integral

$$I=\int_0^2 \frac{x}{\sqrt{4-x^2}}\,dx$$

is improper because the integrand becomes infinite at $x=2$.

Career Avenues

The integral

$$I = \int_0^2 \frac{x}{\sqrt{1-x}}\,dx$$

is improper because the integrand becomes infinite at $x = 1$.

The integral

$$I = \int_0^\infty e^{-t}t\,dt$$

is improper because the interval of integration has an infinite bound.

The integral

$$I = \int_0^\infty \frac{e^{-t}}{\sqrt{1-t}}\,dt$$

is improper because the interval of integration has an infinite bound and the integrand is infinite at $t = 1$.

More Applications of Definite Integrals:-

Arc length:- If f is continuous on $[a,b]$, the length of the curve $y = f(x),\ a \le x \le b$ is given by the formula

$$L = \int_a^b \sqrt{1+[f'(x)]^2}\,dx = \int_a^b \sqrt{1+\left(\frac{dy}{dx}\right)^2}\,dx\,.$$

If g is continuous on $[c,d]$, the length of the curve $x = g(y),\ c \le y \le d$ is given by the formula

$$L = \int_c^d \sqrt{1+[g'(y)]^2}\,dy = \int_c^d \sqrt{1+\left(\frac{dx}{dy}\right)^2}\,dy\,.$$

Surface Area:- If f is positive and has a continuous derivative on $[a,b]$, the area of the surface obtained by revolving the curve $y = f(x),\ a \le x \le b$ about the x-axis is given by the integral

$$S = \int_a^b 2\pi f(x)\sqrt{1+[f'(x)]^2}\,dx = \int_a^b 2\pi y\sqrt{1+\left(\frac{dy}{dx}\right)^2}\,dx\,.$$

If the curve is given in the form $x = g(y),\ c \le y \le d$, then the formula is

$$S = \int_c^d 2\pi y\sqrt{1+\left(\frac{dx}{dy}\right)^2}\,dy\,.$$

If the curve is revolved around the y-axis, the surface area is given by

$$S = \int_a^b 2\pi x\sqrt{1+\left(\frac{dy}{dx}\right)^2}\,dx \qquad \text{or} \qquad S = \int_c^d 2\pi x\sqrt{1+\left(\frac{dx}{dy}\right)^2}\,dy\,.$$

Career Avenues

SET- 4

Limit, Continuity and Differentiability

1. The value of $\lim_{x\to 1}\frac{(x^2-1)}{x-1}$ *is*

a) ∞
b) 0
c) 2
d) 1

2. The value of $\lim_{x\to\infty}\frac{\sin x}{x}$ *is*

a) ∞
b) 0
c) 2
d) 1

3. The value of $\lim_{x\to\infty}\frac{sin^2 x}{x}$ *is*

a) ∞
b) 0
c) 1
d) -1

4. The value of $\lim_{x\to\infty}\frac{x^3-\cos x}{x^2+sin^2 x}$ *is* .

a)∞
b) 0
c) 1
d) 2

5. The value of $\lim_{x\to 0}\frac{e^x+e^{-x}-2}{x^2}$ *is*

a)∞
b) 0
c) 1
d) 2

6. The function $f\ x$ is defined as:

$f\ x = x^2$, $0\le x\le 1/2$

$= x$, $1/2\le x\le 1$

then the value of $\lim_{x\to 1/2} f\ x$ *is*

a) 0
b) 1
c) 2
d) does not exist

Career Avenues

7. *The value of* $\lim_{x \to \pi/2} f\ x$ *is* .....

where $f\ x = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \ldots.$

a) $\pi / 2$
b) 1
c) $\frac{\pi}{3}$
d) $3\pi / 2$

8. Here it is given that

$f\ x = x$ *for* $0 \le x \le 1/2$

$= 1$ *for* $x = \frac{1}{2}$

$= 1 - x$ *for* $1/2 \le x \le 1$

Then $f\ x$ *is*

a) *Continuous at* $x = ½$
b) *Not defined at* $x = ½$
c) *Discontinuous at* $x = ½$
d) *Continuous for all* $x, 0 \le x < 1$

9. For $f\ x = x^2 - 1 \quad 1 < x < \infty$

$= x\ -1 \quad -\infty < x \le 1$

at the point $x = 1$ *is*

a) Continuous and differentiable
b) Continuous and not differentiable
c) Discontinuous and monotonically increasing
d) Discontinuous and monotonically decreasing

10. *The function* $f\ x = |x|$ *is*

P. $f\ x$ *is continuos for all real values of* x

Q. $f\ x$ *is differentiable for all real values of* x

Which of the following is true?
a) **P** is True and **Q** is False
b) **P** is False and **Q** is True
c) Both **P and Q** are True
d) Both **P and Q** are False

Career Avenues

Solutions:

1. Solu: c)

$$\lim_{x\to 1}\frac{x^2-1}{x-1}=\lim_{x\to 1}\frac{x+1\quad x-1}{x-1}$$
$$=\lim_{x\to 1}\ x+1$$
$$=2$$

2. Ans: b)

Here sin x is bounded in range $-1\le x\le 1$ for all values of x. Here numerator is finite and denominator is infinite.

Thus $\lim_{x\to\infty}\frac{\sin x}{x}=0$

3. Ans: b)

$$\lim_{x\to 0}\frac{\sin^2 x}{x}=\lim_{x\to 0}\left(\frac{\sin x}{x}\right)^2 x$$
$$=\ 1^{\ 2}\ \ 0\ =0,\qquad \textit{applying L'Hospital rule}$$

4. Ans: b)

$$\lim_{x\to\infty} f\ \ x\ =\lim_{x\to\infty}\left[\frac{x^3-\cos x}{x^2+\sin^2 x}\right]$$
$$=\lim_{x\to\infty} x\left[\frac{1-\frac{\cos x}{x^3}}{1+\frac{\sin^2 x}{x^2}}\right]$$
$$\lim_{x\to\infty} f\ \ x\ =\lim_{x\to\infty} x\left[\frac{1-0}{1-0}\right]$$
$$=\infty$$

5. Ans: b)

$$l=\lim_{x\to 0}\frac{e^x+e^{-z}-2}{x^2}$$
$$l=\lim_{x\to 0}\frac{e^x-e^{-z}}{2x},$$
$$=\lim_{x\to 0}\frac{e^x+e^{-x}}{2}=1$$

Career Avenues

6. Ans: d)

$$f\ x = \begin{cases} x^2, 0 \le x \le \frac{1}{2} \\ x, \frac{1}{2} < x \le 1 \end{cases}$$

$$\lim_{x \to \frac{1}{2}^-} f\ x = \left(\frac{1}{2}\right)^2 = \frac{1}{4}$$

$$\lim_{x \to \frac{1}{2}^+} f\ x = \frac{1}{2}$$

Both limit are not equal so limit does not exist

7. Ans: b)

The given series is expansion of sin x, thus sin $\frac{\pi}{2} = 1$

8. Ans: c)

$$we\ have, \quad f\ x = \begin{cases} x, & when\ 0 \le x \le \frac{1}{2} \\ 1, & when\ x = \frac{1}{2} \\ 1 - x, & when\ \frac{1}{2} < x < 1 \end{cases}$$

$$At\ x = \frac{1}{2}, \quad Lf\ x = \lim_{h \to 0} f\left(\frac{1}{2} - h\right)$$

$$= \lim_{h \to 0} \frac{1}{2} - h = \frac{1}{2}$$

$$Rf\ x = \lim_{h \to 0}\left(\frac{1}{2} + h\right)$$

$$= \lim_{h \to 0} 1 - \left(\frac{1}{2} + h\right) = \frac{1}{2}$$

$$f\left(\frac{1}{2}\right) = 1$$

Thus f (x) is discontinuous at x = $\frac{1}{2}$

9. Ans: b)

$$We\ have \quad f\ x = \begin{cases} x^2 - 1, & 1 < x < \infty \\ x - 1, & -\infty < x \le 1 \end{cases}$$

$$At\ x = 1 \quad Lf\ x = \lim_{h \to 0} f\ 1 - h$$

$$= \lim_{h \to 0} 1 - h\ ^2 - 1 = 0$$

Career Avenues

At $x=1$, $Rf\ x = \lim_{h\to 0} f\ 1+h$

$= \lim_{h\to 0} 1+h^{\ 2}-1=0$

Since $L\lim_{h\to 0} f\ x = R\lim_{h\to 0} f\ x$

Thus $f\ x$ *is continuous at* $x=1$

Now $Lf\ x = \lim_{h\to 0} \frac{f\ 1-h\ -f\ 1}{-h}$

$= \lim_{h\to 0} \frac{1-h^{\ 2}-1-0}{-h} = 2$

$Rf\ x = \lim_{h\to 0} \frac{f\ 1+h\ -f\ 1}{h}$

$= \lim_{h\to 0} \frac{1+h-1}{h} = 1$

Since Lf' (1) $\neq$ Rf'(1), f(x) is not differentiable at x = 1

10. Ans: a)

f(x) is continuous and differentiable fro all x except 0. We have to check at x = 0

Left limit of f(x) at $x \to 0$, $\lim_{x\to 0-} f\ x = 0$

Right limit of f(x) at $x \to 0$, $\lim_{x\to 0+} f\ x = 0$

Value of f(x) at x = 0, $f\ x_{\ x-0} = 0$

Thus f(x) is continuous at x = 0

Left hand derivative $Lf\ x = \lim_{h\to 0} \frac{f\ 0-h\ -f\ 0}{-h}$

$= \lim_{h\to 0} \frac{-\ 0-h}{-h} = -1$

Right hand derivative $Rf'\ x = \lim_{h\to 0} \frac{f\ 0+h\ -\ 0}{h}$

$= \lim_{h\to 0} \frac{h}{h} = 1$

Since, Lf (x) $\neq$ Rf' (x), there fore, f(x) is not differentiable at x = 0, hence not differentiable for all real values of x.

Career Avenues

SET- 5
Maxima and Minima

1. The function $f\ x = e^{ax} + e^{-ax}$ *where* $a < 0$ *is decreasing for*
   a) $x < 0$
   b) $x > 0$
   c) $x > 1$
   d) $x < 1$

2. The function $f\ x = -2x^3 - 9x^2 - 12x + 1$ is an increasing function in the interval
   a) $-2 < x < -1$
   b) $-2 < x < 1$
   c) $-1 < x < 2$
   d) $1 < x < 2$

3. As x *is increased from* $-\infty$ *to* ∞ *the function* $f\ x = \dfrac{e^x}{1+e^x}$
   a) Monotonicaly increases
   b) Monotonicaly decreases
   c) Increases to a maximum value and then decreases
   d) Decreases to minimum value and then increases

4. If $f\ x = kx^3 - 9x^2 + 9x + 3$ *is increasing in each interval, then*
   a) $k \leq 3$
   b) k < 3
   c) $K \geq 3$
   d) $k > 3$

5. The function $f\ x = x^3 - 6x^2 + 9x + 25$ *has*
   a) *A maxima at* $x = 1$ *and minima at* $x = 3$
   b) *A maxima at* $x = 3$ *and minima at* $x = 1$
   c) *No maxima but a minima at* $x = 3$
   d) *A maxima at* $x = 1$ *but not minima.*

6. The function $f\ x = x^3$ *has*
   a) *a maixima at* $x = 0$
   b) *a minima at* $x = 0$
   c) *neither a maxima nor a minimum at* $x = 0$
   d) *no point of deflection*

Career Avenues

7. $The\ maximum\ value\ of\ f\ x = 1+cosx\ sinx\ is$

a) 3
b) 4
c) $3\sqrt{3}$
d) $3\sqrt{3}$ /4

8. The maxium value of $\left(\frac{1}{x}\right)^{2x^2}$, $x>0$ $is\ equal\ to$

a) $e^{1/e}$
b) e
c) $e\sqrt{e}$
d) $1/e$

9. The minimum value of function $y=x^2$ $in\ the\ interval$ $1,5$ is

a) 0
b) 1
c) 25
d) Undefined

10. The minimum distance from the point (4,2) to the curve $y^2=8x, is$

a) $\sqrt{2}$
b) $2\sqrt{2}$
c) $3\sqrt{2}$
d) 2

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Solutions:

1. Ans: b)

We have $f\ x = e^{az} + e^{-ax} = 2\cosh ax.$

Or $f'\ x = 2a\sinh ax$

$f'\ x < 0\ will\ be\ for\ x > 0\ because\ a < 0$

2. Ans: a)

$f\ x = -2x^3 - 9x^2 - 12x + 1$

$f'\ x = -6x^2 - 18x - 12$

$= -6\ x - 2\ \ x + 1$

The function f (x) is increasing if f'(x) > 0

Or $6\ x + 1\ \ x + 2\ < 0$

Or $-2 < x < -1$

3. Ans: a)

$f\ x = \frac{e^x}{1 + e^x}$

$f'\ x = \frac{1 + e^x\ \ e^x - e^x e^x}{1 + e^x\ ^2}$

$= \frac{e^x}{1 + e^x\ ^2}$

Thus f(x) > 0 for every x, Therefore f (x) is monotonically increasing function

4. Ans: d)

$f\ x = kx^3 - 9x^2 + 9x + 3$

$f'\ x = 3kx^2 - 18x + 9$

$= 3\left[kx^2 - 6x + 3\right]$

This is positive when k > 0 and 36 – 12k < 0 or k >3,

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5. Ans: a)

$f\ x = x^3 - 6x^2 + 9x + 25$

$f'\ x = 3x^2 - 12 + 9$

$f''\ x = 6x - 12 = 6\ x - 2$

Now for maxima and minima, $f'\ x = 0,$

$3x^2 - 12x + 9 = 0$

Or x = 1,3

At x =1 $f'\ 1 = 6\ 1 - 2 = -6\ -ve$

And x = 3 $f''\ 3 = 6\ 3 - 2 = 6\ +ve$

6. Ans: c)

$f\ x = x^3$

$f'\ x = 3x^2$

$f''\ x = 6x$

$f'''\ x = 6$

At x = 0, f'(x) = f'(x) = 0 but f'''(x) $\neq 0$ thus at x = 0, f (x) is neither maximum not minimum.

7. Ans: d)

We have $f\ x = 1 + \cos x\ \sin x$

Or $f'\ x = 2\cos x - 1\ \cos x + 1$

Or $f''\ x = -\sin x\ 1 + \cos x$

Now for maxima and minima $f'\ x = 0$

$\cos x = \frac{1}{2} \qquad or \cos x = -1$

Or $x = \pi/3 or\ x = \pi.$

Now $f''\ \pi/3 = \frac{-3\sqrt{3}}{2}\ -ve\ , So\ x = \pi/3 is\ a\ point\ of\ maxima$

Maximum value = $\left(\sin\frac{\pi}{3}\right)\left(1 + \cos\frac{\pi}{3}\right) = \frac{3\sqrt{3}}{4}$

8. Ans: b)

$y = \left(\frac{1}{x}\right)^{2x^2}$

$\log y = 2x^2 \log\left(\frac{1}{2}\right)$

$\log\ y = -2x^2 \log x$

$\frac{1}{y}\frac{dy}{dx} = -2\ x + 2\log x$

$\frac{dy}{dx} = -2\left(\frac{1}{x}\right)^{2x^2}\ x + 2x\log x$

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For maxiam and minima $\frac{dy}{dx}=0$

Or $1 + 2 \log x = 0$

Or $\log x = -\frac{1}{2}$

Or $x = e^{-1/2}$

Thus maximum value is $e^{1/2\ \ 2/e}$ *Or* $e^{1/e}$

9. Ans: b) Since the given function is montonically increasing in the given interval [1,5], hence 1 will be the minmum value.

10. Ans: b)
Given curve is parabola. Let the point closest to (4, 2) be $(2t^2, 4)$

Now D = $\sqrt{2t^2-4\ ^2+\ 4t-2\ ^2}$ is minimum when $E=\ 2t^2-4\ ^2+\ 4t-2\ ^2$ is minimum

Now, $E = 4t^4-16t+20$

Or $\frac{dE}{dt}=16t^3-16=16\ t-1\ \ t^2+t+1$

Or $\frac{d^2E}{dt^2}=48t^2$

For maxima and minima $\frac{dE}{dt}=0, So\ we\ get\ t=1$

Since, $\left[\frac{d^2E}{dt^2}\right]_{t=1}=48>0,\ \ So\ t=1$ is a point minima

Thus minimum distance D = $\sqrt{2-4\ ^2+\ 4-2\ ^2}=2\sqrt{2}$

Career Avenues

SET- 6
Partial Derivatives and Definite Integrals

1. For $f\ x,y\ = y^3 - 3ax^2 + x^3$.*The value of* $\frac{d^2y}{dx^2}$ *is eqaul to*

a) $-\frac{2a^2x^2}{y^5}$

b) $\frac{2a^2x^2}{y^5}$

c) $-\frac{2a^2x^4}{y^5}$

d) $-\frac{a^2x^2}{y^5}$

2. For given $u = x^2 + y^2 + z^2$, *then* $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} + z\frac{\partial u}{\partial z}$ *is equal to*

a) $4u$
b) $-2u$
c) $-u$
d) $2u$

3. *For* $u = \frac{xy}{x+y}$, *then* $x\frac{\partial y}{\partial x} + y\frac{\partial u}{\partial y}$ *is eqaul to*

a) 1
b) u
c) $-u$
d) 0

4. If $u = \sin^{-1}\left(\frac{x^{1/3} + y^{1/3}}{x^{1/2} + y^{1/2}}\right)^{1/2}$ then

$x\frac{\partial y}{\partial x} + y\frac{\partial u}{\partial y}$ *is equals to*

a) $\frac{1}{12}$ ***tan u***

b) $-\frac{1}{12}$ ***tan u***

c) $\frac{1}{6}$ ***tan u***

d) $-\frac{1}{6}$ ***tan u***

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5. It is given that for $u = \left(\frac{x+y}{\sqrt{x}+\sqrt{y}}\right)$ *.then the valu e of* $x\frac{\partial y}{\partial x} + y\frac{\partial u}{\partial y}$ is equal to

a) $\frac{1}{2}\tan u$

b) $2\cos 2u$

c) $\frac{1}{4}\sin 2u$

d) $2\tan 2u$

6. The value of $\oint_x^0 \frac{x\sin x}{1+\cos^2 x}dx$ *is equal to*

a) π

b) $\pi/4$

c) $\pi^2/4$

d) $\pi^3/4$

7. Let $I = \int_0^1 \frac{e^{\sqrt{1-x^2}}}{\sqrt{1-x^2}} x\,dx$ *then the value of I is*

a) $\frac{e+1}{2}$

b) $\frac{e-1}{2}$

c) $e+1$

d) $e-1$

8. The value of Integral $\int_{-1}^{1} \frac{|x|}{x}dx$ *is*

a) 4

b) 3

c) 2

d) 1

9. The value of Integral $\int_0^{\pi/2} \frac{\sqrt{\cos x}}{\sqrt{\cos x}+\sqrt{\sin x}}dx$ *is*

a) $\pi/8$

b) $\pi/6$

c) $\pi/4$

d) $\pi/2$

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10. The value of Integral $\int_{-\pi/2}^{\pi/2} \frac{\sin 2x}{1+\cos x} dx$ *is*

a) 2 *ln*2
b) 0
c) 2
d) 5

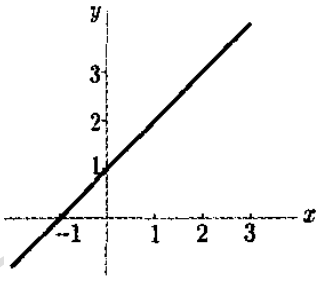
11. Let $I = \int_0^1 \frac{dx}{1-x-x^2}$ *then the value of* I *is*

a) $\frac{\sqrt{3\pi}}{9}$
b) $\frac{2\pi\sqrt{3}}{9}$
c) $\frac{4\pi}{\sqrt{3}}$
d) $\frac{2\pi}{\sqrt{3}}$

12. The value of integral is $\int_0^{2a} \frac{f\ x}{f\ x + f\ 2a-x} dx$ *is*

a) 0
b) *a*
c) 2*a*
d) 3*a*

13. The given diagram have a y *which is a function of* x *then the value of* I *is*



Where $I = \int_1^2 y\, dx$

a) 0
b) 1
c) 2
d) 2.5

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14. The value of $\int_1^2 \int_4^6 xy + e^x \; dy\,dx$ *is*

a) $5 + e^2$

b) $2\ 7 + e^2$

c) $2\ 4 + e^2$

d) $2\ 9 + e^2$

15. The value of Integral $\int_0^1 \int_x^{\sqrt{x}} x^2 + y^2 \; dy\,dx$ *is* is

a) 0

b) $35/3$

c) $3/35$

d) $4/43$

16. The value of the integral $\int_0^\infty \int_0^\infty e^{-x^2} e^{-y^2} \, dx\,dy$ *is*

a) $\sqrt{\pi}$

b) $\sqrt{\pi/2}$

c) $\pi/4$

d) 1

17. R is the region bounded by $x = 0, y = 0 \, and \, x + y = 1$ then

$\int\int_R x^2 + y^2 \; dx\,dy$ *is equal to*

a)1

b)$1/3$

c)$1/5$

d)$1/6$

18. The valu of $\int_0^1 \int_0^1 \int_0^1 e^{x+y+z} \, dx\,dy\,dz$ *is*

a) $2\ e-1^3$

b) $\frac{2}{3}\ e-1$

c) $e-1^2$

d) $e-1^3$

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Answer

1. Ans: b)

$f(x,y) = y^3 - 3ax^2 + x^3 = 0$

$f_x = -6ax + 3x^2,\ f_y = 3y^2,\ f_{xx} = -6a + 6x,\ f_{yy} = 6y,\ and\ f_{xy} = 0$

$$\frac{d^2y}{dx^2} = -\left[\frac{f_{xx}\ f_y{}^2 - 2f_x f_y f_{xy} + f_{yy}\ f_x{}^2}{f_y{}^3}\right]$$

$$= -\left[\frac{6x-6a\ \ 3y^2{}^2 - 0 + 6y\ 3x^2 - 6ax^2}{3y^2{}^3}\right]$$

$$= -\frac{2a^2x^3}{y^5}$$

2. Ans: d) here $u = x^2 + y^2 + z^2$ *is a homogenous function of degree* 2. *Then by Euler's theorem*

$$x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} + z\frac{\partial u}{\partial z} = 2u$$

3. Ans: b)

The given function $u = \frac{xy}{x+y}$ is a homogeneous function of degree 1.

Thus by Euler's theorem

$$x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = u$$

4. Ans: b)

5. Ans: c)

We have $u = \tan^{-1}\left(\frac{x+y}{\sqrt{x}+\sqrt{y}}\right)$

Or $\tan u = \frac{x+y}{\sqrt{x}+\sqrt{y}} = f$

Which is a homogeneous equation of degree ½

By Euler's theorem $x\frac{\partial f}{\partial x} + y\frac{\partial f}{\partial y} = \frac{1}{2}f$

Or $x\frac{\partial \tan u}{\partial x} + y\frac{\partial \tan u}{\partial y} = \frac{1}{2}\tan u$

$$x\sec^2 u\frac{\partial u}{\partial x} + y\sec^2 u\frac{\partial u}{\partial y} = \frac{1}{2}\tan u$$

Thus $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = \frac{1}{2}\sin u\cos u = \frac{1}{4}\sin 2u$

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6. Ans: c)

$$I=\int_0^{\pi}\frac{x\sin x}{1+\cos^2 x}dx$$

$$I=\int_0^{\pi}\frac{\pi-x\ \sin\ \pi-x}{1+\cos^2\ \pi-x}dx$$

$$I=\int_0^{\pi}\frac{\pi-x\ \sin x}{1+\cos^2 x}dx$$

$$2I=\pi\int_0^{\pi}\frac{\sin x}{1+\cos^2 x}dx$$

$$2I=-\pi\int_1^{-1}\frac{1}{1+t^2}dt$$

$$=-\pi\left[\tan^{-1}t\right]_1^{-1}$$

$$=\frac{\pi^2}{2}$$

$$I=\frac{\pi^2}{4}$$

7. Ans: d)

We have $I=\int_0^{I}\frac{e^{\sqrt{1-x^2}}}{\sqrt{1-x^2}}xdx$

Putting $\sqrt{1-x^2}=t\ whcih\ given\ \frac{1}{2\sqrt{1-x^2}}\ -2x\ dx=dt$

We get t = 1 when x = 0 and t = 0 when x = 1

Thus $I=\int_1^0-e^t dt=-\left[e^t\right]_1^0=-\left[e^0-e^1\right]=e-1$

8. Ans: d)

$$I=\int_{-1}^{1}\frac{|x|}{x}dx=\int_{-1}^{0}\frac{-x}{x}dx+\int_0^1\frac{x}{x}dx$$

$$=\int_{-1}^{0}-1dx+\int_0^1 1.dx=-\ x\ {}_{-1}^{0}+\ x\ {}_0^1$$

$$=-\left[0-\ -1\ \right]+1-0\ =0$$

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9. Ans: c)

$$I = \int_0^{\pi/2} \frac{\sqrt{\cos x}}{\sqrt{\cos x} + \sqrt{\sin x}} dx$$

$$= \int_0^{\pi/2} \frac{\sqrt{\cos\ x/2 - x}}{\sqrt{\cos\ x/2 - x} + \sqrt{\sin\ x/2 - x}} dx$$

$$= \int_0^{\pi/2} \frac{\sqrt{\sin x}}{\sqrt{\sin x} + \sqrt{\cos x}} dx$$

$$2I = \int_0^{\pi/2} \frac{\sqrt{\cos x} + \sqrt{\sin x}}{\sqrt{\sin x} + \sqrt{\cos x}} dx$$

$$= \int_0^{\pi/2} 1, dx = \frac{\pi}{2}$$

$$I = \frac{\pi}{4}$$

10. Ans:b) here the integrand function is ODD, so the integration over that interval of time is ZERO.

11. Ans: b)

$$I = \int_0^1 \frac{dx}{1 - x + x^2}$$

$$= \int_0^1 \frac{dx}{\left(x - \frac{1}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2}$$

$$= \frac{1}{\frac{\sqrt{3}}{2}} \left[\tan^{-1} \frac{x - \frac{1}{2}}{\frac{\sqrt{3}}{2}} \right]_0^1$$

$$= \frac{2\pi\sqrt{3}}{9}$$

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12. Ans: b)

$$I=\int_0^{2a}\frac{f\ x}{f\ x+f\ 2a-x}dx$$

$$=\int_0^{2a}\frac{f\ 2a-x}{f\ 2a-x+f\ x}dx$$

$$2I=\int_0^{2a}\frac{f\ x+f\ 2a-x}{f\ x+f\ 2a-x}dx$$

$$=\int_0^{2a}1.dx=x\,{}_0^{2a}=2a$$

$$I=a$$

13. Ans: 25

$$I=\int_1^2 y\,dx$$

$$=2\int_1^2 x+1\ dx$$

$$=\left[\frac{x^2}{2}+x\right]_1^2=25$$

14. Ans: d)

$$\int_1^2\int_4^6 xy+e^x\ dydx=\int_0^2\left[\frac{xy^2}{2}+e^x y\right]_4^6$$

$$=\int_0^2 10x+2e^x\ dx$$

$$=\left[5x^2+2e^x\right]_0^2$$

$$=\ 20+2e^2\ -\ 0+2$$

$$=2\ 9+e^2$$

15. Ans: c)

$$\int_0^1\int_x^{\sqrt{x}} x^2+y^2\ dy\,dx=\int_0^1\left[x^2\ y+\frac{1}{3}y^3\right]_x^{\sqrt{x}}dx$$

$$=\int_0^1\left[x^{5/2}+\frac{1}{3}x^{3/2}-x^3-\frac{1}{3}x^3\right]dx$$

$$=\left[\frac{2}{7}x^{7/2}+\frac{2}{15}x^{5/2}-\frac{1}{3}x^4\right]_0^1=\frac{3}{35}$$

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16. Ans: c)

$$I=\int_0^{\infty}\int_0^{\infty} e^{-x^2}\, e^{-y^2}\, dx\, dy=\int_0^{\infty}\int_0^{\infty} e^{-\left(x^2+y^2\right)}\, dxdy$$

$$=\int_{\theta=0}^{\pi/2}\int_{r=0}^{\infty} e^{-r^2}\; rdr\, d\theta$$

$$=-\frac{1}{2}\int_0^{r/2}\left\{\int_0^{\infty} e^{-r^2}\; -2r\; dr\right\}d\theta$$

$$=-\frac{1}{2}\int_0^{\pi/2}\left|e^{-r^2}\right|_0^{\infty} d\theta$$

$$=\frac{1}{2}\int_0^{\pi/2} d\theta=\frac{\pi}{4}$$

17. Ans: d)

$$\int\int_R x^2+y^2\; dx\, dy=\int_0^1\int_0^{1-x} x^2+y^2\; dy\, dx$$

$$=\int_0^1\left[x^2 y\frac{y^3}{3}\right]_0^{1-x} dx$$

$$=\int_0^1\left[x^2\;1-x\;+\frac{1-x^{\;3}}{3}\right]dx$$

$$=\left[\frac{x^3}{3}-\frac{x^4}{4}-\frac{1-x^{\;4}}{12}\right]_0^1$$

$$=\left[\left(\frac{1}{3}-\frac{1}{4}\right)-\left(-\frac{1}{12}\right)\right]=\frac{1}{6}$$

18. Ans: d)

$$\int_0^1\int_0^1\int_0^1 e^{x+y+z}dx\, dy\, dz=\int_0^1\int_0^1 ex+y+1_0^{\,1}\, dydz$$

$$=\int_0^1\int_0^1\left[e^{1+y+z}-e^{y+z}\right]dy\, dz$$

$$=\int_0^1\left[e^{1+y+z}-e^{y+z}\right]_0^1 dz$$

$$=\int_0^1\left[\;e^{2+z}-e^{1+z}\;-\;e^{1+z}-e^{z}\;\right]dz$$

$$=\int_0^1\;e^{2+z}-2e^{1+z}+e^{z}\;\;dz$$

$$=\left[e^{2+z}-2e^{1+z}+e^{z}\right]_0^1$$

$$=\;e^3-2e^3+e\;-\;e^2-2e+1$$

$$=e^3=-3e^2+3e-1=\;e-1^{\;3}$$

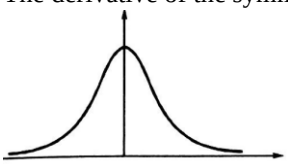
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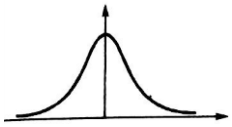
Past GATE Questions

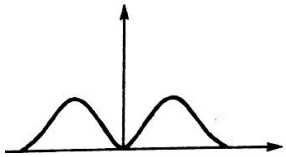
1. The value of the integral $I = \frac{1}{\sqrt{2\pi}} \int_0^{\infty} \exp\left(\frac{-x^2}{8}\right) dx$ *is* **(Gate-05, 1-Mark)**

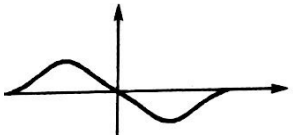
a) 1
b) π
c) 2
d) 2π

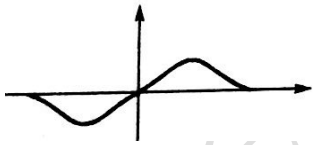
2. The derivative of the symmetric function drawn in given figure will look like **(Gate-05, 2-Mark)**



a) 

b) 

c) 

d) 

3. The Value of $\nabla \times \nabla \times \nabla$ is equals to……., Where P is a vector **(Gate-06, 1-Mark)**

a) $P \times \nabla \times P - \nabla^2 P$
b) $\nabla^2 P + P\nabla \ \nabla \times P$
c) $\nabla^2 P + \nabla \times \mathrm{P}$
d) $\nabla.\ \nabla.P\ - \nabla^2 P$

4. $\iint \nabla \times P\ .ds$ Where P is a vector is equals to **(Gate-06, 1-Mark)**

a) $\oint P.dl$
b) $\oint \nabla \times \nabla \times P.dl$
c) $\oint \nabla \times P.dl$
d) $\iiint \nabla.Pdv$

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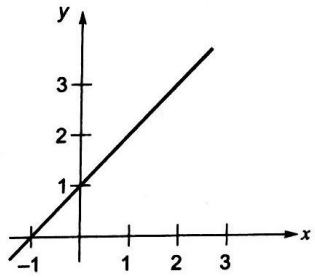
5. The Integral $\int_0^{\pi} \sin^3\theta\, d\theta$ is given by **(Gate-06, 2-Mark)**

a) 1/2 b) 2/3 c) 4/3 d) 8/3

6. As x increased from $-\infty to +\infty$ the function $f\ x\ = \frac{e^x}{1+e^x}$ **(Gate-06, 1-Mark)**

a) Monotonically increases
b) Monotonically decreases
c) Increases to a maximum value and then decreases
d) Decreases to a minimum value and then increases

7. The following plot shows a function y which varies linearly with x. The value of the integral $I = \int_1^2 y\, dx$ is **(Gate-07, 1-Mark)**



a) 1.0 b) 2.5 c) 4.0 d) 5.0

8. For $|x| \ll 1$ cot h(x) can be approximated as **(Gate-07, 1-Mark)**

$a) x$ $b) x^2$ $c) \frac{1}{x}$ $d) \frac{1}{x^2}$

9. The Value of $\lim_{\theta \to 0} \frac{\sin\ \theta/2}{\theta}$ is **(Gate-07, 1-Mark)**

a) 0.5 b) 1 c) 2 d) Not defined

10. Which of the following function is strictly bounded **(Gate-07, 1-Mark)**

a) $\frac{1}{x^2}$ b) e^x c) x^2 d) e^{-x^2}

11. Consider the function f(x)= $x^2 - x - 2$. The maximum value of f(x) in the closed interval [-4,4] is **(Gate-07, 2-Mark)**

a) 18 b) 10 c) -2.25 d) indeterminate

12. Which of the following functions would have only add powers of x in its Taylor series expansion about the point x = 0. **(Gate-08, 1-Mark)**

a) $\sin(x^3)$ b) $\sin(x^2)$ c) $\cos(x^3)$ d) $cos(x^2)$

Career Avenues

13. In Taylor series expansion of exp(x) + sin(x) about the point x= π and the coefficient of (x- π)^2 is **(Gate-08 2-Mark)**

a) exp(π) b) 0.5 exp(π) c) exp(π) +1 d) exp(π) -1

14. Ḟor real values of x, the minimum value of the function value fo the function (x)=exp(x) + exp (-x) is (**Gate-08, 1-Mark)**

a) 2 b) 1 c) 0.5 d) 0

15. The value of the integral of the function g(x,y) = 4 x^3 + 10 y^4 along the straight line segment from the point (0,0) to the point (1,2) in the x-y plane is (**Gate-08, 2-Mark)**

a) 33 b) 35 c) 40 d) 56

16. Consider the point P and Q in the x-y plane, with P =(1,0) and Q=(0,1).The integral $2\int_p^Q xdx + ydy$ along the semicircle with the line segment PQ as its diameter (**Gate-08, 2-Mark)**

a) is -1
b) is 0
c) 1
d) depend on the direction (clockwise or anti-clockwise) o the semi circle

17. The Taylor series expansion of $\frac{\sin x}{x-\pi}$ at x= π is given by (**Gate-09, 2-Mark)**

a) $1+\frac{x-\pi}{3!}^2+\ldots$

b) $-1-\frac{x-\pi}{3!}^2+\ldots$

c) $1-\frac{x-\pi}{3!}^2+\ldots$

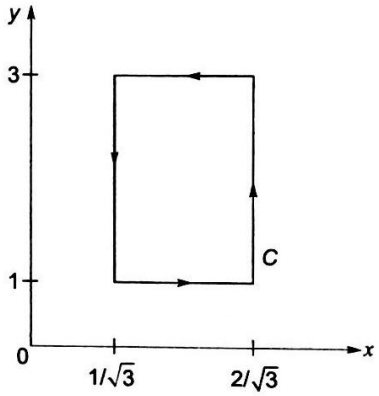
d) $-1+\frac{x-\pi}{3!}^2+\ldots$

18. If $e^y = x^{1/x}$, then y has a (**Gate-10, 2-Mark)**

a) Maximum at x=e
b) Minimum at x=e
c) Maximum at x=1/e
d) Minimum at x=1/e

Career Avenues

19. If $\vec{A} = xya_x + x^2 a_y$. $\oint_C \vec{A}.\vec{dl}$ over the path shown in the figure is **(Gate-10, 2-Mark)**



a) 0 b) $2/\sqrt{3}$ c) 1 d) $2\sqrt{3}$

20. The maximum value of f $x = x^3 - 9x^2 + 24x + 5$ in the interval [1,6] is **(Gate-12, 2-Mark)**

a) 21 b) 25 c) 41 d) 46

21. Consider a vector field $\vec{A}$ ($\vec{r}$). The closed loop line integral $\oint \vec{A}.\vec{dl}$ can be expressed as **(Gate-13, 1-Mark)**

a) $\iint \nabla x\vec{A}.\vec{ds}$ over the closed surface bounded by the loop.

b) $\iiint \nabla.\vec{A}\ dv$ over the closed volume bounded by the top

c) $\iiint \nabla.\vec{A}\ dv$ over the open volume bounded by the loop

d) $\iint \nabla x\vec{A}.\vec{ds}$ over the open surface bounded by the loop

22. The divergence of the vector field $\vec{A} = x\hat{a}_x + y\hat{a}_y + z\hat{a}_z$ is **(Gate-13, 1-Mark)**

a) 0 b) 1/3 c) 1 d) 3

23. The Taylor series expansion of 3 sinx + 2 cosx is ……. **(Gate-14, 1-Mark)**

a) $2 + 3x - x^2 - \frac{x^3}{2} + \ldots$

b) $2 - 3x + x^2 - \frac{x^3}{2} + \ldots$

c) $2 + 3x + x^2 \mp \frac{x^3}{2} + \ldots$

d) $2 - 3x - x^2 + \frac{x^3}{2} + \ldots$

24. The volume under the surface z (x,y) = x+y and above the triangle in the x-y plane defined by $0 \le y \le x \text{ and } 0 \le x \le 12$ is…… **(Gate-14, 2-Mark)**

25. For $0 \le t < \infty$, the maximum value of the unction f(t)= $e^{-t} - 2e^{-2t}$ occurs at **(Gate-14, 1-Mark)**

a) t = $\log_e 4$ b) t = $\log_e 2$ c) t = 0 d) t = $\log_e 8$

Career Avenues

26. The value of $\lim_{x\to\infty}\left(1+\frac{1}{x}\right)$ is **(Gate-14, 1-Mark)**

a) $\log_e 2$ b) 1.0 c) e d) ∞

27. The maximum value of the function **(Gate-14, 1-Mark)**
f(x) = $\log_e\ 1+x\ -x$ (where $x>-1$) *occurs at* $x=$ …….

28. The maximum value of $f(x)=2x^3\ -9x^2+12x-3$ in the interval $0\le x\le 3$ *is* ……
(Gate-14, 2-Mark)

29. The Series $\sum_{n=0}^{\infty}\frac{1}{n!}$ converges to **(Gate-14, 1-Mark)**

a) 2 $\log_e 2$ b) $\sqrt{2}$ c) 2 d) e

30. Ḟor a right triangle I the sum o the lengths o the hypotenuse and a side is kept constant,in order to have maximum area o the triangle, the angle between the hypotenuse and the side is **(Gate-14, 2-Mark)**
a) 12° b) 36° c) 60° d) 45°

31. If $\vec{r}=x\hat{a}_x+y\hat{a}_y+z\hat{a}_z$ *and* $|\vec{r}|=r$ then *div* $r^2\nabla\ \ln r\ =$____ **(Gate-14, 2-Mark)**

32. Given the vector $\vec{A}=\ \cos x\ \ \sin y\ \hat{a}_x+\ \sin x\ \ \cos y\ \hat{a}_y$ *where* $\hat{a}_x\,\hat{a}_y$ denotes unit vectors along x, y directions, respectively. The magnitude of curl of $\vec{A}$ is…… **(Gate-14, 2-Mark)**

33. The magnitude of gradient for then function $f\ x,y,z\ =\ x^2+3\ y^2+\ z^3$ at the *point* 1,1,1 *is* ….
(Gate-14, 1-Mark)

34. The directional derivative of f(x,y) = $\frac{xy}{\sqrt{2}}\ x+y$ at (1,1) in the direction of the unit vector an angle of $\frac{\pi}{4}$ with y-axis, is given by…….. **(Gate-14, 1- Mark)**

35. Given $\vec{A}=z\hat{a}_x+x\hat{a}_y+y\hat{a}_z$. **If** S represents the portion of the sphere $x^2+y^2+\ z^2=1$ for $z\ge 0$, then $\int_s \nabla X\,\vec{F}.\vec{ds}$ ……. **(Gate-14, 1- Mark)**

36. If $\boldsymbol{z=xyln\ \ xy}$. ***then*** **(Gate-14, 1-Mark)**

a) $x\frac{\partial z}{\partial x}+\mathrm{y}\frac{\partial z}{\partial y}=0$

b) $y\frac{\partial z}{\partial x}=x\frac{\partial z}{\partial y}$

c) $x\frac{\partial z}{\partial x}=y\frac{\partial z}{\partial y}$

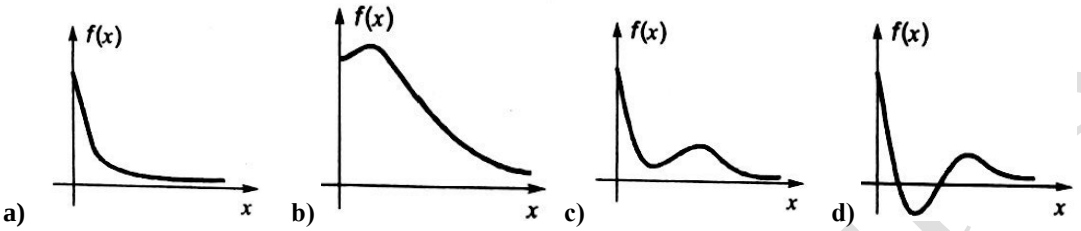
d) $y\frac{\partial z}{\partial x}+x\frac{\partial z}{\partial y}=0$

Career Avenues

37. A function $f\ x = 1 - x^2 + x^3$ is defined in the closed interval [-1,1]. The value of x in the open interval (-1,1) for which the mean value theorem is satisfied, is **(Gate-15,1-Mark)**

a) -1/2 b) -1/3 c) 1/3 d) ½

38. Which of the following graphs describe the function $f\ x = e^{-x}\ x^2 + x + 1$? **(Gate-15,2-Mark)**



a) **b)** **c)** **d)**

39. The maximum area (in square unit) of a rectangle whose vertices on the ellipse $x^2 + 4y^2 = 1$ *is*…….. **(Gate-15, 2-Mark)**

40. The value of the integral $\int_{-\infty}^{\infty} 12\cos\ 2\pi t\ \frac{\sin\ 4\pi t}{4\pi t} dt$ *is*…… **(Gate-15, 2-Mark)**

41. The contour on the x- y plane, where the partial derivative of $x^2 + y^2$ with respect to y is equal to the partial derivative of 6y+ 4x with respect to x, is **(Gate-15, 1-Mark)**

a) y=2 b) x=2 c) x=y=4 d) x=y=0

42. Given the following statements about a function f: R→ R, select the right option: **(Gate-16, 1-Mark)**

P: if f(x) is continuous at $x = x_0$ then it differential at $x = x_0$

Q: if f(x) is continuous at $x = x_0$ then it may not be differentiable at $x = x_0$

R: If f(x) is differentiable at $x = x_0$ then it is also continuous at $x = x_0$

a) P is true, Q is false, R is false

b) P is false, Q is true, R is true

c) P is false, Q is true, R is false

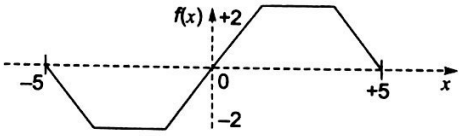
d) P is true, Q is false, R is true

43. The integral $\frac{1}{2\pi}\iint_D x + y + 10\ dxdy, where\ D\ denotes\ the\ disc: x^2 + y^2 \le 4$, evaluates to……. **(Gate-16, 2-Mark)**

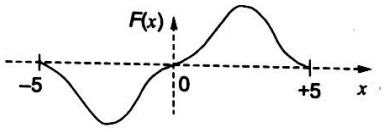
44. The region specified by $\left\{ \rho, \varphi, z\ : 3 \le \rho \le 5, \frac{\pi}{8} \le \varphi \le \frac{\pi}{4}, 3 \le z \le 4.5 \right\}$ in the cylindrical coordinates has volume of ……. **(Gate-16, 2-Mark)**

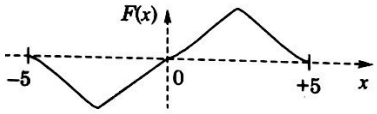
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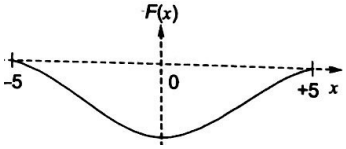
45. Consider the plot f(x) verses x as shown below **(Gate-16, 1-Mark)**

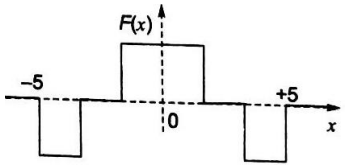


Suppose $F(x) = \int_{-5}^{x} f\ y\ dy$. Which on eof the following is a graph of F(x) ?

a)


b)


c)


d)


46. Suppose C is the closed curve defined as the circle $x^2 + y^2 = 1$ with C oriented anti clockwise. The value of $\oint xy^2\,dx + x^2 ydy$ over the curve C equals …… **(Gate-16, 2-Mark)**

47. As x varies from -1 to +3, which one of the following describes the behavior of the function $f\ x = x^3 - 3x^2 + 1$? (**Gate-16, 1-Mark)**

a) f(x) increases monotonically
b) f(x) increases, then decreases and increases again
c) f(x) decreases, then increases and decreases again
d) f(x) increases and then decreases

48. How many distinct values of x satisfy the equation sin(x) = x/2, where x is in radians? (**Gate-16, 1-Mark)**

a) 1 b) 2 c) 3 d) 4 or more

49. Cosider the time-varying vector $I = \hat{x}15\cos\ \omega t\ +\ y\ 5\sin\ \omega t\ in$ Cartesian coordinates, where $\omega > 0 is a constant.$ When the vector magnitude $|I|$ is at its minimum value, the angle $\theta that I\ makes$ with the x-axis (in degrees, such that $0 \le \theta \le 180\ is$ …….

(**Gate-16, 1-Mark)**

Career Avenues

50. The Integral $\int_0^1 \frac{dx}{\sqrt{1-X}}$ is equal to …… **(Gate-16, 1-Mark)**

51. If the vectors $e_1 = 1,0,2$, $e_2 = 0,1,0$ and $e_3 = -2,0,1$ form an orthogonal basis of the three- dimensional real; space R^3, then the vector $u = (4,3,-3)\ \epsilon\ R^3$ can be expressed as .**(Gate-16, 2-Mark)**

a) $u = -\frac{2}{5}e_1 - 3e_2 - \frac{11}{5}e_3$

b) $u = -\frac{2}{5}e_1 - 3e_2 + \frac{11}{5}e_3$

c) $u = -\frac{2}{5}e_1 + 3e_2 + \frac{11}{5}e_3$

d) $u = -\frac{2}{5}e_1 + 3e_2 - \frac{11}{5}e_3$

52. A triangle in the x-y plane is bounded by the staright lines 2x=3y, y=0 and x=3. The volume above the triangle and under the plane x + y + z =6 is ……. **(Gate-16, 2-Mark)**

53. Consider the following statements about the linear dependence of the real valued functions $y_1 = 1$, $y_2 = x$ and $y_3 = x^2$, over the field of real numbers. **(Gate-17, 1-Mark)**

I. y_1, y_2 and y_3 are linearly independent on $-1 \le x \le 0$.

II. y_1, y_2 and y_3 are linearly dependent on $0 \le x \le 1$.

III. y_1, y_2 and y_3 are linearly independent on $0 \le x \le 1$.

IV. y_1, y_2 and y_3 are linearly dependent on $-1 \le x \le 0$.

Which one of the following is correct?

a) Both I and II are true
b) Both I and III are true
c) Both II and IV are true
d) Both III and IV are true

54. Let $I = \int_C 2z\,dx + 2y\,dy + 2x\,dz$ where x, y, z are real, and let C be the straight line segment from point A: (0,2,1) to point B: (4,1,-1). The value of I is …….. **(Gate-17, 2-Mark)**

55. Let $f\ x = e^{x+x^2}$ for real x. From among the following, choose the Taylor series approximation of f(x) around x=0, which includes all powers of x less than or equal to 3. **(Gate-17, 2-Mark)**

a) $1 + x + x^2 + x^3$

b) $1 + x + \frac{3}{2}x^2 + x^3$

c) $1 + x + \frac{3}{2}x^2 + \frac{7}{6}x^3$

d) $1 + x + 3x^2 + 7x^3$

56. A three dimensional region R of finite volume is described by $x^2 + y^2 \ge z^3; 0 \le z \le 1 where\ x, y\ , z\ are\ real.$ the volume of R (up to two decimal places) is…….

(Gate-17, 2-Mark)

Career Avenues

57. The smaller angle (in degrees) between the planes $x + y + z = 1$ *and* $2x - y + 2z = 0$ *is* ... ... **(Gate-17, 1-Mark)**

58. The minium value of the function $f\ x = \frac{1}{3}x\ x^2 - 3$ in the interval $-100 \leq x \leq 100$ occurs at $x = \ldots\ldots\ldots\ldots$ **(Gate-17, 2-Mark)**

59. The value of the integrals $\int_0^1 (\int_0^1 \frac{x-y}{x+y^{\ 3}} dy) dx$ *and* $\int_0^1 (\int_0^1 \frac{x-y}{x+y^{\ 3}} dx) dy$ are **(Gate-17, 2-Mark)**

a) Same and equal to 0.5
b) Same and equal to -0.5
c) 0.5 and -0.5 respectively
d) -0.5 and 0.5 respectively

Career Avenues

Answers

1)a	2) c	3) d	4) a	5) c	6) a	7)b	8) c	9) a	10)d	11) a	12) a	13) b
14) a	15) a	16) b	17) b	18)a	19) c	20) c	21) d	22) d	23) a	24) 864	25) a	26) c
27) 0	28) 6	29) d	30) c	31) 3	32) 0	33) 7	34) 3	35) 6.28	36) c	37) b	38) b	39) 1
40) 3	41) a	42) b	43) 20	44) 4.712		45) c	46) 0	47) b	48) c	49) 90°	50) 2	51) d
52) 10	53)b	54) -11	55) c	56) 0.7853		57) 54.73°		58) -100		59) c		

Solutions

1. Answer is [a]

$$I = \frac{1}{\sqrt{2\pi}}\int_0^\infty e^{-x^2/8}dx$$

Put $\frac{x^2}{8} = t \Rightarrow x^2 = 8t$

$\Rightarrow \quad x = 2\sqrt{2t}$

$\therefore \quad dx = 4\frac{dt}{x} = \frac{4}{2\sqrt{2}}\frac{dt}{t}$

or $dx = \frac{\sqrt{2}}{\sqrt{t}}dt$

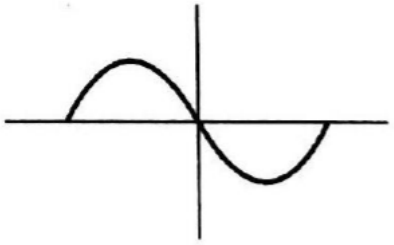
$$I = \frac{1}{\sqrt{2\pi}}\int_0^\infty e^{-t}\frac{\sqrt{2}}{\sqrt{t}}dt = \frac{1}{\sqrt{\pi}}\int_0^\infty e^{-t}t^{\frac{1}{2}-1}dt$$

$$\left[\because \lceil n = \int_0^\infty e^{-x}x^{n-1}dx \text{ and } \left\lceil\frac{1}{2} = \sqrt{\pi}\right.\right]$$

$\therefore \quad I = \frac{1}{\sqrt{\pi}}\left\lceil\frac{1}{2} = \frac{1}{\sqrt{\pi}}\cdot\sqrt{\pi} = 1\right.$

2. Answer is [c]

Given function has negative slope in +ve half and +ve slope in –ve half. So its differentiation curve is satisfied by [c]



3. Answer is [d]

From vector triple product.

$A \times (B \times C) = B(A.C) - C(A.B)$

$A = \bar{V}$, $B = \bar{V}$, $C = P$

$\bar{V}\times\bar{V}\times P = \bar{V}P - P\,\bar{V}\bar{V} = \bar{V}\,\bar{V}P - \bar{V}^2P$

4. Answer is [a]

$$\iint \Delta\times P\cdot ds = \oint P\cdot dl$$ (strokes Theorem)

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5. Answer is [c]

Let $\cos = t$

$= -\int_{+1}^{-1} 1-t^2 \ dt$

$-\sin\theta \ d\ \theta = dt,\ \theta = 0, \cos 0 = 1 = t$

$\theta = \pi, \cos\pi = -1 = t$

$= -\int_{-1}^{+1} 1-t^2 \ dt$

$= \left| t - \frac{t^3}{3} \right|_{-1}^{1}$

$I = \int_0^{\pi} \sin^3\theta \cdot d\theta$ $= \left(1-\frac{1}{3}\right) - \left(-1+\frac{1}{3}\right)$

$\int_0^{\pi} 1-\cos^2\theta \ \sin\theta d\theta,$ $= I = \frac{2}{3}+\frac{2}{3} = \frac{4}{3}$

6.

$$f'\ x\ = \frac{e^x\ 1+e^x\ -e^{2x}}{1+e^x\ ^2} = \frac{e^x}{1+e^x\ ^2}$$

$\because$ for all x, f'(x) is positive

So, $x \to \infty$, function's value increases.

7.

$y = x + 1$

$I = \int_1^2 y\ dx$

$= \int_1^2 x+1\ dx = \frac{x+1\ ^2}{2}\Big|_1^2$

$= \frac{1}{2}\ 9-4\ = 2.5$

8.

$\cot hx = \frac{\cos\ hx}{\sin\ hx} = \frac{1}{x}$

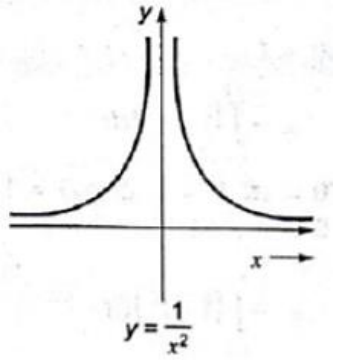
9.

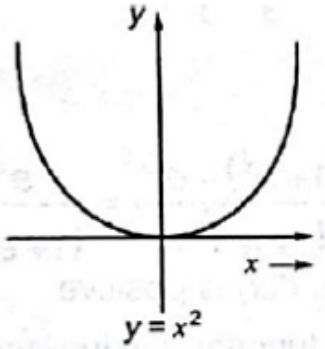
$\lim_{\theta\to 0} \frac{\frac{1}{2}\times\sin\left(\frac{\theta}{2}\right)}{\theta\times\frac{1}{2}} = \frac{1}{2}\lim_{\theta\to 0}\frac{\sin\frac{\theta}{2}}{\frac{\theta}{2}}$

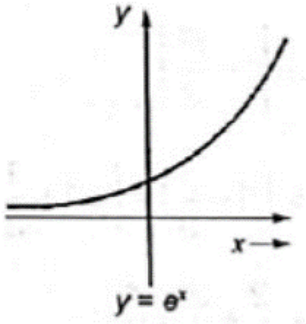
$= \frac{1}{2} = 0.5$

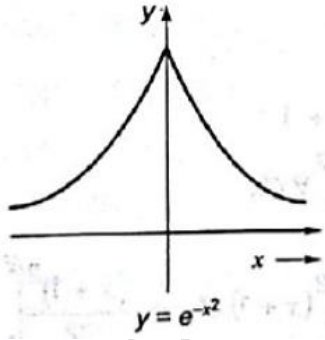
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10. Answer is [d]









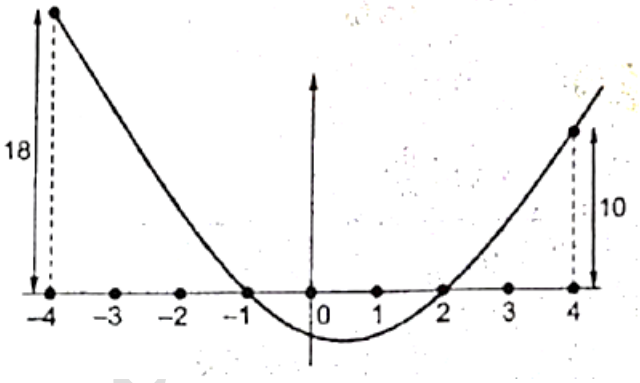
11. Answer is [a]

f(x) $= x^2 - x - 2$

$= (x + 1)(x - 2)$

f(– 4) = 18

f(+ 4) = 10



f(x) is maximum in interval [– 4, 4] at x = – 4

12. Answer is [a]

$\sin x = x - \frac{x^3}{\lfloor 3} + \frac{x^5}{\lfloor 5} - \frac{x^7}{\lfloor 7} + \ldots$

$\cos x = 1 - \frac{x^2}{\lfloor 2} + \frac{x^4}{\lfloor 4} - \frac{x^6}{\lfloor 6} + \ldots$

13. Answer is [b]

$f(x) = e^x + \sin x$

We wish to expand about $x = \pi$

Taylor's series expansion about x = a is

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$f(x) = f(a) + (x - a)\, f'(a)$

$$+\frac{x-a}{2!}^{2} f''\ a\ +\frac{x-a}{3!}^{3} f'''\ a\ \ldots$$

Now about $x = \pi$

Now about $x = \pi$

$$f(x) = f\ \pi\ +\ x-\pi\ \ f'\ \pi\ +\frac{x-\pi}{2!}^{2} f''\ \pi\ +\ldots$$

The coefficient of $x-\pi^{2}\ is\frac{f'''\ \pi}{2!}$

Here $f(x) = e^x + \sin x$

$f'(x) = e^x + \cos x$

$f''(x) = e^x - \sin x$

$f''(\pi) = e^{\pi} - \sin\pi = e^{\pi} - 0 = e^{\pi}$

The coefficient of $x-\pi^{2}$ is therefore

$$\frac{e^{\pi}}{2!} = 0.5\exp\ \pi$$

14. Answer is [a]

$$f(x) = e^x + e^{-x} = e^x + \frac{1}{e^x}$$

Arithmetic mean of e^x and $\frac{1}{e^x}$ is

$$\text{A.M.} = \frac{e^x + \frac{1}{e^x}}{2}$$

Geometric mean of e^x and $\frac{1}{e^x}$ is

$$\text{G.M.} = e^{\pi}\cdot\frac{1}{e^x} = 1$$

It is known that A.M. $\geq$ G.M.

$$\frac{\left(e^x + \frac{1}{e^x}\right)}{2} \geq 1$$

$$e^x + \frac{1}{e^x} \geq 2$$

Therefore, $(e^x + e^{-x})_{min} = 2$

15. Answer is [a]

Equation straight line from point (0, 0) to (1, 2) is

$$y - 0 = \frac{2-0}{1-0}\ x-0$$

or $y = 2x$

$g(x, y) = 4x^3 + 10y^4$

$= 4x^3 + 10(2x)^4$

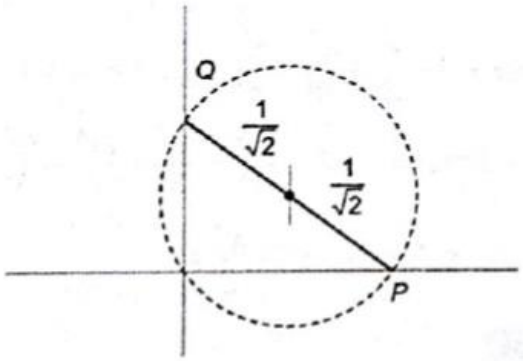
$= 4x^3 + 160\ x^4$

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$$\int_0^1 4x^3 + 160x^4 \, dx = \left(\frac{4x^4}{4} + \frac{160x^5}{5} \right)\Big|_0^1$$

$$= 1 + 32 = 33$$

16. Answer is [b]



$$= 2\int_P^O x\,dx + y\,dx$$

$$= 2\int_1^0 x\,dx + 2\int_0^1 y\,dy$$

$$= 2\left[\frac{x^2}{2}\Big|_1^0 + \frac{y^2}{2}\Big|_0^1 \right] = 0$$

17. Answer is [b]

$$f(x) = f(\pi) + \frac{x-\pi}{1!} f'(\pi) + \frac{(x-\pi)^2}{2!} f''(\pi) \ldots$$

Now $f(\pi) = \lim_{x\to\pi} \frac{\sin x}{x-\pi}$

$$= \lim_{x\to\pi} \frac{\cos x}{1} = -1$$

$f'(\pi) = 0$

$f''(\pi) = -\frac{1}{6}$

So the expansion is

$$f(x) = -1 + \left(-\frac{1}{6}\right)(x-\pi)^2 + \ldots$$

$$f(x) = -1 - \frac{(x-\pi)^2}{3!} + \ldots$$

18. Answer is [a]

$$e^y = x^{1/x}$$

Taking natural log on both sides,

$$y = \frac{1}{x}\ln x$$

Differentiating on both sides,

$$\frac{dy}{dx} = \frac{1}{x}\cdot\frac{1}{x} - \frac{\ln x}{x^2} = \frac{1}{x^2}(1 - \ln x)$$

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$$\frac{dy}{dx} = 0$$

$\Rightarrow \quad 1 - \ln x = 0$

$\Rightarrow \quad \ln x = 1$

$\Rightarrow \quad x = e$

$$\frac{d^2y}{dx^2} = \frac{-2}{x^3}\,1-\ln x + \frac{1}{x^2}\left(\frac{-1}{x}\right)$$

$$= -\frac{1}{x^3}\,2-\ln x+1$$

$$\frac{d^2y}{dx^2} = -\frac{1}{x^3}[3-\ln x]$$

Putting, x = e,

$$\left.\frac{d^2y}{dx^2}\right|_{x=e} = \frac{-1}{e^3}\,3-1 = \frac{-2}{e^3} < 0$$

Since, second derivative is negative, therefore y has maxima at x = e.

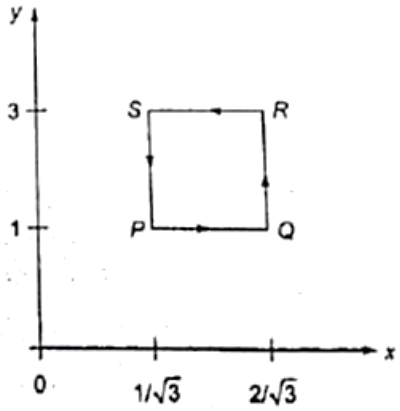
19. Answer is [c]

$$\vec{A} = xy\hat{a}_x + x^2\hat{a}y$$

$$\vec{l} = x\hat{a}_x + y\hat{a}_y$$

$$\vec{dl} = dx\hat{a}_x + dy\hat{a}_y$$

$\vec{A}\cdot\vec{dl}$ = xy dx + x^2 dy



P – Q:

y = 1, dy = 0

$$\int_P^Q \vec{A}\cdot\vec{dl} = \int_{1/\sqrt{3}}^{2/\sqrt{3}} x\,dx = \frac{x^2}{2}\Big|_{1/\sqrt{3}}^{2/\sqrt{3}} = \frac{1}{2}$$

Q – R

$$x = \frac{2}{\sqrt{3}},\ dx = 0$$

$$\int_Q^R \vec{A}\cdot\vec{dl} = \int_1^3 \left(\frac{2}{\sqrt{3}}\right)^2 dy = \frac{4}{3}\,3-1 = \frac{8}{3}$$

R – S

y = 3, dy = 0

$$\int_R^S \vec{A}\cdot\vec{dl} = \int_{2/\sqrt{3}}^{1/\sqrt{3}} 3x\,dx = \frac{3}{2}x^2\Big|_{2/\sqrt{3}}^{1/\sqrt{3}}$$

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$$= \frac{3}{2}\left(\frac{1}{3}-\frac{4}{3}\right)=\frac{-3}{2}$$

S – P

$$x = \frac{1}{\sqrt{3}},\ dx = 0$$

$$\int_S^P \vec{A}\cdot\vec{dl} = \int_3^1 \left(\frac{1}{\sqrt{3}}\right)^2 dy = \frac{1}{3}\,1-3 = \frac{-2}{3}$$

So, $$\oint_C \vec{A}\cdot\vec{dl} = \int_P^Q \vec{A}\cdot\vec{dl} + \int_Q^R \vec{A}\cdot\vec{dl} + \int_R^S \vec{A}\cdot\vec{dl} + \int_S^P \vec{A}\cdot\vec{dl}$$

$$= \frac{1}{2}+\frac{8}{3}-\frac{3}{2}-\frac{2}{3}=1$$

20. Answer is [c]
We need absolute maximum of
$f(x) = x^3 - 9x^2 + 24x + 5$ in the interval [1, 6]
First find local maximum if any by putting $f'(x) = 0$.
i.e. $f'(x) = 3x^2 - 18x + 24 = 0$
i.e. $x^2 - 6x + 8 = 0$

$$x = 2, 4$$

Now $f'(x) = 6x - 18$
$f''(2) = 12 - 18 = -6 < 0$
(So x = 2 is a point of local maximum)
and $f'(4) = 24 - 18 = +6 > 0$
(So x = 4 is a point of local minimum)
Now tabulate the values of f at end point of interval and at local maximum point, to find absolute maximum in given range, as shown below:

x	f(x)
1	21
2	25
6	41

Clearly the absolute maxima is at x = 6 and absolute maximum value is 41.

21. Answer is [d]
According to Stoke's theorem

$$\oint_C \vec{A}\cdot\vec{dl} = \iint_s \vec{V}\times\vec{A}\cdot\vec{ds}$$

22. Answer is [d]

$$\vec{V}\cdot\vec{A} = \frac{\partial A_x}{\partial x}+\frac{\partial A_y}{\partial y}+\frac{\partial A_z}{\partial z}$$

$$\vec{V}\cdot\vec{A} = \frac{\partial}{\partial x}\,x + \frac{\partial}{\partial y}\,y + \frac{\partial}{\partial z}\,z = 1+1+1$$

$$\vec{V}\cdot\vec{A} = 3$$

23. Answer is [a]
The Taylor's series expansion for

$$\sin x = x-\frac{x^3}{3!}+\frac{x^5}{5!}-\frac{x^7}{7!}+\cdots \qquad -\infty < x < \infty$$

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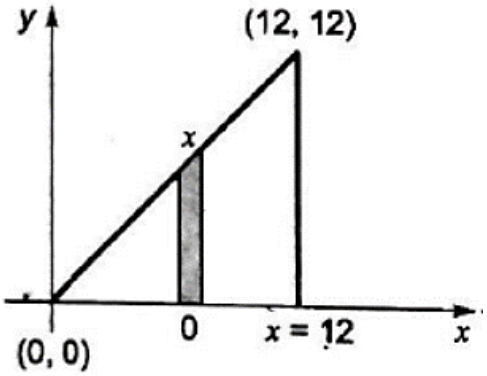
and

$\cos x = 1-\frac{x^2}{2!}+\frac{x^4}{4!}-\frac{x^6}{6!}\cdots \qquad -\infty < x < \infty$

$\therefore \quad 3\sin x + 2\cos x = 2+3x-\frac{2x^2}{2!}-\frac{3x^3}{3!}+\frac{2x^4}{4!}+\frac{3x^5}{5!}\cdots$

$= 2+3x-x^2-\frac{x^3}{2}+\cdots$

24.



Volume $= \iint z\,dxdy = \iint x+y\;dxdy$

$= \int_0^{12}\int_0^{x} x+y\;dxdy$

$= \int_0^{12}\left(xy+\frac{y^2}{2}\right)\Big|_0^{x} dx$

$= \int_0^{12}\left(x^2+\frac{x^2}{2}\right)dx$

$= \frac{3}{2}\int_0^{12} x^2dx = \frac{3}{2}\times\frac{x^3}{3}\Big|_0^{12}$

$= 864 \text{ unit}^3$

25. Answer is [a]

$f(t) = e^{-1} - 2e^{-2t}$

$f'(t) = -e^{-1} + 4e^{-2t}$

For maximum value $f'(t) = 0$

$f'(t) = 0 = -e^{-1} + 4e^{-2t}$

$\Rightarrow \quad 4e^{-2t} = e^{-t}$

$4e^{-1} = 4$

$\therefore \quad t = \log_e 4$

26. Answer is [c]

$\lim_{x\to\infty}\left(1+\frac{1}{x}\right)^x = e^{\lim_{x\to\infty}\frac{1}{x}\cdot x} = e^1 = e$

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27.

$f'(x) = \frac{1}{1+x} - 1 = 0$

$\frac{1-1-x}{1+x} = 0$

$\frac{x}{1+x} = 0$

$x = 0$

$f'(x) = \frac{-1}{1+x^2}$

$f'(0) = -1 < 0$

f(x) have maximum value at x = 0

$f(0) = \ln(1+0) - 0 = 0$

$f_{max} = 0$

28.

$f(x) = 2x^3 - 9x^2 + 12x - 3$

$f'(x) = 6x^2 - 18x + 12$

$f'(x) = 0$

$6x^2 - 18x + 12 = 0$

$x^2 - 3x + 2 = 0$

$x = 1, 2$

Hence critical points are {0, 1, 2, 3,}.

f(x) attains its maximum value at one of these points.

f(0) = – 3; f(1) = 2; f(2) = 1; f(3) = 6

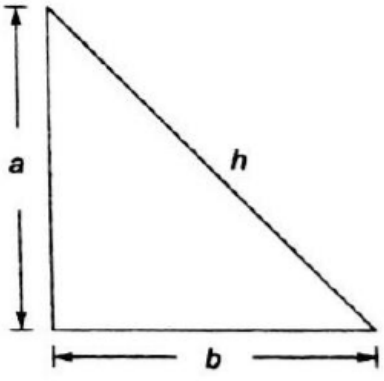
29. Answer is [d]

Given

Let $x(n) = \sum_{n=0}^{\infty} \frac{1}{n!}$

$= \frac{1}{0!} + \frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \frac{1}{4!} + \ldots$

30. Answer is [c]



Let a, b and h be the sides of Δ such that

$h^2 = a^2 + b^2$

Given,

a + h = constant (K) say

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$$\text{Area of } \Delta = \frac{1}{2}\times a\times b = \frac{1}{2}\times a\sqrt{h^2-a^2}$$

$$= \frac{1}{2}a\sqrt{K^2-2aK}$$

$$\frac{d\Delta}{da} = \frac{1}{2}\sqrt{K^2-2aK}+\frac{1}{2}a\times\frac{1}{2}\frac{-2K}{\sqrt{K^2-2ak}}=0$$

$$\frac{1}{2}\left[\frac{K^2-2aK-aK}{\sqrt{K^2-2aK}}\right] = 0$$

$$K(K-3a) = 0$$

$$K\neq 0 \Rightarrow \quad K = 3a$$

$$h = 2a$$

$$\theta = \cos^{-1}\left(\frac{a}{2a}\right)=\cos^{-1}\left(\frac{1}{2}\right)=60^o$$

31.

$$div\ r^2\bar{V}\ \ln r$$

$$= div\left(r^2\cdot\frac{1}{r}\left(\frac{\partial r}{\partial}\hat{a}_x+\frac{\partial r}{\partial y}\hat{a}_y+\frac{\partial r}{\partial z}\hat{a}_z\right)\right)$$

$$= div\left(r\frac{\partial r}{\partial x}\hat{a}_x+\partial\frac{\partial r}{\partial y}\hat{a}_y+r\frac{\partial r}{\partial z}\hat{a}_z\right) \quad \ldots \text{(i)}$$

Putting value of $\bar{r} = n\hat{a}_x+y\hat{a}_y+z\hat{a}_z$ in equation (i)

$$= div\left[r\ \hat{a}_x+\hat{a}_y+\hat{a}_z\ \right]$$

$$= div\ r\hat{a}_x+r\hat{a}_y+r\hat{a}_z$$

$$= \frac{\partial}{\partial x}r+\frac{\partial}{\partial y}r+\frac{\partial}{\partial z}r$$

$$= 1 + 1 + 1 = 3$$

32.

$$\vec{A} = \cos x\ \sin y\ \hat{a}_x+\ \sin x\ \ \cos y\ \hat{a}_y$$

$$\vec{A} = \cos x\ \sin y\ \hat{a}_x+\ \sin x\ \ \cos y\ \hat{a}_y+0\hat{a}_z$$

$$\bar{V}\times\vec{A} = \begin{vmatrix} \hat{a}_x & \hat{a}_y & \hat{a}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \cos x\sin y & \sin x\cos y & 0 \end{vmatrix}$$

$$= \hat{a}_z\ \cos x\cos y-\cos x\cos y\ =0$$

$$\therefore \quad \left|\bar{V}\times\vec{A}\right| = 0$$

33.

$$f(x, y, z) = x^2 + 3y^2 + z^3$$

$$\bar{V}f = \frac{\partial f}{\partial x}\hat{i}+\frac{\partial f}{\partial y}\hat{j}+\frac{\partial f}{\partial z}\hat{k}$$

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$$= 2x\hat{i} + 6y\hat{j} + 3z^2\hat{k}$$

$$\bar{V}f_{1,1,1} = 2\hat{i} + 6\hat{j} + 3\hat{k}$$

$$\left|\bar{V}f\right|_{1,1,1} = \sqrt{4+36+9} = \sqrt{49} = 7$$

34. Answer is [a]

$$f(x, y) = \frac{xy}{\sqrt{2}}\ x, y\ = \frac{x^2y + xy^2}{\sqrt{2}}$$

$$\bar{V}f = \frac{\partial f}{\partial x}\hat{i} + \frac{\partial f}{\partial y}\hat{j}$$

$$= \frac{2xy + y^2}{\sqrt{2}}\hat{i} + \frac{x^2 + 2xy}{\sqrt{2}}\hat{j}$$

The direction is $\hat{n} = \frac{1}{\sqrt{2}}\hat{i} + \frac{1}{\sqrt{2}}\hat{j}$

Directional derivative in direction of $\hat{n}$ is

$$= \left(\frac{2xy + y^2}{2} + \frac{x^2 + 2xy}{2}\right)_{1,1}$$

$$= \frac{3}{2} + \frac{3}{2} = 3$$

35.

$$\bar{V} \times \vec{f} = \begin{vmatrix} \hat{a}_x & \hat{a}_y & \hat{a}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ z & x & y \end{vmatrix} = \hat{a}_x + \hat{a}_y + \hat{a}_z$$

Now,

$$\int_s \bar{V} \times \vec{F} \cdot ds = \int \hat{a}_x + \hat{a}_y + \hat{a}_z \cdot \hat{a}_z \cdot dxdy$$

$$= \int_s dxdy = 2\pi\ 1^{\ 2}$$

$$= 6.28$$

36. Answer is [c]

$$\frac{\partial z}{\partial x} = y \ln\ xy\ + \frac{xy}{xy} y$$

$$\frac{\partial z}{\partial x} = y\left[\ln\ xy\ + 1\right]$$

$$\frac{\partial z}{\partial y} = x \ln\ xy\ + \frac{xy}{xy} \times x$$

$$\frac{\partial z}{\partial y} = x\left[\ln\ xy\ + 1\right]$$

Here $\boxed{x\frac{\partial z}{\partial x} = y\frac{\partial z}{\partial y}}$

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37. Answer is [b]
Since $f(1) \neq f(-1)$, Rolle's mean value theorem does not apply.
By Lagrange mean value theorem

$$f'(x) = \frac{f\ 1 - f\ -1}{1 - -1} = \frac{2}{2} = 1$$

$$-2x + 3x^2 = 1$$

$$x = 1, -\frac{1}{3}$$

x lies in (–1, 1)

$$\Rightarrow \quad x = -\frac{1}{3}$$

38. Answer is [b]

$f(x) = e^{-x}(x^2 + x + 1)$

$f'(x) = e^{x}(2x + 1) - e^{-x}(x^2 + x + 1)$

$= e^{-x}(x - x^2) = e^{-x}(x)\ (1 - x)$

Putting $f'(x) = 0$, we get

$x = 0$ or $x = 1$

$f'(x) = e^{-x}(1 - 2x) - e^{-x}(x - x^2)$

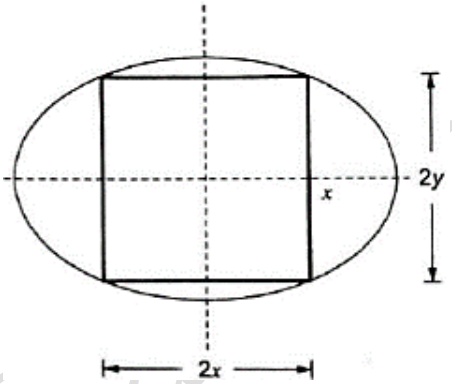
$= e^{-x}(1 - 3x + x^2)$

At $x = 0$, $f'(x) = 1$ (so we have a minimum).

At $x = 1$, $f'(x) = -\frac{1}{e}$

(so we have a maximum).
Only curve (b) shows a single local minimum at x =0 and a single local maximum at x = 1.

39. Explanation:



$x^2 + 4y^2 = 1$
Area of rectangle

$= 2x \cdot 2y = 4xy$

Let $f = (\text{Area})^2$

$= 16x^2y^2$

$= 4x^2(1 - x^2)$ $(\because \quad 1 - x^2 = 4y^2)$

$f'(x) = 0$

$$\frac{d}{dx}\left[4\ x^2 - x^4\ \right] = 0$$

$4(2x - 4x^3) = 0$

We get, $x = \pm\frac{1}{\sqrt{2}}$

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$$y = \pm\frac{1}{\sqrt{8}}$$

$$\text{Area} = 4xy = 4\times\frac{1}{\sqrt{2}}\times\frac{1}{\sqrt{8}} = 1$$

40. Explanation:

$$\int 12\cos 2\pi t \ \frac{\sin 4\pi t}{4\pi t}dt$$

$$= 12F.T.\left[\cos 2\pi t \ \frac{\sin 4\pi t}{4\pi t}\right]_{\omega=0}$$

$$= 12\times\frac{1}{4} = 3$$

41. Answer is [a]

Partial derivative w.r.t. y

$$\frac{\partial}{\partial y}\ x^2 + y^2 = 2y$$

Partial derivative w.r.t. x

$$\frac{\partial}{\partial x}\ 6y + 4x = 4$$

From given condition

$$2y = 4$$

$$\Rightarrow \quad y = 2$$

42. Answer is [b]

P: If f(x) is continuous at $x = x_0$, then it is also differentiable at $x = x_0$.

Q: If f(x) is continuous at $x = x_0$, then it may or may not be derivable at $x = x_0$.

R: If f(x) is differentiable at $x = x_0$, then it is also continuous $x = x_0$.

P is false

Q is true

R is true

43. Explanation:

Put $x = r\cos\theta$

$y = r\sin\theta$

$dxdy = rdrd\ \theta$

$$= \frac{1}{2\pi}\int_0^{2\pi}\int_0^2 r\ \cos\theta + \sin\theta\ + 10\ rdrd\ \theta$$

$$= \frac{1}{2\pi}\int_0^{2\pi}\int_0^2 r^2\ \cos\theta + \sin\theta\ + 10r\ drd\ \theta$$

$$= \frac{1}{2\pi}\left(\int_0^{2\pi}\cos\theta + \sin\theta\ \left(\frac{r^3}{3}\right)\Big|_0^2 d\theta + 10\int_0^{2\pi}\left(\frac{r^2}{2}\right)\Big|_0^2 d\theta\right)$$

$$= \frac{1}{2\pi}\int_0^{2\pi}\frac{8}{3}\ \cos\theta + \sin\theta\ d\theta + \frac{1}{2\pi}\int_0^{2\pi}5\cdot\ 4\ d\theta$$

$$= \frac{1}{2\pi}\left[\frac{8}{3}\ \sin\theta - \cos\theta\ \right]_0^{2\pi} + \frac{1}{2\pi}\cdot 20\ 2\pi$$

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$$= \frac{1}{2\pi}\left(\frac{8}{3}\ 0-1\ -\ 0-1\ +20\right) = 0+20 = 20$$

44. Answer is [a]

$$V = \int_{\rho=3}^{5}\int_{\phi=\frac{\pi}{8}}^{\frac{\pi}{4}}\int_{z=3}^{4.5} \rho d\rho\ d\phi\ dz$$

$$= \int_{3}^{4.5}\int_{\pi/8}^{\pi/4}\left(\frac{\rho^2}{2}\right)\Big|_3^5 d\phi dz$$

$$= \int_{3}^{4.5}\int_{\pi/8}^{\pi/4} 8\cdot d\phi dz = 8\phi\Big|_{\pi/8}^{\pi/4}\cdot z\Big|_3^{4.5}$$

$$= 8\left(\frac{\pi}{4}-\frac{\pi}{8}\right)\ 4.5-3$$

$$= 8\cdot\frac{\pi}{8}\cdot\ 1.5\ = 4.712$$

45. Answer is [c]

F'(x) = f(x) which is density function
F'(x) = f(x) < 0 when x < 0
∴ F(x) is decreasing for x < 0
F'(x) = f(x) > 0 when x > 0
∴ F(x) is increasing for x > 0

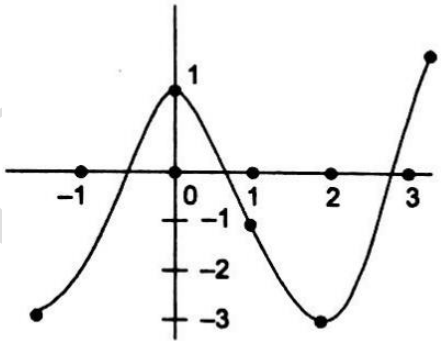
46. Explanation:

$$\int xy^2dx + x^2ydy = \int\int_R\left(\frac{d}{dx}\ x^2y\ -\frac{d}{dy}\ xy^2\ \right)dxdy$$

$$= \int\int_R\ 2xy-2xy\ = 0$$

47. Answer is [b]

f(x) = $x^3 - 3x^2 + 1$
f'(x) = $3x^2 - 6x$
f'(x) = 0

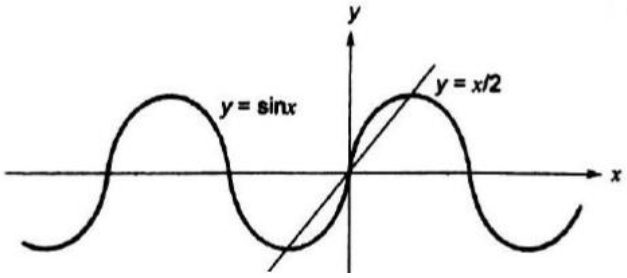


$3x^2 - 6x = 0$
$3x(x - 2) = 0$
x = 0, 2
f''(x) = 6x – 6

At x = 0 f''(0) = – 6 maxima
x = 2 f''(2) = 6 minima

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48. Answer is [c]



Hence 3 solutions.

49. Explanation:

$I = \hat{x}15\cos\omega t + \hat{y}5\sin\omega t$

$|I| = \sqrt{15\cos\omega t^{\,2} + 5\sin\omega t^{\,2}}$

$= \sqrt{225\cos^2\omega t + 25\sin^2\omega t}$

$= \sqrt{25 + 200\cos^2\omega t}$

$|I|$ is minimum when $\cos^2\omega t = 0$

or $\theta = \omega t = 90^o$

50. Explanation:

$$\int_0^1 \frac{1}{\sqrt{1-x}}dx = -2\int_0^1 \frac{1}{2\sqrt{1-x}}dx$$

$$= -2\sqrt{1-x}\Big|_0^1$$

$$= -2(0-1) = 2$$

51. Answer is [d]

$$\begin{bmatrix}4\\3\\-3\end{bmatrix} = a\begin{bmatrix}1\\0\\2\end{bmatrix} + b\begin{bmatrix}0\\1\\0\end{bmatrix} + c\begin{bmatrix}-2\\0\\1\end{bmatrix}$$

a – 2c = 4

b = 3

2a + c = –3

from here $a = -\frac{2}{5}$

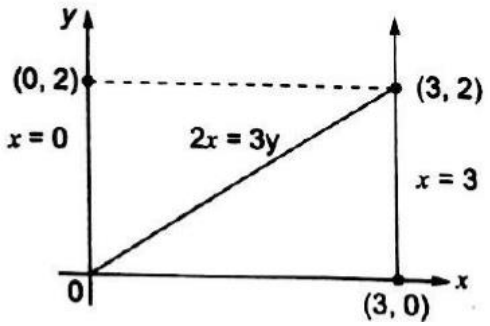
b = 3

$c = -\frac{11}{5}$

$u = -\frac{2}{5}e_1 + 3e_2 - \frac{11}{5}e_3$

52. Explanation:

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Volume $= \iiint dzdxdy = \iint zdydx$

$= \int_0^3 \int_0^{2/3x} 6 - x - y \; dydx$

$= \int_0^3 \left(6y - xy - \frac{y^2}{2} \right)\Big|_0^{2/3x} dx$

$= \int_0^3 \left(4x - \frac{8}{9}x^2 \right) dx$

$= \left(4\frac{x^2}{2} - \frac{8}{9} \cdot \frac{x^3}{3} \right)\Big|_0^3$

$= \left[2x^2 - \frac{8}{9}\left(\frac{x^3}{3} \right) \right]\Big|_0^3$

$= 18 - 8 = 10$

53. Answer is [b]
Any of the given three functions cannot be written as the linear combination of other two functions.
Hence, the statements I and III are correct.

54. Explanation:
A(0, 2, 1) and B(4, 1, – 1)
The equation of the line AB is

$\frac{x-0}{4-0} = \frac{y-2}{1-2} = \frac{3-1}{-1-1}$ = t say

x = 4t; y = – t + 2; z = – 2t + 1

dx = 4dt; dy = – dt; dz = – 2 dt

t varies from 0 to 1

$I = \int 2\ -2t+1\ 4dt + 2\ -t+2\ \ -dt\ \ +2\ 4t\ \ -2dt$

$= \int\ -16t+8+2t-4-16t\ dt = \int_0^1\ -30t+4\ dt$

$= \left(-30\frac{t^2}{2} + 4t \right)\Big|_0^1 = -15+4 = -11$

55. Answer is [d]
About x = 0

$f(x) = e^x e^{x^2}$

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$$= \left(1+x+\frac{x^2}{2!}+\frac{x^3}{3!}+....\right)\left(1+x^2+\frac{x^4}{2!}+\frac{x^6}{3!}+....\right)$$

$$= 1+x^2+\frac{x^4}{2}+\frac{x^6}{6}+...+x+x^3+\frac{x^5}{2}+\frac{x^7}{6}+....$$

$$+\frac{x^2}{2}+\frac{x^4}{2}+\frac{x^6}{4}+\frac{x^8}{12}....+\frac{x^3}{6}+\frac{x^5}{6}+\frac{x^7}{12}+\frac{x^9}{36}+....$$

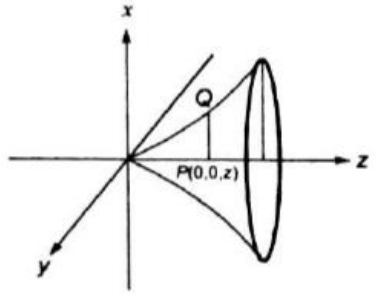
$$= 1+x+\frac{3}{2}x^2+\frac{7}{6}x^3$$

56. Explanation:

Let $x^2 + y^2 = t^2$

$t^2 = z^3$

Here revolution is about z axis



Volume of region $R = \int_0^1 \pi \ PQ^{\ 2}\, dz$

Here PQ is radius of circle at some z, which is given by

$PQ = \sqrt{x^2+y^2}$

$(PQ)^2 = x^2 + y^2 = z^3$

So, volume of region R

$$= \int_0^1 \pi t^2 dz = \int_0^1 \pi z^3 dz = \frac{\pi z^4}{4}\Big|_0^1$$

$$= \frac{\pi}{4} = 0.7853$$

57. Answer is [a]

If π is the angle between

$a_1x + b_1y + c_1z + d_1 = 0$

$a_2x + b_2y + c_2z + d_2 = 0$ then

$$\cos\theta = \frac{a_1a_2+b_1b_2+c_1c_2}{\sqrt{a_1^2+b_1^2+c_1^2}\sqrt{a_2^2+b_2^2+c_2^2}}$$

$a_1 = 1 \quad b_1 = 1 \quad c_1 = 1 \quad d_1 = 1$

$a_2 = 2 \quad b_2 = 2 \quad c_2 = 2 \quad d_2 = 0$

$$\cos\theta = \frac{1\ \ 2\ +\ 1\ \ -1\ +\ 1\ \ 2}{\sqrt{1^{\ 2}+1^{\ 2}+1^{\ 2}}\sqrt{2^{\ 2}+-1^{\ 2}+2^{\ 2}}}$$

$$= \frac{2-1+2}{\sqrt{3}\sqrt{9}} = \frac{3}{3\sqrt{3}} = \frac{1}{\sqrt{3}}$$

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$$\theta = \cos^{-1}\left(\frac{1}{\sqrt{3}}\right) = 54.73^{o}$$

58. Explanation:

$$f(x) = \frac{1}{3}x\ x^2 - 3$$

$$= \frac{1}{3}\ x^3 - 3x$$

$$f'(x) = \frac{1}{3}\ 3x^2 - 3\ = x^2 - 1$$

$$f'(x) = x^2 - 1 = 0$$

$$\Rightarrow \quad x = \pm 1$$

$$f''(x) = 2x$$

At x = 1, f''(1) = 2 = 0 $\Rightarrow$ minima

At x = – 1, f''(– 1) = – 2 < 0 $\Rightarrow$ maxima

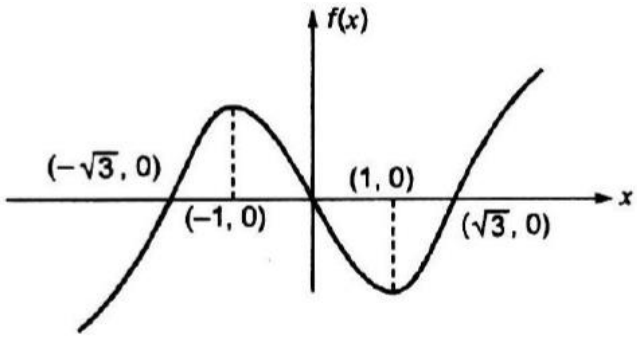
Minimum value of f(x) in [– 100, 100] is given by

Minimum {*f*(– 100), *f*(100), *f*(1)}

Minimum {– 333433.3, 333233.3, – 0.666} = – 3335433.3

Hence the minimum value occurs at x = – 100

Also graph of the function will be like



59. Answer is [c]

Integral $I_1 = \int_0^1 \int_0^1 \frac{x-y}{x+y^3}\,dy\,dx$

$$= \int_0^1 \left[\int_0^1 \left(\frac{2x}{x+y^3} - \frac{1}{x+y^2} \right) dy \right] dx$$

$$= \int \left[2x \left(\frac{-1}{2\ x+y^2} \right)\Bigg|_0^1 + \left(\frac{1}{x+y} \right)\Bigg|_0^1 \right] dx$$

$$= \int \frac{1}{x+1^2}\,dx = -\left[\frac{1}{x+1} \right]\Bigg|_0^1 = 0.5$$

and Integral $I_2 = I_2 = \int_0^1 \left(\int_0^1 \frac{x-y}{x+y^3}\,dx \right) dy$

$$= \int_0^1 \left(\int_0^1 \frac{1}{x+y^2} - \frac{2y}{x+y^3}\,dx \right) dy$$

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$$= \int_0^1 \left[\frac{-1}{x+y} + \frac{2y}{2\ x+y} \right]_0^1 dy$$

$$= \int_0^1 \frac{-1}{1+y^{\ 2}} dy = -\left[\frac{-1}{y+1} \right]_0^1 = -0.5$$

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